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Master Thesis

Modeling and Analysis of Tritium Trapping in Breeding Blankets

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Abstract

Currently, the most promising fuel for fusion reactors is deuterium–tritium. However, the global inventory of tritium is limited, and this hydrogen isotope has a short half-life; consequently, its availability decreases over time.

A key objective of fusion research is, therefore, the development of a self-sufficient reactor. In such a system, the tritium produced within the device can be used to sustain the deuterium–tritium fusion reaction.

Tritium production is achieved through a dedicated component, the breeding blanket, by interaction between neutrons and lithium, the breeding blanket generates a tritium surplus with respect to the burned fuel. Several breeding blanket concepts have been proposed and investigated. For example, the ARC design adopts a liquid breeder, whereas the DEMO design considers both solid and liquid breeder solutions. The produced tritium is extracted from the breeder by means of a tritium carrier, which is subsequently processed within the fuel cycle to separate tritium from the carrier. Nevertheless, all materials involved in breeding blankets are affected by the presence of trapping sites, which capture tritium and reduce the amount of tritium that can be extracted. The aim of this thesis is to investigate tritium transport with trapping, to model trapping phenomena in two simplified breeding blanket configurations, and to estimate the tritium concentration lost in traps. For the first time, a study on trapping in ceramic breeders has been performed.

The thesis begins with an introduction to the relevant physical context, followed by a description of the physical models implemented in FESTIM. The subsequent chapters focus on tritium trapping in breeding blankets. First, the modeling and analysis of an ARC breeding blanket are presented and used as a benchmark for the FESTIM models, allowing comparison with results available in the literature. The second part of the work is dedicated to the simulation of a ceramic breeding blanket, representing a novel study of tritium trapping in ceramic breeders. Finally, a comparison between the liquid immersion blanket and ceramic breeders reveals that solid breeders exhibit higher tritium trapping rates.

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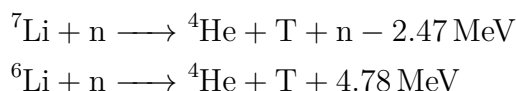
Chapter 1

Introduction

One of the most challenging aspects of fusion engineering is the fuel cycle. Fusion energy production is based on the deuterium–tritium reaction; however, global tritium reserves are limited due to the isotope’s short half-life. For this reason, fusion reactors must be designed not only to produce power but also to achieve tritium self-sufficiency by breeding more tritium than is consumed [1].

The most promising strategy for tritium production involves the interaction between neutrons released during fusion and lithium. Natural lithium consists of approximately 92.5% ${}^7\text{Li}$ and 7.5% ${}^6\text{Li}$ [1].

The interaction between a neutron and ${}^7\text{Li}$ is an endothermic reaction that produces a tritium atom and a secondary neutron. Conversely, the reaction with ${}^6\text{Li}$ consumes a neutron to produce tritium. The secondary neutron generated by the ${}^7\text{Li}$ reaction can subsequently be captured by another lithium nucleus. They are transmuted to tritium in following reactions:



The dedicated component for tritium production is the Breeding Blanket. Once tritium is generated, the tritium is removed from the materials by a tritium carrier. The figure of merit use for tritium breeding is the Tritium Breeding Ratio (TBR), defined as the ratio between the tritium produced in the blanket and the tritium consumed (burnt) in the plasma.

The fusion fuel cycle is the system designed to recover the maximum amount of tritium from the reactor. Its primary objective is to extract tritium produced in the breeding blanket via a carrier medium and subsequently separate it from other compounds by exploiting specific chemical and physical properties [1]. Additionally, the fuel cycle recovers unburnt fuel from the vacuum vessel. Since the vacuum pumps extract a mixture of fuel, He, and other impurities, a separation process is required to isolate the tritium from the other gasses. Once purified, the tritium is stored and recirculated for reuse [2] [3].

However, because the recovery processes and material interactions involve unavoidable losses, the system must produce a surplus of fuel to compensate. To ensure self-sufficiency, the TBR must be greater than unity ($\text{TBR} > 1$). This is necessary because the tritium extraction efficiency is always lower than 100 %, a fraction of the tritium must be stored as a reserve, and a significant amount can be trapped within the materials[1].

This inefficiency is evident during the tritium extraction from solid breeders. Once produced, tritium permeates through the solid materials to reach the carrier.

However, the presence of traps leads to tritium retention, thereby creating a mismatch between the tritium produced and the amount actually extracted.

Tritium is captured within materials due to *intrinsic defects* (existing in the material lattice) or *extrinsic traps* (defects induced by neutron irradiation). This trapping effect reduces the quantity of tritium that can be successfully extracted and processed during the fuel cycle, thereby increasing the tritium inventory within the reactor components [4].

Current research is still focused on the implantation and trapping of tritium in plasma-facing components (PFCs), which are in direct contact with the plasma. These studies highlight the detrimental effect of trapping within the bulk of Tungsten (W) [5] [6]. Nevertheless, recent efforts are also shifting toward investigating trapping phenomena within the breeding blankets. In fact, trapping studies have been conducted for the liquid immersion blanket of the ARC fusion reactor [4], as well as for the Water-Cooled Lithium-Lead (WCLL) design in DEMO [7]. Building on these works, the objective of this thesis is improving modeling and increasing the detail of the analysis of trapping phenomena in breeding blanket designs.

The first implementation involves the ARC breeding blanket. This is initially simulated in a 1D geometry, followed by a 2D study that incorporates the advection of the liquid breeder. Since tritium trapping in ARC components has already been documented in the literature, this model serves as a benchmark for the tool and provides reference data for trapping phenomena.

The second part of the analysis focuses on the simulation of a ceramic breeding blanket, representing a novel study of tritium trapping in ceramic materials. Ceramic materials are particularly of interest for solid breeder designs as the Helium-Cooled Pebble Bed (HCPB). This task is particularly challenging due to the limited availability of diffusion and solubility data for both ceramic breeders and neutron multiplier materials.

The initial 2D model represents a simplified pre-conceptual design of the DEMO breeding blanket, developed to explore trapping phenomenology within the ceramic breeder and neutron multiplier materials. Subsequently, a more sophisticated model was designed based on the final breeder pin concept, incorporating two distinct coolants: helium and water.

All models involving the ceramic breeder are evaluated under both surface-limited and diffusion-limited configurations. This approach ensures generality and allows for a comprehensive comparison between the two regimes.

Chapter 2

Methodology

2.1 Hydrogen transport model

Hydrogen transport in materials is modeled using the McNabb & Foster equations [8]. When tritium penetrates a material, it diffuses through the lattice; this tritium population is referred to as the mobile concentration, c_m [m^{-3}]. Tritium atoms can be captured by defects, forming the trapped concentration, c_t . However, if an atom gains sufficient thermal energy, it can escape, leading to a dynamic balance between trapping and detrapping processes [9].

The diffusion of mobile tritium is governed by Fick's law, where the hydrogen flux J [m^{-2}] is defined as:

$$J = -D \nabla c_m \quad (2.1)$$

The diffusion coefficient D [$\text{m}^2 \text{s}^{-1}$] follows an Arrhenius dependence:

$$D = D_0 \exp\left(\frac{-E_D}{k_B T}\right) \quad (2.2)$$

where D_0 is the pre-exponential factor, E_D [eV] is the activation energy for diffusion, T [K] is the temperature and $k_B = 8.617 \cdot 10^{-5} \text{ eV K}^{-1}$ is Boltzmann's constant [9]. Increasing the temperature leads to higher diffusivity; conversely, an increase in activation energy results in a decrease in diffusivity.

Each trap type i is characterized by a trap density n_i [m^{-3}], and associated with trapping and detrapping coefficients, k_i [$\text{m}^3 \text{s}^{-1}$] and ν_i [s^{-1}], respectively. The temporal evolution of c_m and c_t is described by the following system [9]:

$$\frac{\partial c_m}{\partial t} = \nabla \cdot (D \nabla c_m) + S - \sum_i \frac{\partial c_{t,i}}{\partial t} \quad (2.3)$$

$$\frac{\partial c_{t,i}}{\partial t} = k_i c_m (n_i - c_{t,i}) - \nu_i c_{t,i} \quad (2.4)$$

where:

- S [$\text{m}^{-3} \text{s}^{-1}$] is the volumetric source term for mobile tritium.
- $k_i c_m (n_i - c_{t,i})$ represents the trapping rate; the factor $(n_i - c_{t,i})$ accounts for the availability of unoccupied sites, ensuring the flux vanishes as traps approach saturation.
- $\nu_i c_{t,i}$ represents the detrapping rate, accounting for the release (due to thermal motion) of hydrogen from the traps.

Both $k_i = k_{0,i} \exp(-E_{k,i}/(k_B T))$ and $\nu_i = \nu_{0,i} \exp(-E_{\nu,i}/(k_B T))$ coefficients are governed by Arrhenius-type relations [9].

2.1.1 Trapping phenomenon

Traps are formed by microscopic and macroscopic defects that create deep potential wells (Figure 2.1), effectively hindering the diffusion process [10].

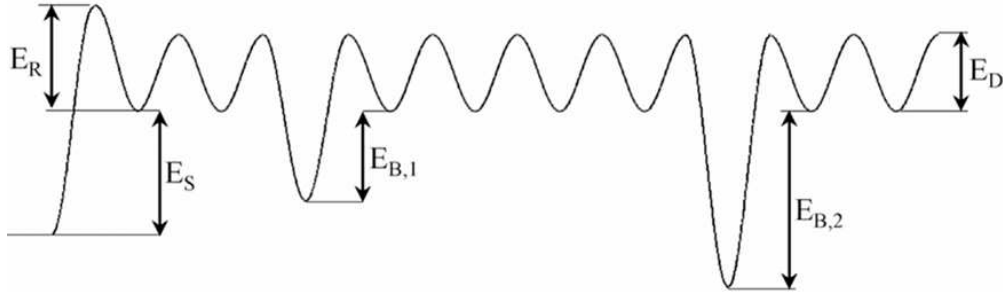


Figure 2.1: Potential energy [eV] diagram for a hydrogen atom [10].

Where:

- $E_R + E_S$ represents the total energy barrier for an atom to penetrate the lattice from the surface [10].
- E_D is the diffusion (or migration) energy barrier that atoms must overcome to move between interstitial sites [10].
- $E_{B,1}$ and $E_{B,2}$ are the binding energies associated with two different types of traps [10].
- $E_{T,1} = E_{B,1} + E_D$ and $E_{T,2} = E_{B,2} + E_D$ represent the detrapping energy barriers required for an atom to escape from those traps [10].

The presence of traps within the material lattice can significantly influence the tritium inventory required for reactor start-up. Figures 2.2 and 2.3 compare tritium inventories calculated using a model without traps (blue curve), one with intrinsic traps (green curve), and one including damage-induced or extrinsic traps (red curve). The two categories of traps are described below.

- **Intrinsic traps:** They are present due to fabrication process that produces defects inside the material. Intrinsic traps are present even if the material is not damaged and the trapping energy is usually < 0.5 eV [10]. Due to their low trapping energy and density, intrinsic traps have a negligible impact on the required Tritium Breeding Ratio (TBR) and the startup concentration (Figure 2.2). Consequently, the fuel cycle parameters remain virtually unchanged compared to a scenario where intrinsic traps are not considered [4].

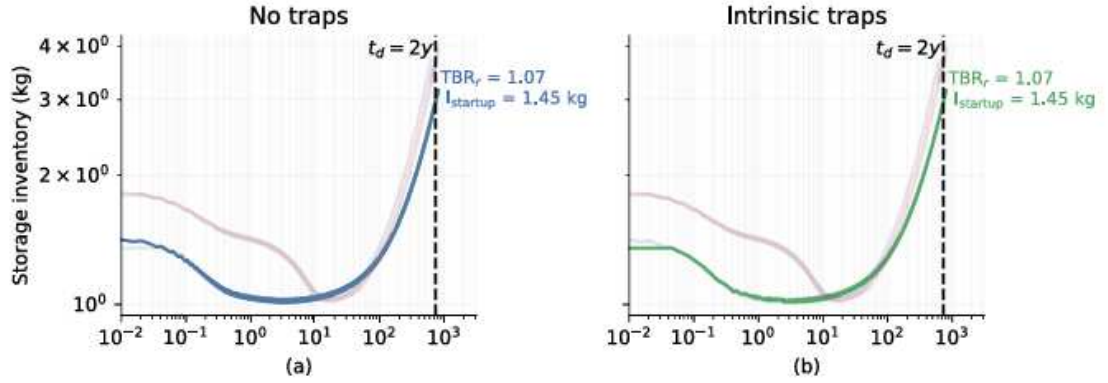


Figure 2.2: Evolution of storage concentration at a fixed doubling time. Case (a) is without traps instead case (b) considers the intrinsic traps. Reproduced from [4].

- **Extrinsic traps:** These traps are caused by irradiation damage, with associated trapping energies ranging between 1–2 eV [10]. Such damage can be produced by both ion and neutron irradiation. Ion-induced traps are primarily found in PFCs and are localized near the material surface; this is because ions, being charged particles, slow down within a very short range. In contrast, neutrons penetrate deeply through the material thickness, causing bulk damage throughout the Breeding Blanket. The effect of extrinsic traps is significantly more severe, impacting both the required TBR and the startup inventory (Figure 2.3) [4].

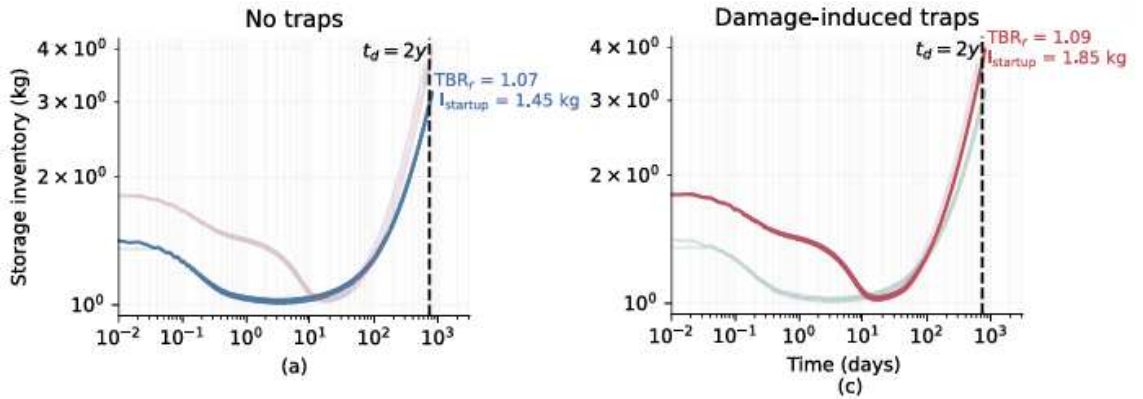


Figure 2.3: Evolution of storage inventory at a fixed doubling time. Case (a) is without traps instead case (c) considers the extrinsic traps. Reproduced from [4].

The density of extrinsic traps evolves over time and can be modeled as a balance between trap generation and annealing processes. Furthermore, higher temperatures limit the overall trapping effect, as they provide the atoms with sufficient thermal energy to overcome the associated potential wells [7]. How shown in Figure 2.4

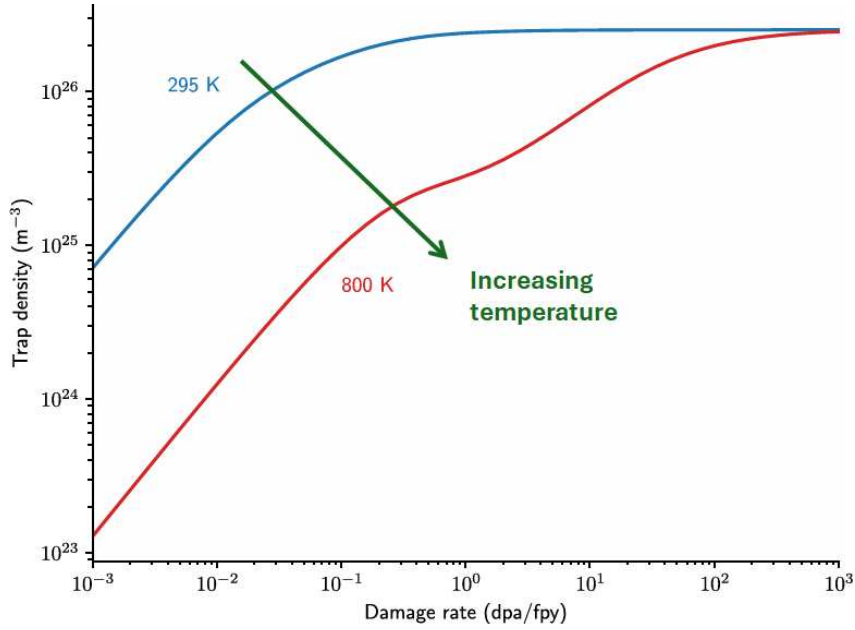


Figure 2.4: Trap density variation with temperature [7].

2.2 Local partial pressure continuity

The relationship between gas pressure and the concentration of dissolved species depends on the material involved. For materials where tritium undergoes dissociation upon entry (atomic dissolution), the partial pressure p [Pa] is described by Sieverts' law [9]:

$$p = \left(\frac{c_m}{K_S} \right)^2 \quad (2.5)$$

where K_S [$\text{m}^{-3} \text{Pa}^{-0.5}$] is the Sieverts' solubility constant. This behavior is characteristic of most metals. Conversely, if tritium remains in its molecular form within the lattice, the solubility follows Henry's law [9]:

$$p = \frac{c_m}{K_H} \quad (2.6)$$

where K_H [$\text{m}^{-3} \text{Pa}^{-1}$] represents the Henry's constant. This is typically observed in materials such as molten salts. The Sievert and Henry's constants follow the Arrhenius law [9].

In multi-material simulations, the primary objective is to ensure partial pressure continuity at the interface between different materials. This condition is essential to correctly model the concentration jump that occurs when tritium moves across a boundary [9].

If "a" and "b" are chosen arbitrarily as sides of an interface between two Sievert or two Henry materials, the partial pressure continuity is model as [9]:

$$\frac{c_m^a}{K_S^a} = \frac{c_m^b}{K_S^b} \quad (2.7)$$

$$\frac{c_m^a}{K_H^a} = \frac{c_m^b}{K_H^b} \quad (2.8)$$

The pressure continuity also needs to be ensured in an interface between a Sievert and a Henry material or vice versa. However, this continuity introduces a concentration jump at interface [9].

$$\left(\frac{c_m^a}{K_S^a} \right)^2 = \frac{c_m^b}{K_H^b} \quad (2.9)$$

2.3 Boundary conditions and Kinetic surface

To correctly close the McNabb & Foster model, a set of boundary conditions must be defined. The most frequently employed are Dirichlet, Neumann, and Robin conditions.

- Dirichlet's condition: imposition of the value of the solution at a boundary.

$$c_m = f(x, y, z, t) \text{ on } \partial\Omega \quad (2.10)$$

$$c_m = K_S \sqrt{p} \text{ on } \partial\Omega \quad (2.11)$$

$$c_m = K_H p \text{ on } \partial\Omega \quad (2.12)$$

this condition can also be used to approximate plasma implantation in near surface regions [9]. Where $\partial\Omega$ represents the boundary surface where the condition is applied.

- Neumann's condition: enforcing the value of the gradient of the solution [9].

$$J \cdot \mathbf{n} = -D \nabla c_m \cdot \mathbf{n} \text{ on } \partial\Omega \quad (2.13)$$

where $\mathbf{n}$ is the normal vector of the boundary.

- Robin's condition: Defining the gradient as a function of the local solution value. These conditions are used to simulate recombination and dissociation fluxes:

$$J \cdot \mathbf{n} = -D \nabla c_m \cdot \mathbf{n} = K_d p \quad (2.14)$$

$$J \cdot \mathbf{n} = -D \nabla c_m \cdot \mathbf{n} = -K_r c_m^n \quad (2.15)$$

In the recombination equation, the exponent of c_m is either $n = 1$ or $n = 2$, depending on the reaction order. The constants K_d and K_r denote the dissociation coefficient [$\text{m}^{-2}\text{s}^{-1}\text{Pa}^{-1}$] and the recombination coefficient [m^4s^{-1}], respectively. These fall under the category of Robin boundary conditions because the boundary value is not fixed, but rather explicitly dependent on the solution variable c_m [9].

2.3.1 Plasma implantation approximation

If the plasma implantation flux is approximated as a volumetric hydrogen source $\Gamma = \varphi_{\text{imp}} f(x)$, where $f(x)$ is a narrow Gaussian distribution, the mobile concentration profile can be further approximated by a triangular shape with its maximum at $x = R_p$ [m] [11]. As reported in Figure 2.5.

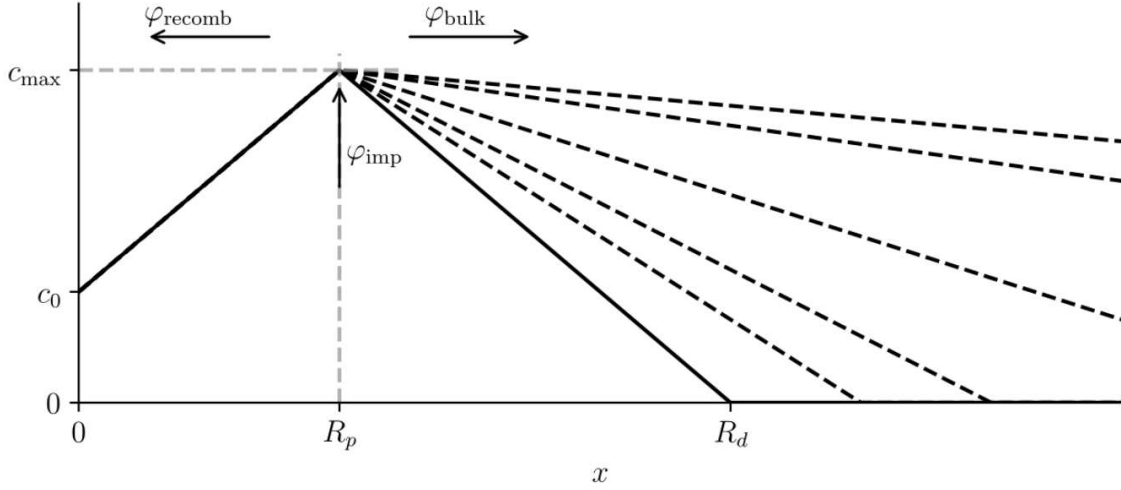


Figure 2.5: Concentration profile with a volumetric source (triangular shaped) with a maximum at $x = R_p$ [11].

The implantation flux is evaluated as:

$$\varphi_{\text{imp}} = \varphi_{\text{recomb}} + \varphi_{\text{bulk}} \quad (2.16)$$

where:

- φ_{recomb} [m^{-2}] is the recombination flux.
- φ_{bulk} is the flux of hydrogen atoms directed toward the bulk of the material, expressed as:

$$\varphi_{\text{bulk}} = D \cdot \frac{c_m}{R_d(t) - R_p} \quad (2.17)$$

- R_d is the diffusion depth.
- R_p is the implantation range.

The recombination flux can be evaluated considering $R_d \gg R_p$ and $\varphi_{\text{bulk}} \rightarrow 0$. φ_{recomb} is expressed as:

$$\varphi_{\text{recomb}} = \varphi_{\text{imp}} = D \cdot \frac{c_m - c_0}{R_p} \Leftrightarrow c_m = \frac{\varphi_{\text{imp}} R_p}{D} + c_0 \quad (2.18)$$

if a second order recombination is assumed, φ_{recomb} can also be expressed as a recombination coefficient K_r :

$$\varphi_{\text{recomb}} = \varphi_{\text{imp}} = K_r c_0^2 \Leftrightarrow c_0 = \sqrt{\frac{\varphi_{\text{imp}}}{K_r}} \quad (2.19)$$

substituting the expression of c_0 in the previous equation, the mobile concentration is directly retrieved for recombination as:

$$c_m = \frac{\varphi_{imp} R_p}{D} + \sqrt{\frac{\varphi_{imp}}{K_r}} \quad (2.20)$$

if the recombination is fast: $K_r \rightarrow \infty$, the mobile concentration becomes

$$c_m = \frac{\varphi_{imp} R_p}{D} \quad (2.21)$$

The same procedure can be followed in case of dissociation and the mobile concentration is expressed as:

$$c_m = \frac{\varphi_{imp} R_p}{D} + \sqrt{\frac{\varphi_{imp} + K_d p}{K_r}} \quad (2.22)$$

In conclusion, this condition here is a Dirichlet boundary condition [11].

2.3.2 Surface physics

For the scope of this work, we focus on the recombination process, which occurs when tritium atoms reach the surface and recombine upon exiting the surface. This specific type of recombination is known as Langmuir-Hinshelwood recombination (Figure 2.6) [12]; it occurs, for example, when tritium is released into a helium purge gas.

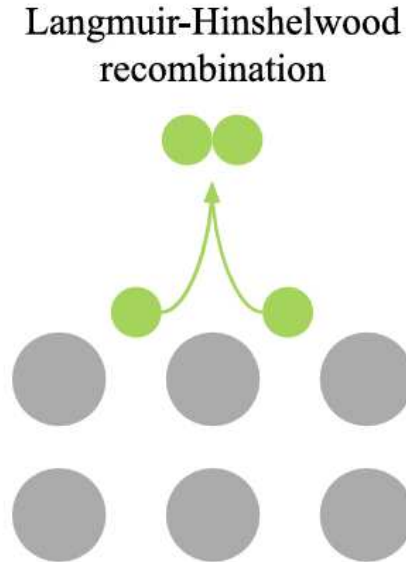


Figure 2.6: Langmuir-Hinshelwood recombination. Gray circles represent the metal and green circles the adsorbed atoms [12].

When tritium is released in water, it does not recombine or dissociate. In this case, the physical phenomenon involved is the isotopic exchange. The reactions involved are the following [13]:



Isotopic exchange is a physical phenomenon that occurs when water is used to cool down the breeding blanket, leading to the uptake and transport of permeated tritium [13].

2.3.3 Diffusion limited model VS surface limited model

Tritium transport models can be categorized into diffusion limited models (DLM) or surface limited models (SLM). The distinction between these approaches lies in whether surface effects are taken into account. The (dimensionless) permeation parameter W is defined as:

$$W = \frac{k_d t \sqrt{p}}{D K_S} \quad (2.25)$$

Where:

- $k_d [\text{m}^{-2} \text{Pa}^{-1} \text{s}^{-1}]$ is the dissociation constant
- $t [\text{m}]$ is the material thickness
- p is the partial pressure
- D is the tritium diffusion coefficient
- K_S is the Sievert constant.

If $W \gg 1$, surface effects become negligible and the DLM can be applied using the mass transport coefficient; otherwise, the SLM must be used to account for surface kinetics [14].

Mass transfer can be evaluated using an analogy with heat transfer. Specifically, both the thermal and mass boundary layers are related to the velocity field through the Reynolds number, with the Prandtl (Pr) and Schmidt (Sc) numbers playing analogous roles. Consequently, the temperature and concentration profiles within the boundary layers take the same functional form. The Nusselt (Nu) and Sherwood (Sh) numbers are similarly analogous [15]. D_{AB} represents the binary diffusion coefficient of species A in medium B at a constant temperature.

Where:

$$Sc = \frac{\nu}{D_{AB}} \Rightarrow \text{Ratio between momentum and mass diffusivity.} \quad (2.26)$$

$$Sh_L = \frac{h_m L}{D_{AB}} \Rightarrow \text{Dimensionless concentration gradient at surface.} \quad (2.27)$$

2.4 FESTIM

FESTIM (Finite Element Simulation of Tritium In Materials) is an open-source finite element code developed in Python for simulating hydrogen transport in materials based on the McNabb & Foster model [9].

Built on the FEniCS library (a finite element framework written in Python and C++) [16], FESTIM enables the transformation of a problem’s strong formulation into its weak formulation. This variational form is then discretized; in FEniCS, the underlying linear algebra is handled by PETSc [17]. While non-linear equations are solved using Newton’s method, users can choose between direct and iterative linear solvers [9].

FESTIM has been validated against thermal desorption experiments, profilometry data, and hydrogen diffusion models. Regarding verification, the tool was tested using the Method of Manufactured Solutions (MMS)—a mathematical technique employed to verify numerical codes. The resulting solutions ensure correct implementation of the code’s underlying physics [9].

FESTIM provides built-in meshing capabilities; specifically, one-dimensional meshes are generated directly within the code. However, for the complex geometries investigated in this thesis, the domains were designed and discretized using SALOME [18]. These meshes were subsequently converted into the XDMF format using the meshio package [19] to ensure compatibility with the solver.

All simulations are conducted in both steady-state and transient regimes, primarily to check the model’s consistency and robustness. However, to maintain conciseness, the results presented hereafter focus on the steady-state solutions, with transient evolutions included only where essential for the discussion.

Chapter 3

ARC breeding blanket

The ARC (Affordable Robust Compact) reactor is a fusion pilot power plant concept. Originally proposed by the Plasma Science and Fusion Center (PSFC) at the Massachusetts Institute of Technology (MIT), it is a Tokamak based on D-T reactions. The strengths of this design lie in its compactness, achieved through the use of high magnetic fields produced by REBCO superconductors, as well as the implementation of a molten salt (FLiBe) that acts as a breeding blanket and as a neutron multiplier and reduces radiation damage from high-energy neutrons [20] [21].

In this thesis, the focus is on the materials composing the first wall and the breeding blanket. The design features a 1 mm thick Tungsten layer as the first wall, supported by 10 mm of Inconel 718 [22]. This is followed by a 20 mm thick FLiBe channel within a double-wall vacuum vessel (VV), where the second wall consists of a 30 mm thick Inconel 718 layer [20]. A 10 mm thick Beryllium layer is placed between the channel and the outer VV wall. The double vacuum vessel cross section is reported in the picture 3.1.

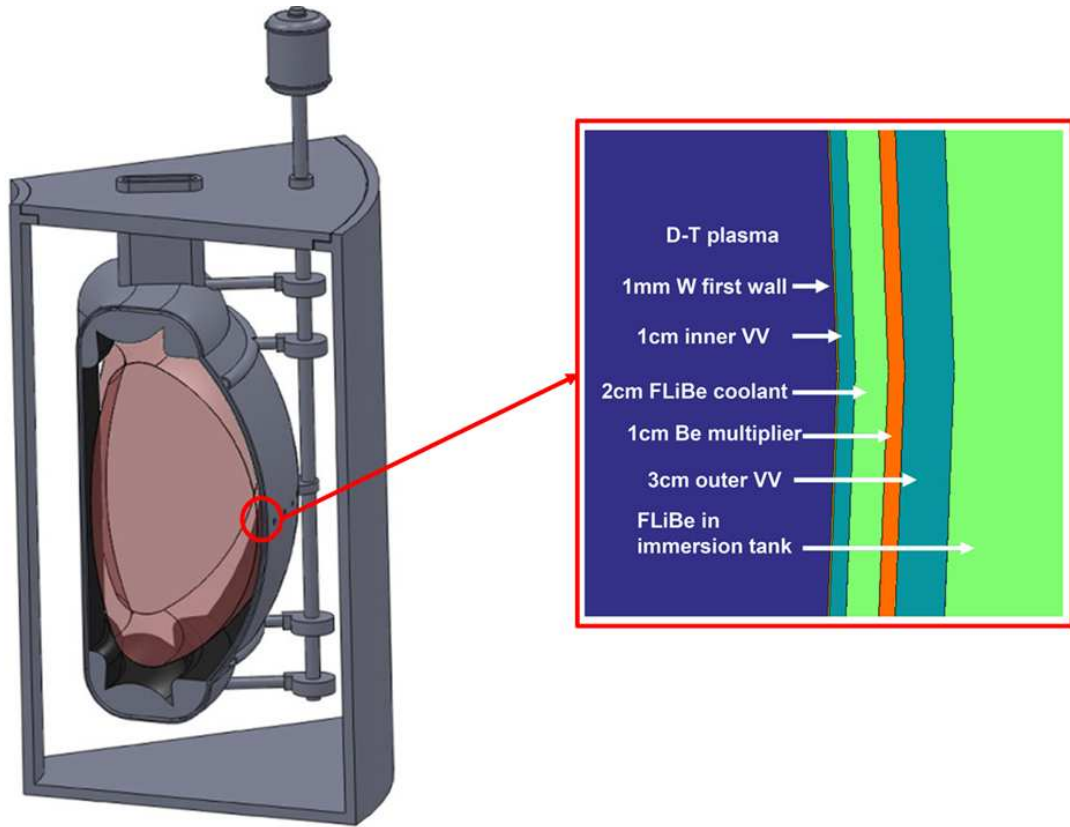


Figure 3.1: Vacuum vessel design parameters with FLiBe coolant channel [20], [23].

The geometry adopted for this model is a simplified representation of the one previously detailed. Since Inconel 718 is poorly characterized from the tritium trapping standpoint, it is replaced by Eurofer. In fact, the goal is not to assess tritium trapping in an ARC breeding blanket, which was already done in [4]. Instead, this model provides an effective benchmark case (against results presented in [4]) before modeling solid breeder designs. Specifically, the plasma is in contact with a 10 mm thick Tungsten first wall, supported by a 30 mm Eurofer [24] layer and a 20 mm FLiBe layer. The domain has a height of 10 mm (Figure 3.2). The decision to replace Inconel 718 with Eurofer is due to a greater understanding of trapping phenomenon.

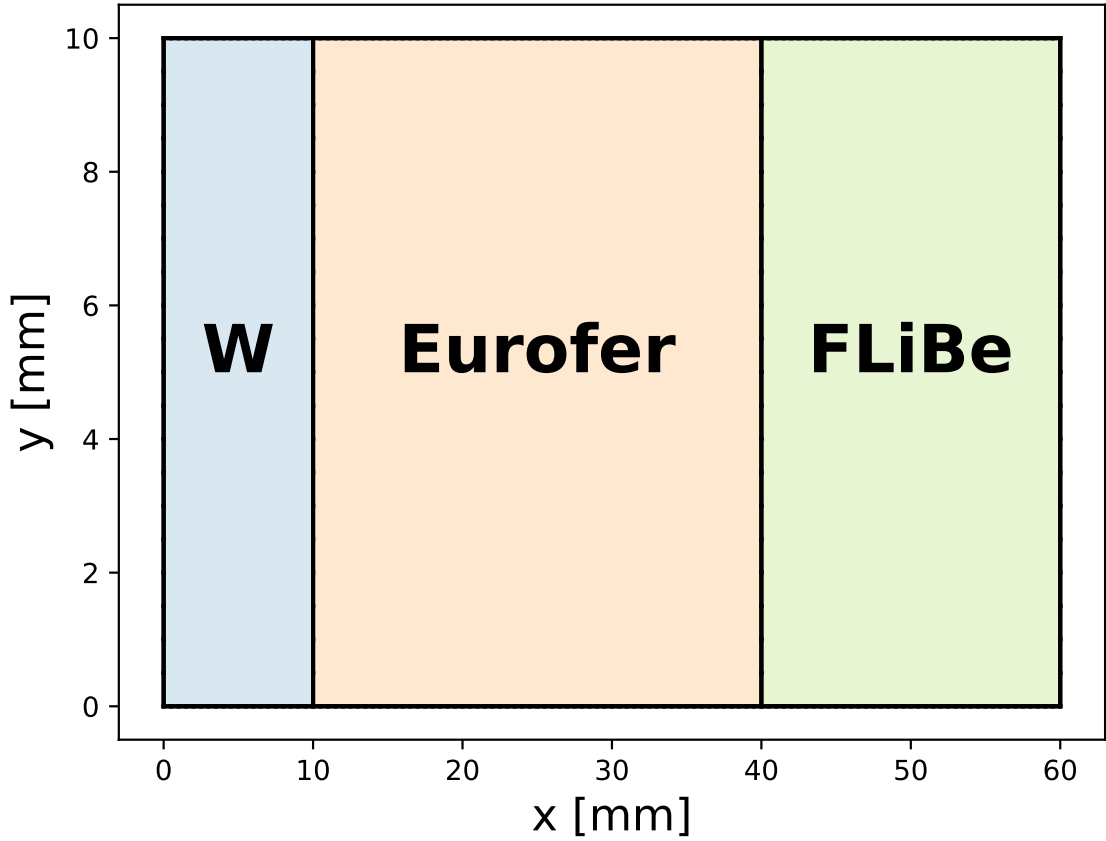


Figure 3.2: Simplified domain of ARC breeding blanket. ■ Tungsten, ■ Structural material, ■ Molten salt.

3.1 Physical Parameters

Physical parameters common to both 1D and 2D models are now introduced. For the model implementation, the quantities required for tritium transport include Sievert's and Henry's laws, as well as the Arrhenius parameters for the diffusion coefficient. Table 3.1 summarizes the values used. The equations are subject to specific temperature ranges; however, the model occasionally operates outside these bounds. These equations were used regardless, as they constitute the only data currently available in the literature.

Table 3.1: Transport Properties for ARC Materials.

Domain	Sievert's and Henry's laws	Diffusion law
FLiBe [25]	$k_H = 4.75 \cdot 10^{22} \exp\left(\frac{-0.36 \text{ eV}}{k_B \cdot T}\right)$	$9.3 \cdot 10^{-7} \exp\left(\frac{-0.44 \text{ eV}}{k_B \cdot T}\right)$
Tungsten [26] [27]	$k_S = 1.83 \cdot 10^{24} \exp\left(\frac{-1.04 \text{ eV}}{k_B \cdot T}\right)$	$4.1 \cdot 10^{-7} \exp\left(\frac{-0.39 \text{ eV}}{k_B \cdot T}\right)$
Eurofer [28] [27]	$k_S = 1.35 \cdot 10^{22} \exp\left(\frac{-0.16 \text{ eV}}{k_B \cdot T}\right)$	$4.57 \cdot 10^{-7} \exp\left(\frac{-0.23 \text{ eV}}{k_B \cdot T}\right)$

Regarding the materials involved, traps are present in Tungsten and Eurofer. Their properties are reported in Tables 3.2 and 3.3. Due to the lack of specific experimental data in the literature regarding extrinsic traps in Eurofer, the same trap parameters as Tungsten were assumed. This approach is justified by the similar nature of radiation-induced defects expected in both materials under the operating conditions of the ARC reactor, as already done in [4]. All k_0 values are evaluated using the formula from [4]:

$$k_0 = \frac{D_0}{\beta d^2} \quad (3.1)$$

where β is the interstitial site density [m^{-3}], D_0 is the diffusion pre-exponential factor [$\text{m}^2 \text{s}^{-1}$], and d is the average distance between two interstitial sites [m]. The trapping activation energy E_k is assumed to be equal to the diffusion activation energy E_D . This assumes that the rate-limiting step for trapping is the interstitial migration of the solute to the trap site, with no additional energy barrier for capture.

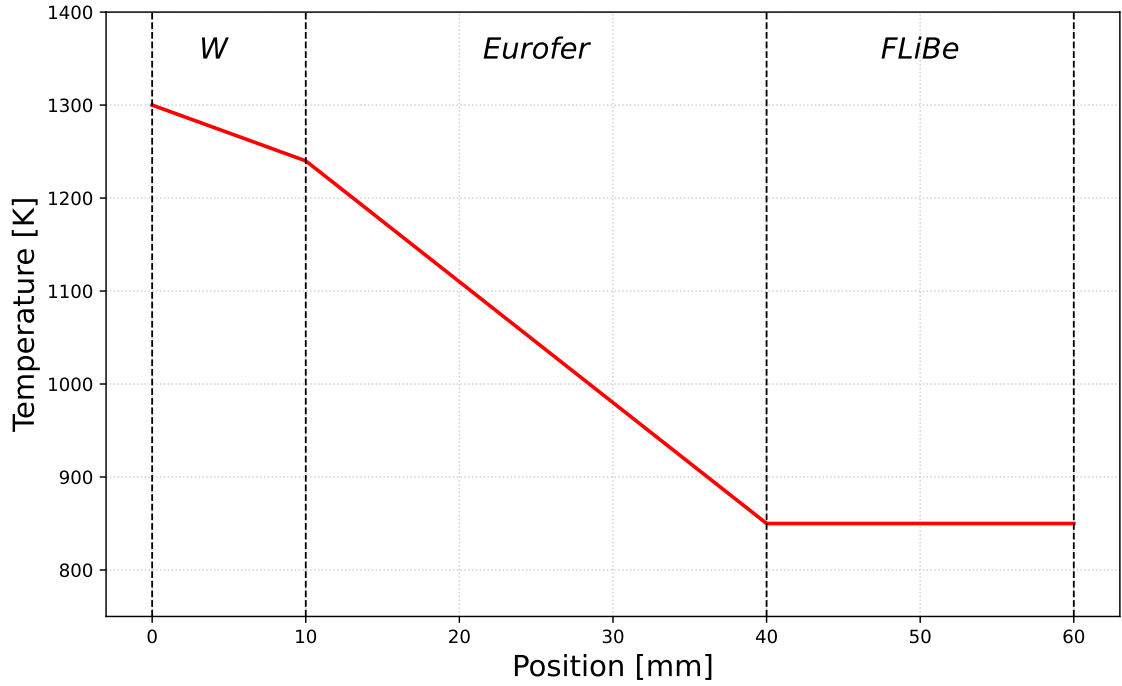
Table 3.2: Trap Properties for Tungsten [7].

Trap	k_0 [$\text{m}^3 \text{s}^{-1}$]	E_k [eV]	ν_0 [s^{-1}]	E_ν [eV]	Trap density [m^{-3}]
Intrinsic Trap #1	$8.96 \cdot 10^{-17}$	0.28	$1 \cdot 10^{13}$	1.04	$2.4 \cdot 10^{22}$
Extrinsic Trap #2	$8.96 \cdot 10^{-17}$	0.39	$1 \cdot 10^{13}$	1.15	$6.9 \cdot 10^{25}$
Extrinsic Trap #3	$8.96 \cdot 10^{-17}$	0.39	$1 \cdot 10^{13}$	1.35	$7 \cdot 10^{25}$
Extrinsic Trap #4	$8.96 \cdot 10^{-17}$	0.39	$1 \cdot 10^{13}$	1.65	$6 \cdot 10^{25}$
Extrinsic Trap #5	$8.96 \cdot 10^{-17}$	0.39	$1 \cdot 10^{13}$	1.85	$4.7 \cdot 10^{25}$
Extrinsic Trap #6	$8.96 \cdot 10^{-17}$	0.39	$1 \cdot 10^{13}$	2.05	$2 \cdot 10^{25}$

Table 3.3: Trap Properties for Eurofer [7].

Trap	k_0 [m ³ s ⁻¹]	E_k [eV]	ν_0 [s ⁻¹]	E_ν [eV]	Trap density [m ⁻³]
Extrinsic Trap #7	$8.43 \cdot 10^{-17}$	0.45	$1 \cdot 10^{13}$	1.15	$6.9 \cdot 10^{25}$
Extrinsic Trap #8	$8.43 \cdot 10^{-17}$	0.45	$1 \cdot 10^{13}$	1.35	$7 \cdot 10^{25}$
Extrinsic Trap #9	$8.43 \cdot 10^{-17}$	0.45	$1 \cdot 10^{13}$	1.65	$6 \cdot 10^{25}$
Extrinsic Trap #10	$8.43 \cdot 10^{-17}$	0.45	$1 \cdot 10^{13}$	1.85	$4.7 \cdot 10^{25}$
Extrinsic Trap #11	$8.43 \cdot 10^{-17}$	0.45	$1 \cdot 10^{13}$	2.05	$2 \cdot 10^{25}$

The temperature profile is assumed and modeled with a linear decrease across the Tungsten and Eurofer layers, while a constant temperature is maintained within the FLiBe. Specifically, the Tungsten temperature decreases from 1300 K to 1240 K, whereas the Eurofer temperature drops from 1240 K to 850 K. The FLiBe temperature is fixed at 850 K, representing the average between the inlet (800 K) and outlet (900 K) temperatures [27]. The temperature profile is shown in Figure 3.3; however, it does not account for any thermomechanical stresses resulting from these thermal conditions.

**Figure 3.3:** Temperature profile set in the ARC models.

3.2 1D model

The 1D model is shown in Figure 3.4; it is based on a nodal discretization of the geometry illustrated in Figure 3.2. The following boundary conditions are applied:

- **Implantation boundary condition** on the left surface, with a flux of $7 \cdot 10^{15} \text{ m}^{-2} \text{ s}^{-1}$ and an implantation depth of $9.52 \cdot 10^{-10} \text{ m}$ [4].
- **Dirichlet boundary condition** on the right surface, with a concentration value of $2.77 \cdot 10^{19} \text{ m}^{-3}$ retrieved by previous calculation.
- **Volumetric source** applied to the FLiBe region to account for tritium generation, with a value of $1.1 \cdot 10^{19} \text{ m}^{-3} \text{ s}^{-1}$ taken from [29].

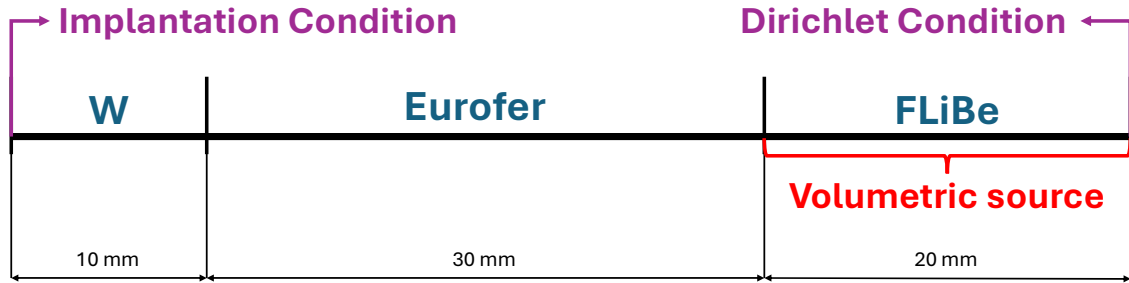


Figure 3.4: 1D model of simplify ARC breeding blanket.

3.3 1D model results

The simulation results are presented in this section. The simulation has been performed in steady state and transient mode.

Mobile concentration refers to the concentration of tritium capable of permeating through the materials. Figure 3.5 illustrates the evolution of the mobile concentration profile over time.

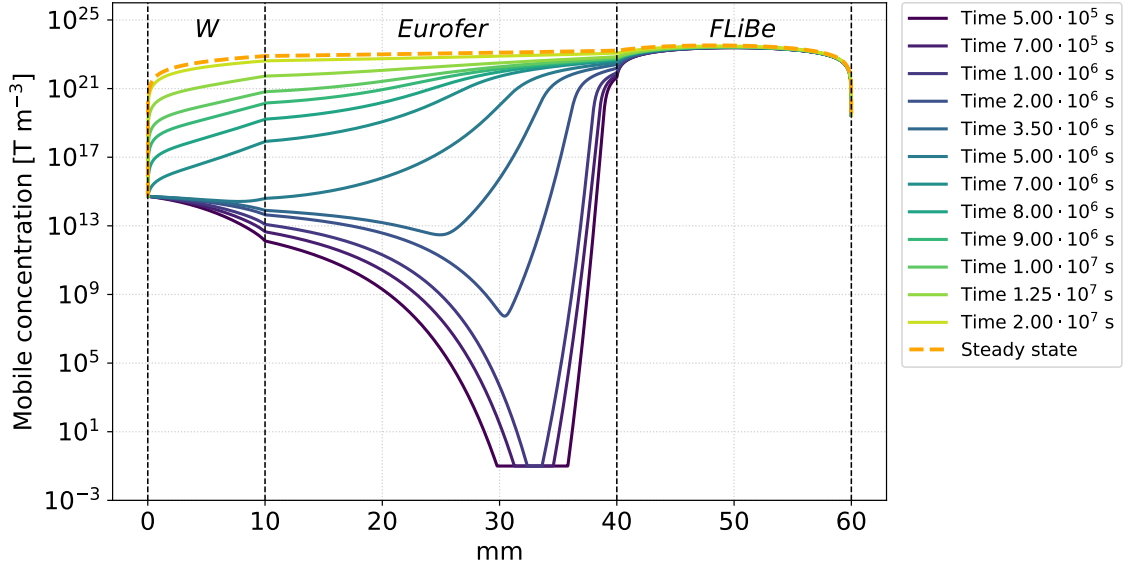


Figure 3.5: Mobile concentration evolution in 1D domain.

Figure 3.5 shows that tritium permeates from the plasma through the Tungsten into the Eurofer, while also migrating from the FLiBe toward the Eurofer. While the path from the plasma is expected, the latter is driven by the tritium breeding within the FLiBe, which creates a concentration gradient that results in a net flux toward the Eurofer.

Trapped concentration is shown in Figure 3.6, representing the evolution over time of the tritium trapped within the Tungsten and Eurofer layers. For the sake of completeness, FLiBe is also included in the plot; however, since no traps are defined for this material, the trapped concentration is zero.

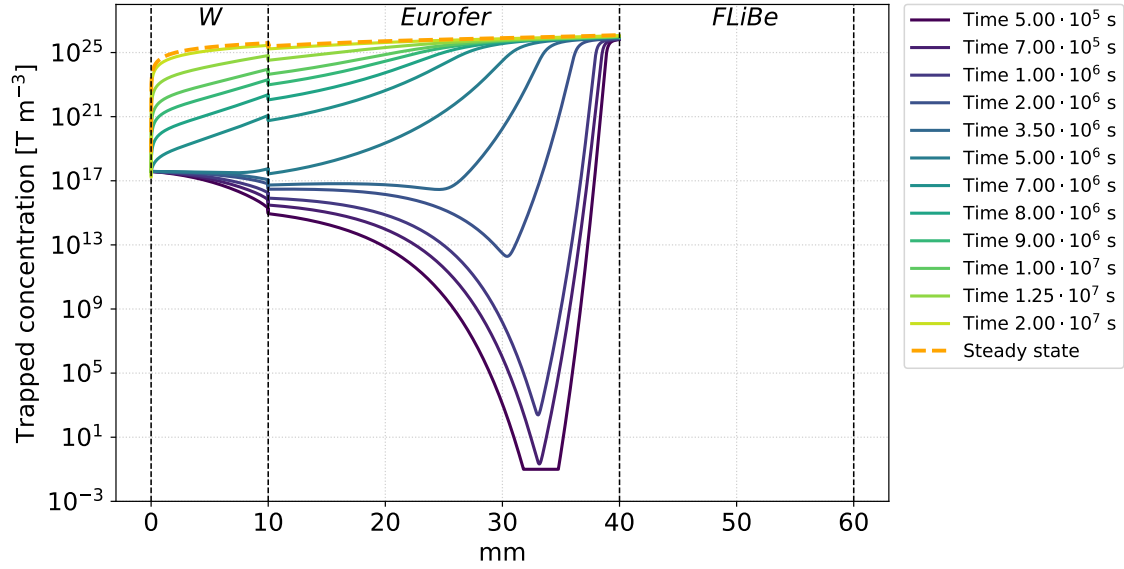


Figure 3.6: Trapped concentration evolution across the 1D multi-layer modelain.

Figures 3.7 and 3.8 show the evolution of the trapped concentration for each individual trap. Figure 3.7 refers to the traps within the Tungsten layer (table 3.2), while Figure 3.8 refers to those within the Eurofer layer (table 3.3).

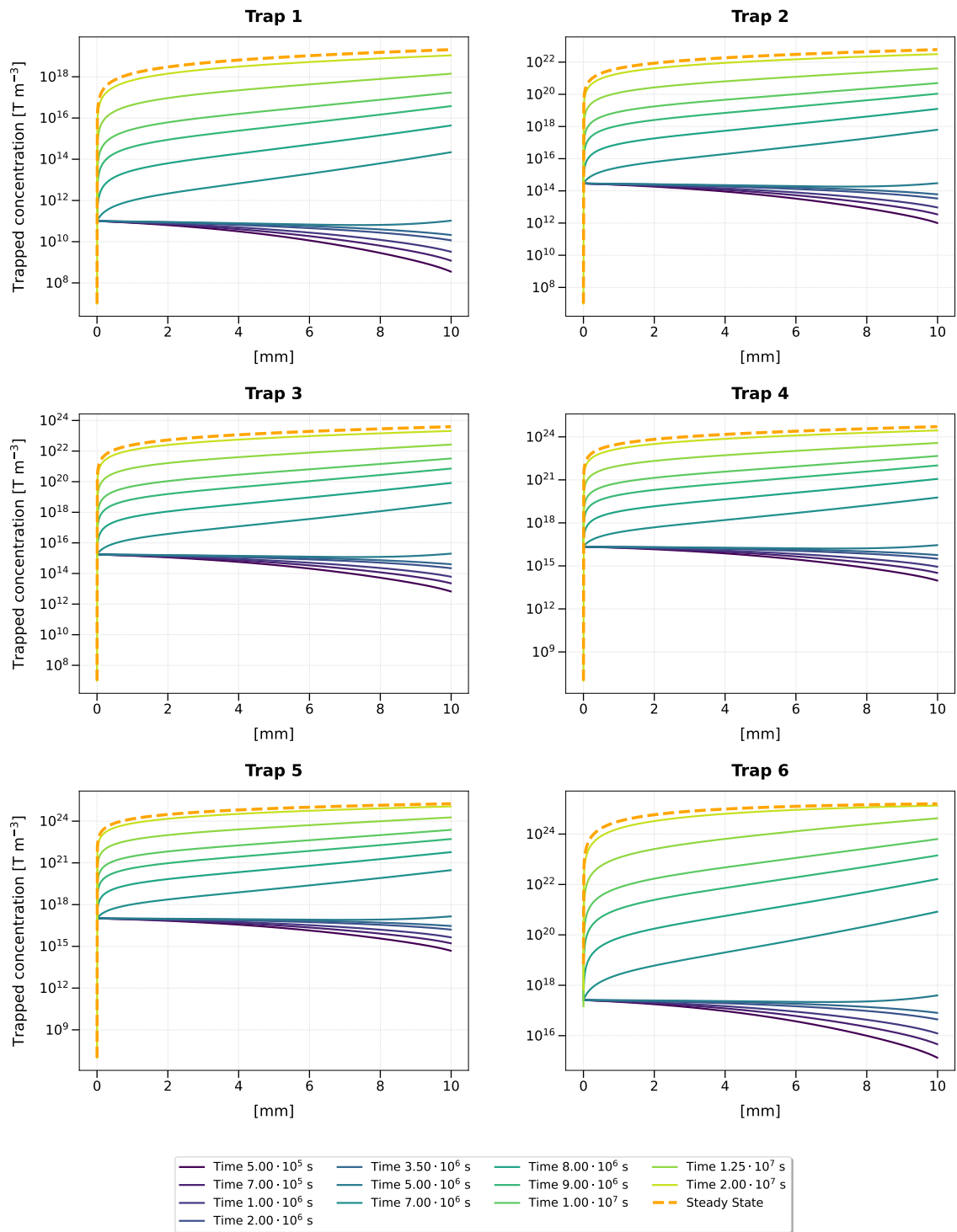


Figure 3.7: Trapped concentration evolution in Tungsten across the 1D multi-layer model.

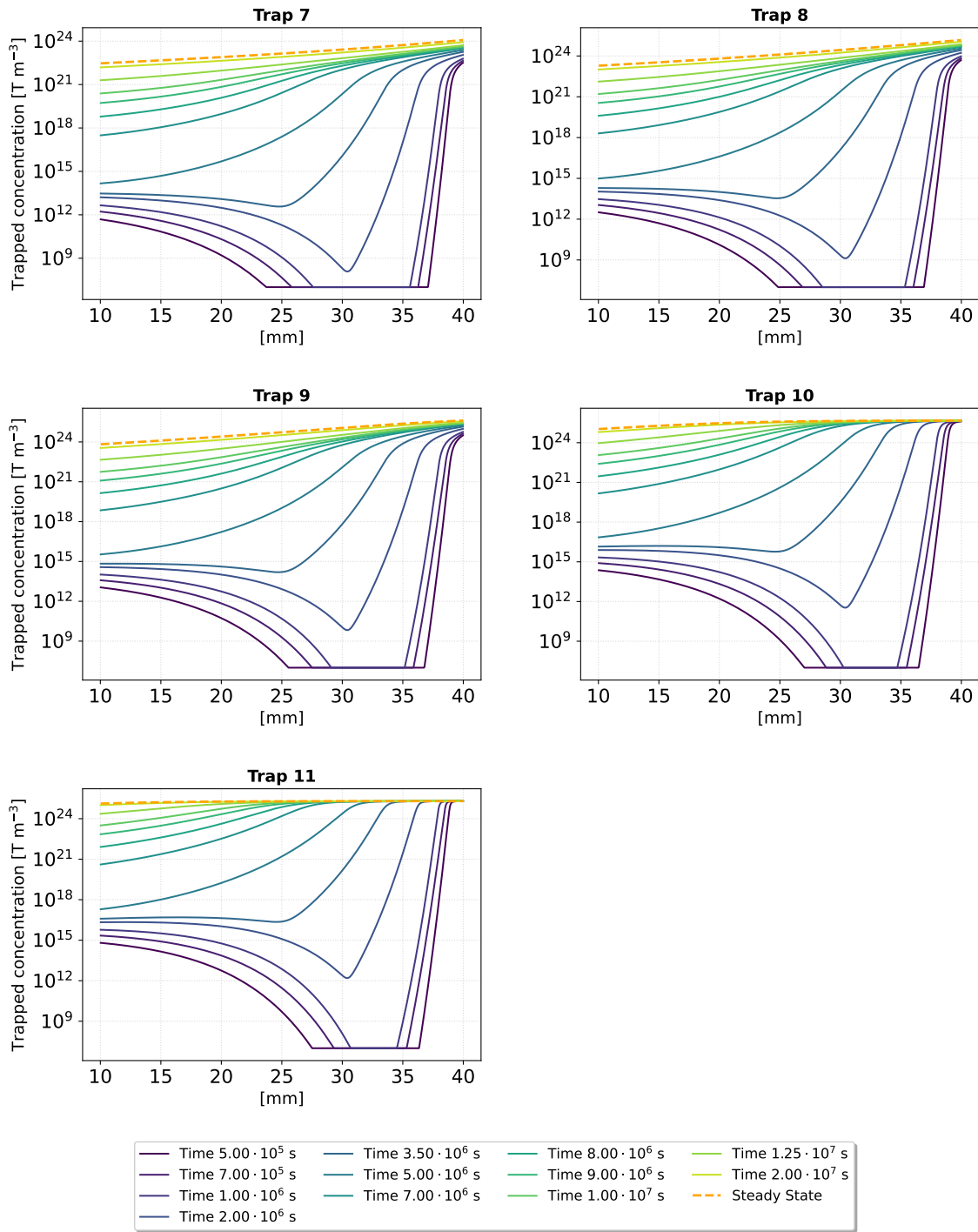


Figure 3.8: Trapped concentration evolution in Eurofer across the 1D multi-layer model.

Table 3.4 summarizes the integral value of tritium captured in each individual trap, as obtained from the steady-state simulation.

It is observed that Eurofer traps exhibit a higher concentration of tritium compared to Tungsten traps. This phenomenon could be explained by the fact that tritium also permeates from FLiBe into Eurofer, as previously discussed.

Table 3.4: Total amount of tritium trapped in each individual trap and the quantity trapped in the Tungsten and Eurofer layers.

Material	Trap	Trapped concentration [m ⁻³]
Tungsten	Intrinsic #1	$9.12 \cdot 10^{18}$
	Extrinsic #2	$2.62 \cdot 10^{22}$
	Extrinsic #3	$1.68 \cdot 10^{23}$
	Extrinsic #4	$2.17 \cdot 10^{24}$
	Extrinsic #5	$8.44 \cdot 10^{24}$
	Extrinsic #6	$1.02 \cdot 10^{25}$
Eurofer	Extrinsic #7	$2.63 \cdot 10^{23}$
	Extrinsic #8	$2.94 \cdot 10^{24}$
	Extrinsic #9	$1.06 \cdot 10^{25}$
	Extrinsic #10	$3.45 \cdot 10^{25}$
	Extrinsic #11	$1.86 \cdot 10^{25}$

The final figure, Figure 3.9, illustrates the tritium trapped concentration profile at steady state, expressed in $[\text{g m}^{-3}]$.

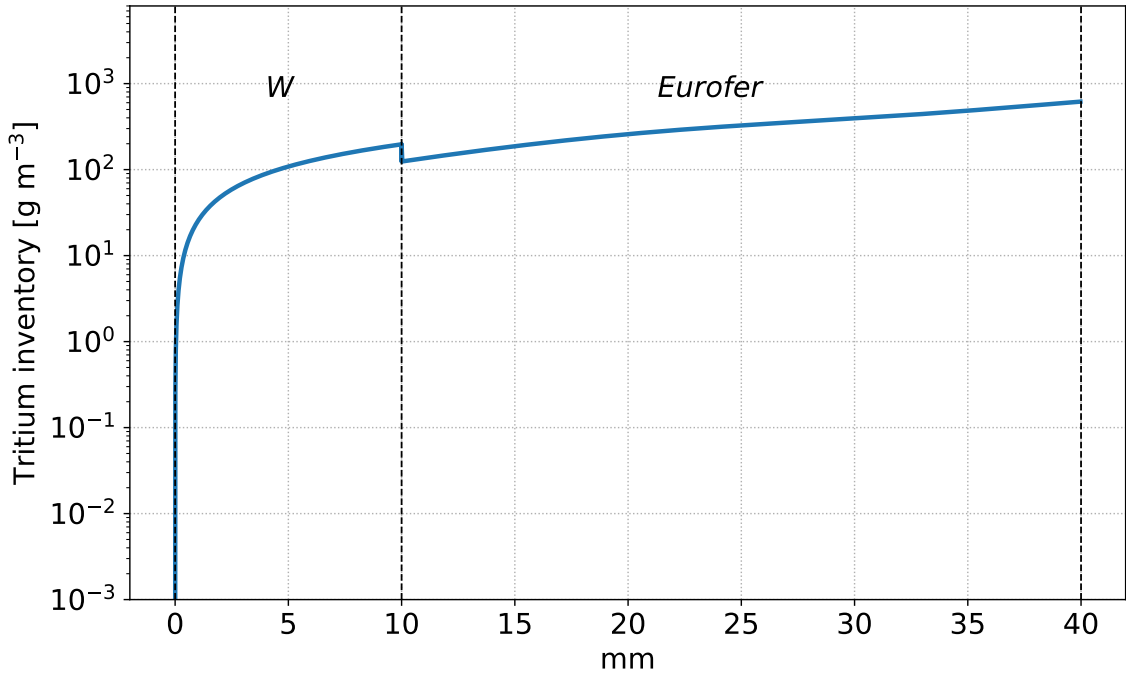


Figure 3.9: Tritium concentration profile across the 1D multi-layer model.

From a technical standpoint, it is essential to determine the integral average tritium concentration and the corresponding activity within both material layers. These values are reported in table 3.5 and are expressed in $[\text{g m}^{-3}]$ and $[\text{Bq m}^{-3}]$ (or $[\text{Ci m}^{-3}]$), respectively.

Table 3.5: Concentration and activity of trapped concentration.

Material	Concentration $[\text{g m}^{-3}]$	Activity $[\text{Bq m}^{-3}]$	Activity $[\text{Ci m}^{-3}]$
Tungsten	$1.05 \cdot 10^2$	$3.75 \cdot 10^{16}$	$1.01 \cdot 10^6$
Eurofer	$3.35 \cdot 10^2$	$1.19 \cdot 10^{17}$	$3.21 \cdot 10^6$

3.4 2D model

The 2D model is illustrated in Figure 3.10, utilizing the same domain as Figure 3.2. While the boundary conditions remain consistent with the 1D case, the implantation boundary condition has been converted into a Neumann boundary condition to prevent numerical instabilities caused by the mesh and the triangular approximation

of the implantation profile. Furthermore, the Dirichlet boundary condition is applied to the bottom surface of the FLiBe; this allows for the simulation of the tritium entering the domain from the bottom.

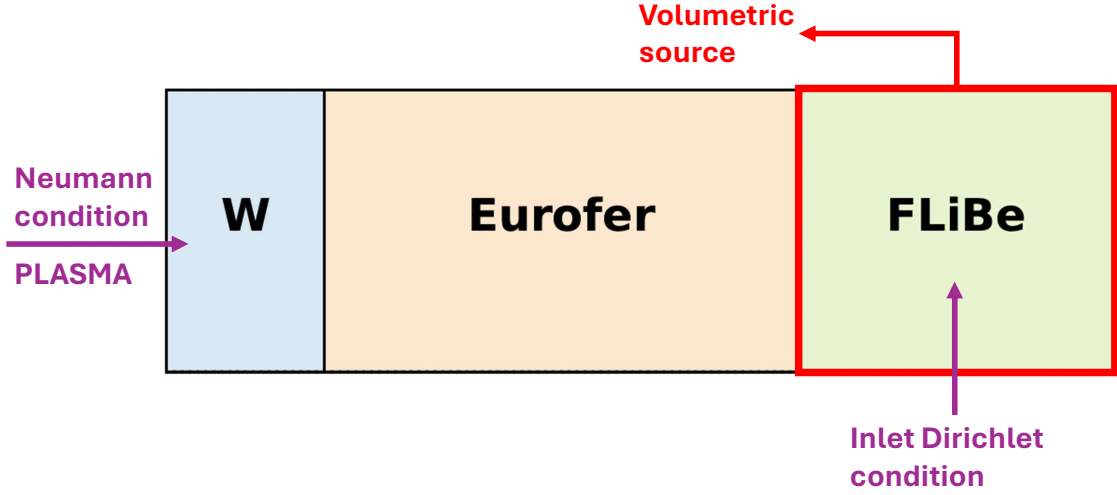


Figure 3.10: 2D model of simplified ARC breeding blanket.

While the model is 2D, the temperature distribution varies only in the radial (horizontal) direction, as reported in figure 3.3. The vertical temperature profile remains uniform, as the distribution is strictly a function of x .

In this model, advection is incorporated into the FLiBe simulation. The velocity is set to zero at the FLiBe wall boundaries to satisfy the no-slip condition. The implemented velocity distribution is shown in Figure 3.11.

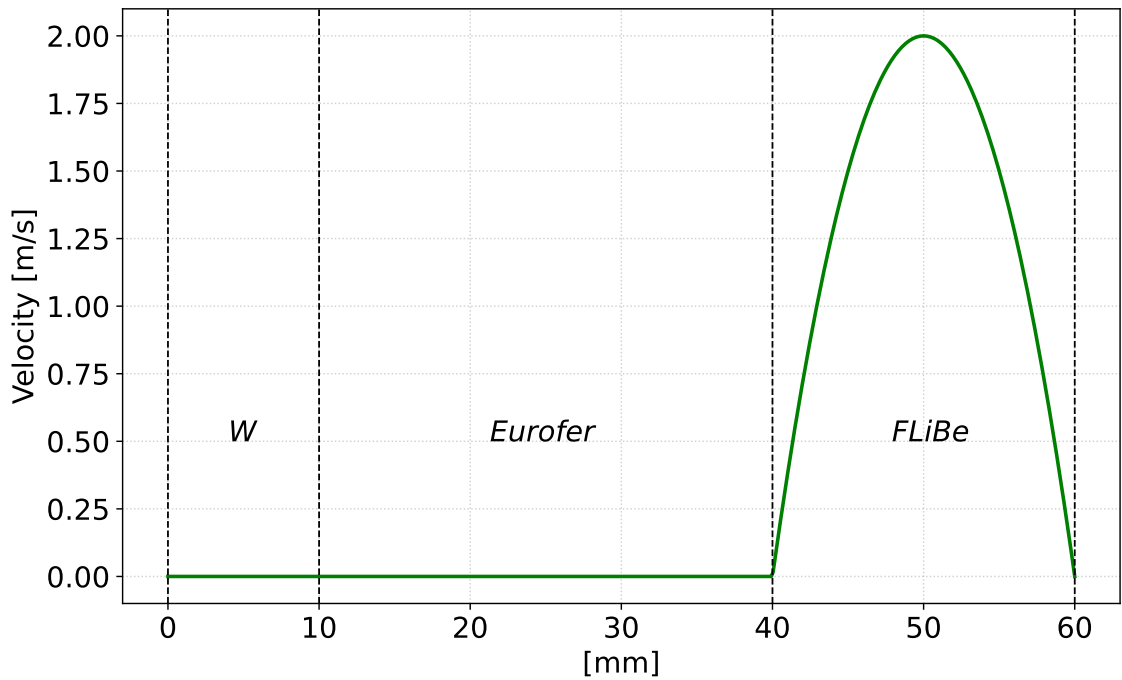


Figure 3.11: Horizontal velocity profile.

3.5 2D model results

The results are presented below, including both the 2D distributions and the corresponding radial profiles. Radial data were extracted along the entire length of the domain at a height of 5 mm.

The 2D mobile concentration is reported in the figure 3.12.

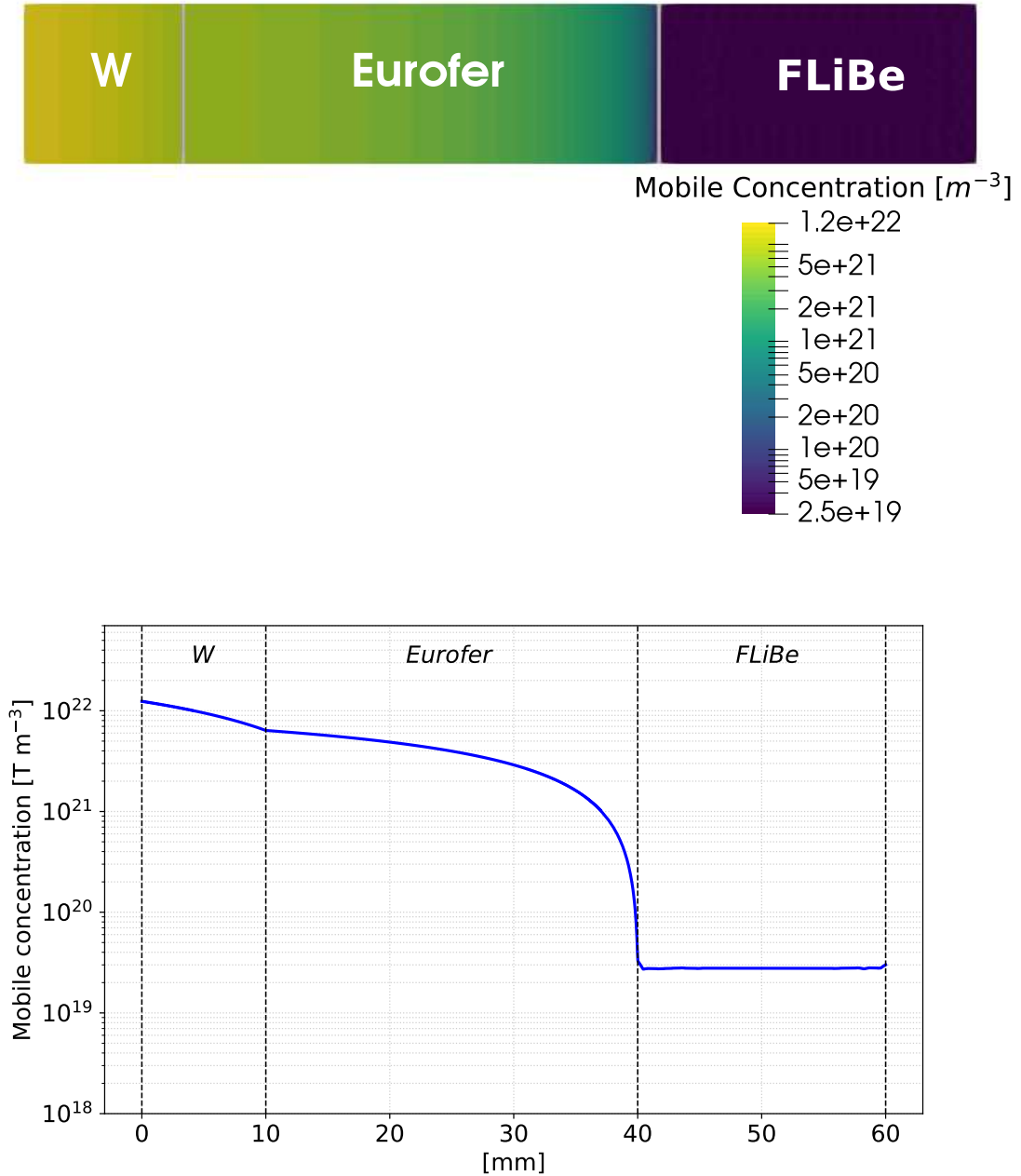


Figure 3.12: 2D model and radial profile of mobile concentration.

Compared to the 1D model, the mobile tritium profile in the FLiBe layer is altered by advective effects. Advection homogenizes the distribution, resulting in a flatter gradient caused by convective transport.

Figure 3.13 illustrates the trapped tritium concentration. As the FLiBe layer does not contain traps, the concentration vanishes within this specific domain. For this reason, the radial profile does not show the FLiBe sub-domain.

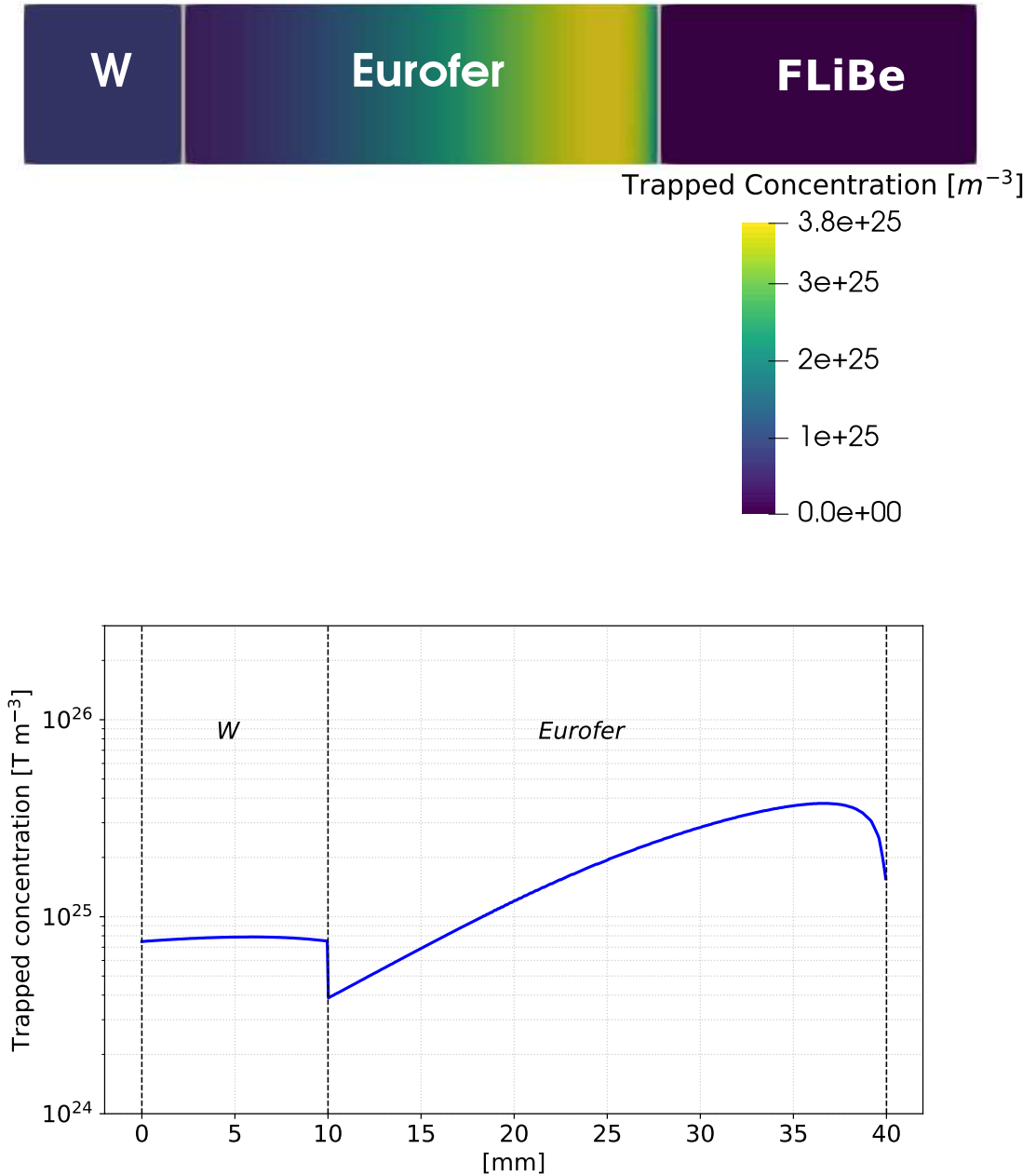


Figure 3.13: 2D model and radial profile of trapped concentration.

The amount of tritium trapped in the Tungsten layer reaches near-saturation levels (Figure 3.14). The distribution within the Eurofer is non-uniform, with the highest concentration observed in the final portion of the layer. This behavior can likely be attributed to the trapping-detraping kinetics: tritium atoms are released from the initial traps (near the plasma-facing surface) but lack sufficient thermal energy to escape the traps located deeper in the material, where temperatures might be lower or the concentration gradient favors accumulation (Figure 3.15). The

high trapped concentration in the final Eurofer section is further exacerbated by tritium permeation from the liquid breeder. Despite FLiBe's role in carrying tritium, diffusion into the structural walls remains unavoidable. Notably, the flow velocity cannot be increased to mitigate this effect without encountering critical corrosion limits in the structural components.

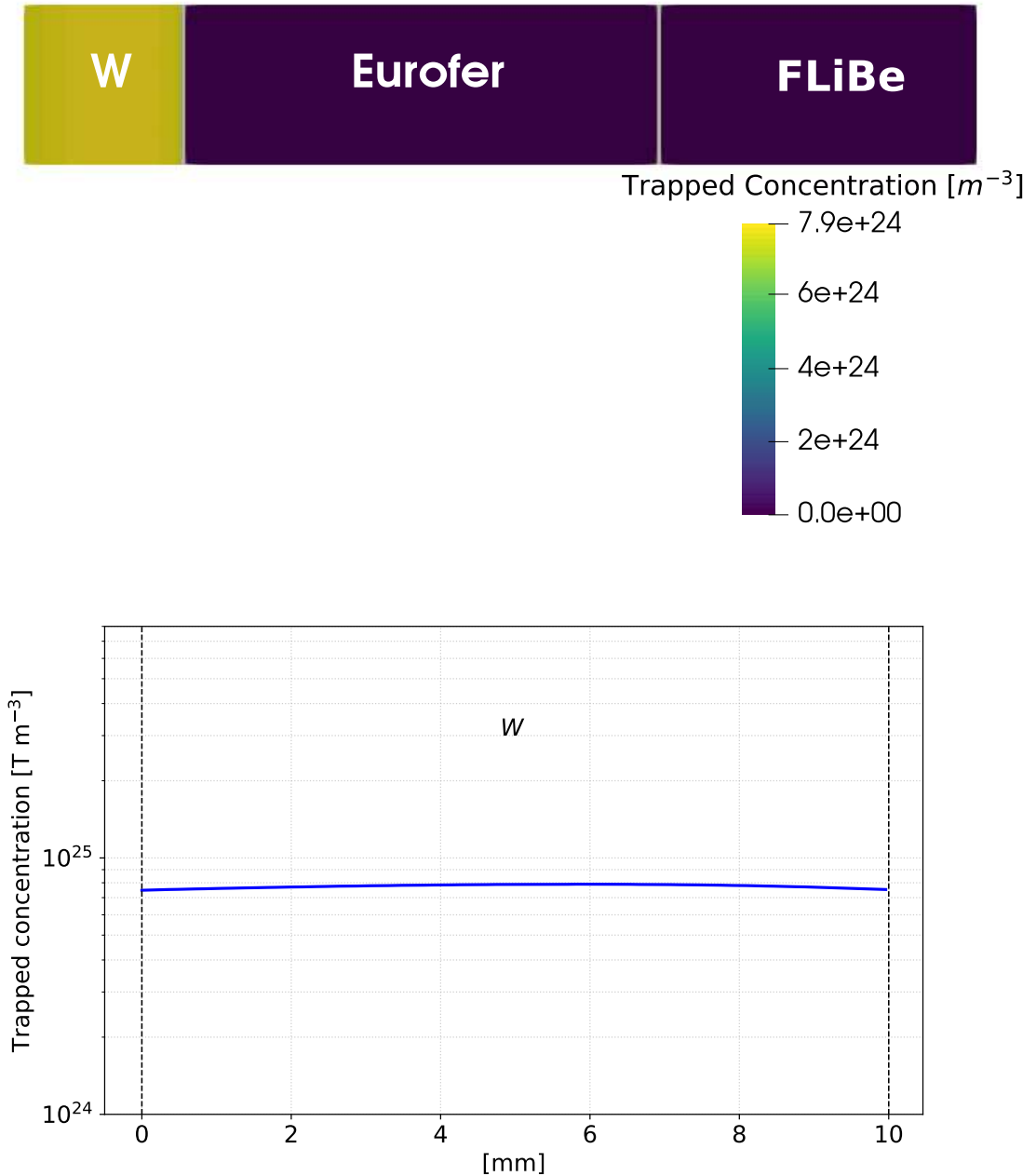


Figure 3.14: 2D model and radial profile of trapped concentration in Tungsten.

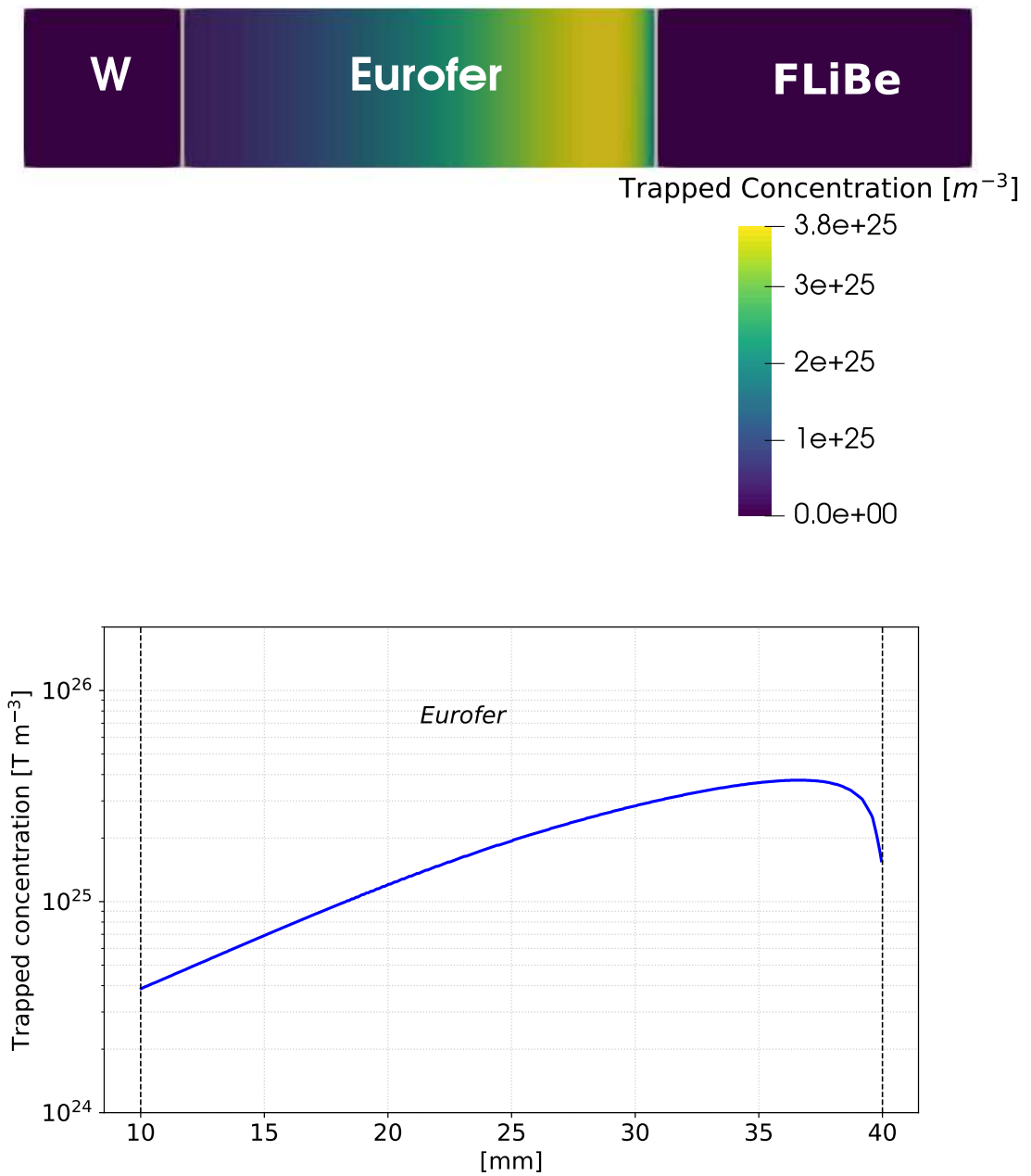


Figure 3.15: 2D model and radial profile of trapped concentration in Eurofer.

The first mean value theorem for integrals is applied to determine the average tritium concentration and activity per unit of volume within the materials; these values are summarized in Table 3.6. As previously analyzed, the advection transport reduces the total tritium inventory, consequently lowering both the concentration and activity levels.

Table 3.6: Concentration and activity per unit of volume of trapped concentration.

Material	Concentration [g m ⁻³]	Activity [Bq m ⁻³]	Activity [Ci m ⁻³]
Tungsten	$3.89 \cdot 10^1$	$1.38 \cdot 10^{16}$	$3.74 \cdot 10^5$
Eurofer	$1.01 \cdot 10^2$	$3.59 \cdot 10^{16}$	$9.71 \cdot 10^5$

Chapter 4

DEMO breeding blanket

The European and Switzerland fusion communities started a collaboration on fusion research. They worked on a Demonstration Fusion Power Plant (DEMO) based on a tokamak layout while ITER still ongoing construction. The Pre-Concept Design (PCD) called "DEMO BL2017 configuration" is shown in figure 4.1.

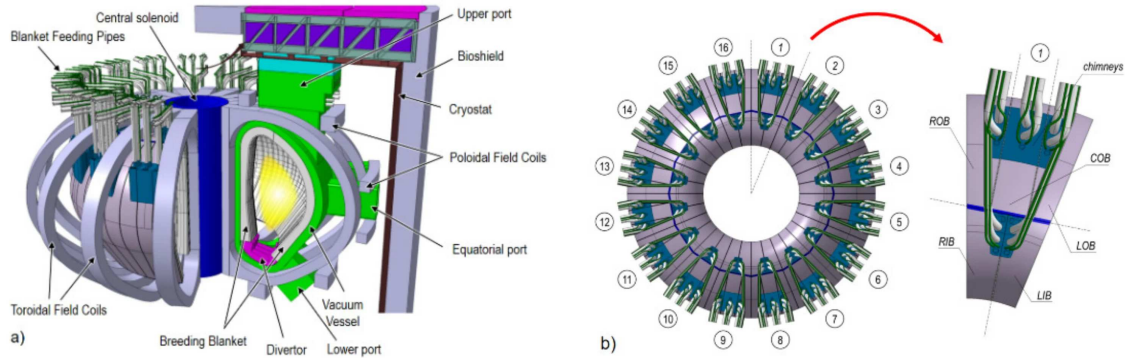


Figure 4.1: DEMO BL2017 configuration: (a) 3D view of DEMO BL2017; (b) top view of DEMO BL2017 [30].

This thesis focuses on the design of the breeding blanket, specifically the Helium Cooled Pebble Bed (HCPB) concept. In this configuration, the pebble bed serves as the breeding material, while helium acts as both the purge gas and the coolant. A Beryllium material is selected as the neutron multiplier [30].

Over time, the design concepts for the breeding blanket have evolved. Evolving from the initial pre-conceptual models, the first HCPB layout was based on the "beer box" concept (Figure 4.2); while it guaranteed horizontal and vertical structural robustness, the excessive amount of steel in the breeding zone compromised the tritium breeding performance. To address this, a second layout, known as the "sandwich" concept (Figure 4.2), was developed. This design alternated layers of tritium breeder pebbles with Beryllium multiplier pebbles, separated by Eurofer cooling plates. However, the manufacturing complexity was significant due to the small channels embedded in the plates and the resulting high pressure drops [30].

Consequently, the HCPB design was further refined, leading to the "fuel-breeder pin" (Figure 4.2) concept released in 2017. This approach is based on two concentric tubes surrounding the ceramic breeding material, which are then inserted into a cladding surrounded by the neutron multiplier. Additionally, advanced materials have been introduced to enhance both the performance and the safety of the reactor [30]. This thesis focuses on the trapping phenomena within the 'fuel-breeder pin'

configuration. Although KIT has proposed several designs for this configuration, this work focuses on the final version. However, before modeling the full geometry, a preliminary case is analyzed to introduce and understand the tritium trapping phenomena in the ceramic breeder and Beryllium.

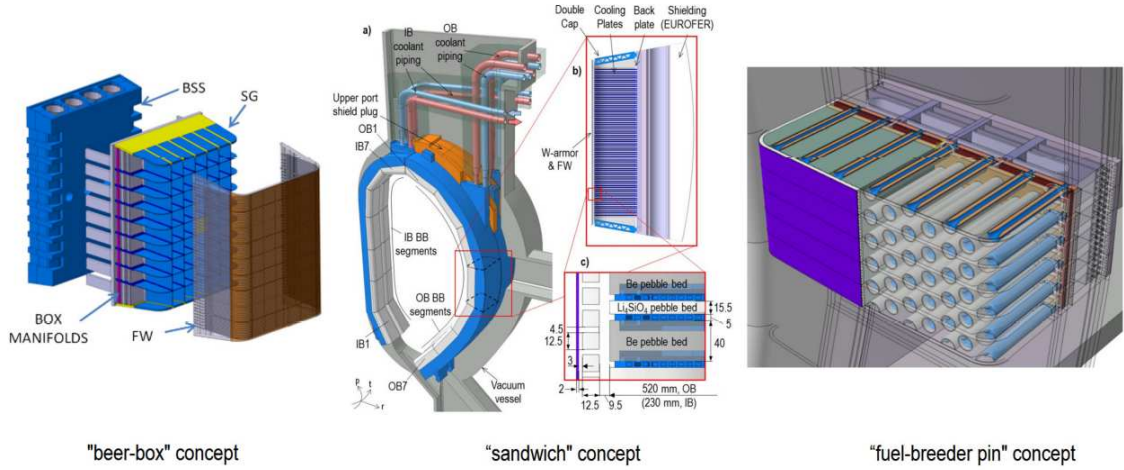


Figure 4.2: Design evolution of HCPB [30].

4.1 Preliminary design of fuel-breeder pin concept

The preliminary design of the fuel-breeder pin is a simplified version of the concepts illustrated in Figures 4.4 and 4.3.

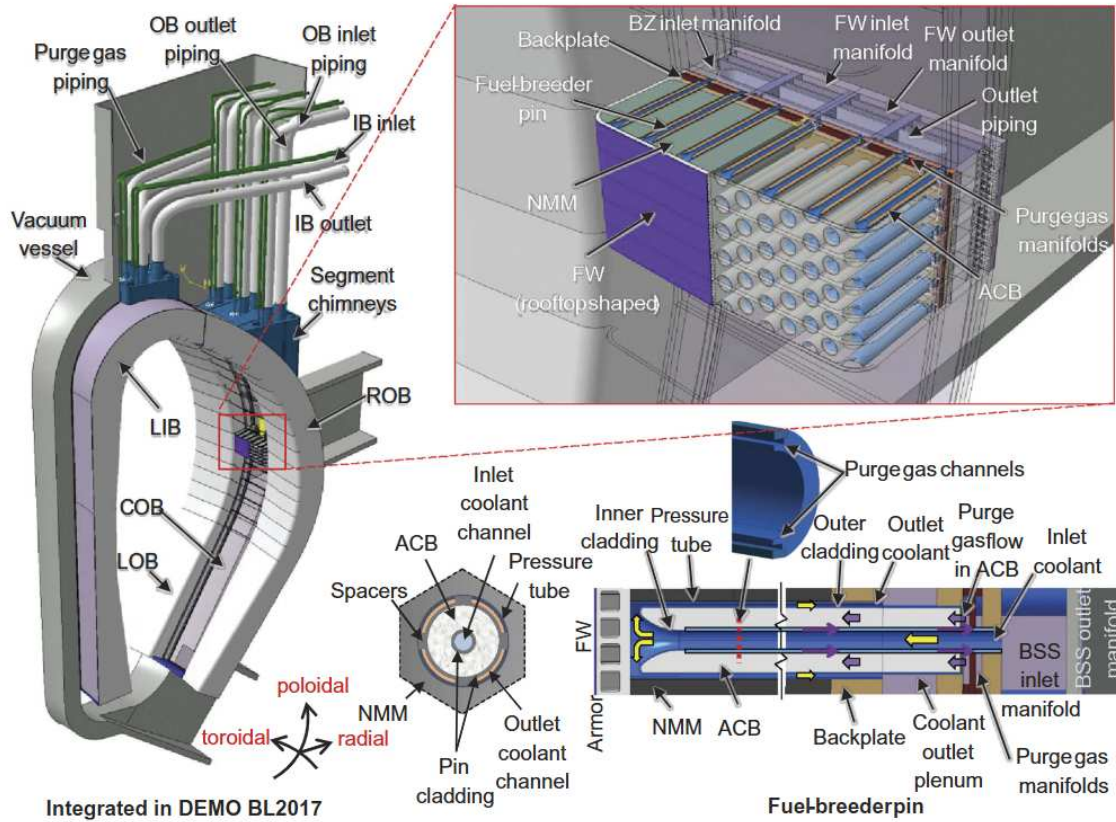


Figure 4.3: Integrated HCPB in DEMO BL2017 [31].

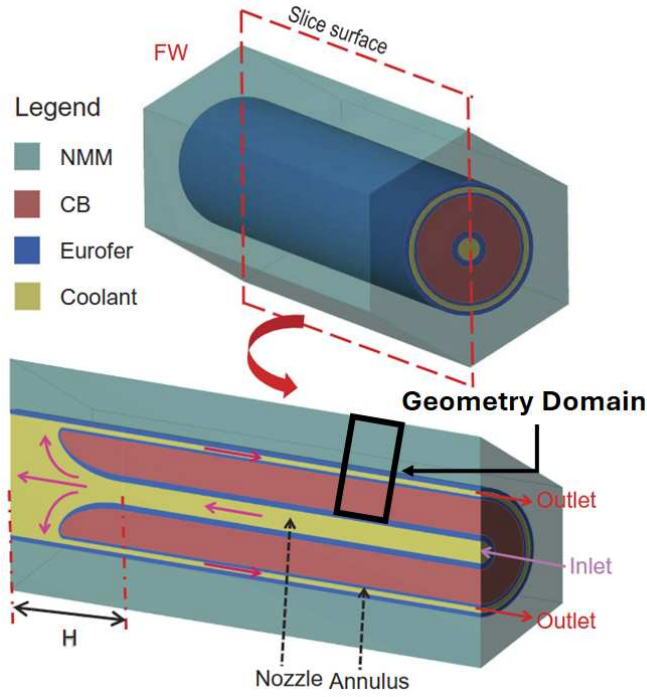


Figure 4.4: schematization of HCPB in DEMO BL2017 [31].

The first model is based on the design proposed in [31]. The domain, shown in Figure 4.4, represents half of the cross-sectional surface. This preliminary model omits the annular channel between the breeder and the Eurofer cladding ('annulus' in Figure 4.3), as well as the Eurofer layer separating the coolant from the breeder. While Hernandez et al. [30] specify the use of KALOS—a two-phase solid solution of Li_2SiO_4 and Li_2TiO_3 developed at KIT—the following models utilize Li_2TiO_3 as the sole breeding material. This choice is motivated by the availability of literature regarding trapping phenomena in Li_2TiO_3 , which is available for this material. On the contrary, trapping data is not available for KALOS. A similar approach was adopted for the neutron multiplier.

A further assumption in the model is the exclusion of tortuosity in the diffusion calculation. While tortuosity is typically employed to model radionuclide leaching through porous media (such as soil), it was deemed unnecessary for the specific hydrogen transport phenomena investigated here.

Although Hernandez et al. [30] utilize TiBe_{12} , the models presented here employ pure Beryllium. This remains a reasonable approximation, as the Titanium content in TiBe_{12} is significantly lower than that of Beryllium, making the latter the dominant functional component interacting with hydrogen.

The final geometry layout for the preliminary model is presented in Figure 4.5. In this configuration, the Li_2TiO_3 breeder has a thickness of 16 mm [32] and is contained within a 4 mm thick Eurofer pressure tube [32]. Furthermore, a thickness of 20 mm is assumed for the Beryllium multiplier. The length of the domain is 10 mm.

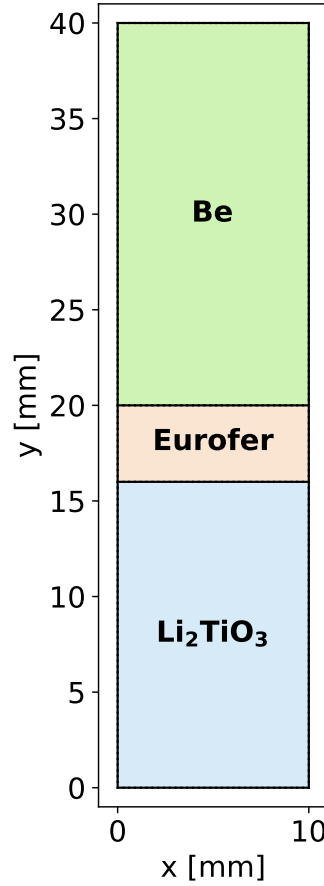
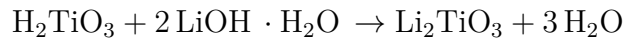


Figure 4.5: Preliminary domain of the fuel-breeder pin. ■ Breeder material, ■ Structural material, ■ Neutron multiplier material.

4.1.1 Physical Parameters

The physical parameters used across all subsequent models are described below. As previously mentioned in Section 4.1, the three materials involved are Li_2TiO_3 , Eurofer, and Beryllium. Since the properties of Eurofer have already been discussed in Section 3.1, they will not be repeated here. Various manufacturing methods exist for the fabrication of Li_2TiO_3 ceramic pebbles; for this study, the solid-state method was selected [33]. This process involves a chemical reaction between H_2TiO_3 and $\text{LiOH} \cdot \text{H}_2\text{O}$ as follows:



At an industrial level, several organizations are involved in Li_2TiO_3 production [33]. In this framework, French pebble beds were chosen. These are characterized by a theoretical density (TD) of 90%, resulting in an effective density for Li_2TiO_3 of $3.09 \cdot 10^6 \text{ g m}^{-3}$ [34].

The diffusion laws for Li_2TiO_3 and Beryllium are reported in Table 4.1. Conversely, the solubility of Li_2TiO_3 is assumed to be negligible. Under this assumption, while the lattice solubility is treated as zero, the trapping phenomena are correctly modeled. The reason is the following. The solubility provides a link between the gas

concentration and its pressure inside the lattice. Equation 2.3, which describes the evolution of the mobile concentration, is independent on the solubility¹. Hence, the evolution of the mobile and trapped concentration is correctly modeled even if a zero solubility is assigned to a material. This hypothesis is due to the lack of available experimental data regarding hydrogen solubility in Li_2TiO_3 . Instead, the Beryllium solubility is taken as a temperature independent value.

Table 4.1: Transport Properties for preliminary design of fuel-breeder pin.

Domain	Sievert's and Henry's laws	Diffusion law
Li_2TiO_3 [35]	—	$6.9 \cdot 10^{-7} \exp\left(\frac{-1.07 \text{ eV}}{k_B \cdot T}\right)$
Beryllium [27]	$k_S = 9.03 \cdot 10^{20}$	$48 \cdot 10^{-9} \exp\left(\frac{-0.36 \text{ eV}}{k_B \cdot T}\right)$

Regarding the trapping phenomenon, the Eurofer trap parameters are listed in Table 3.3. Trapping data for Li_2TiO_3 are sourced from [35], where samples were irradiated with fission neutrons (2 MeV). Due to the lack of experimental data, the pre-exponential factor for the trapping rate in Beryllium is assumed to be equal to that of Eurofer. This assumption also accounts for scenarios where the neutron multiplier might be replaced by structural material. Furthermore, the activation energy for trapping is assumed to be equal to the diffusion activation energy, while the detrapping properties are taken from [36]. For completeness, all trap parameters are listed in Tables 4.2, 4.3 and 4.4.

Table 4.2: Trap Properties for Li_2TiO_3 [35].

Trap	k_0 [$\text{m}^3 \text{s}^{-1}$]	E_k [eV]	ν_0 [s^{-1}]	E_ν [eV]	Trap density [m^{-3}]
Extrinsic Trap #1	$8.55 \cdot 10^{-16}$	1.04	$4.1 \cdot 10^6$	1.19	$4.91 \cdot 10^{23}$

¹Setting the solubility to zero in the breeder domain requires a careful analysis of the breeder-Eurofer interface. Since the partial pressure variable θ becomes mathematically undefined when solubility is zero, FESTIM appears to bypass the thermodynamic equilibrium condition. As a result, the standard partial pressure continuity is substituted by numerical concentration continuity.

Table 4.3: Trap Properties for Eurofer [7].

Trap	k_0 [m ³ s ⁻¹]	E_k [eV]	ν_0 [s ⁻¹]	E_ν [eV]	Trap density [m ⁻³]
Extrinsic Trap #2	$8.43 \cdot 10^{-17}$	0.45	$1 \cdot 10^{13}$	1.15	$6.9 \cdot 10^{25}$
Extrinsic Trap #3	$8.43 \cdot 10^{-17}$	0.45	$1 \cdot 10^{13}$	1.35	$7 \cdot 10^{25}$
Extrinsic Trap #4	$8.43 \cdot 10^{-17}$	0.45	$1 \cdot 10^{13}$	1.65	$6 \cdot 10^{25}$
Extrinsic Trap #5	$8.43 \cdot 10^{-17}$	0.45	$1 \cdot 10^{13}$	1.85	$4.7 \cdot 10^{25}$
Extrinsic Trap #6	$8.43 \cdot 10^{-17}$	0.45	$1 \cdot 10^{13}$	2.05	$2 \cdot 10^{25}$

Table 4.4: Trap Properties for Beryllium [36].

Trap	k_0 [m ³ s ⁻¹]	E_k [eV]	ν_0 [s ⁻¹]	E_ν [eV]	Trap density [m ⁻³]
Extrinsic Trap #7	$8.43 \cdot 10^{-17}$	0.45	$1 \cdot 10^{13}$	1.8	$1.24 \cdot 10^{26}$

As discussed in Section 2.3.3, the interaction between a fluid and a material surface can be characterized by two distinct regimes: recombination and mass exchange. To ensure a comprehensive analysis, results for both configurations are reported. The paper [37] describes two helium channels: the first is used to cool the breeder and remove heat, while the second acts as the purge gas to remove and carry tritium. The fluid parameters are reported in table 4.5.

Table 4.5: Velocity parameters of the fluids.

Fluid	Pressure [bar]	Velocity [m s ⁻¹]	Regime
Coolant [37]	80	20	Turbulent
Purge gas [37]	2	$1 \cdot 10^{-2}$	Laminar

The recombination reaction is expressed using a pre-exponential factor [m⁶⁻ⁿ s⁻¹] where n indicates the order of the reaction, and an activation energy for recombination [eV] [9]. The values implemented in the model are reported in Table 4.6

Table 4.6: Recombination equations.

Interaction	Recombination reaction	Order
	$[\text{m}^4\text{s}^{-1}]$	
Li_2TiO_3 and Coolant [37]	$9.83 \cdot 10^{-29} \exp\left(\frac{9.33 \cdot 10^{-3} \text{ eV}}{k_B \cdot T}\right)$	2
Be and Purge gas [38]	$3.4 \cdot 10^{-25} \exp\left(\frac{-0.28 \text{ eV}}{k_B \cdot T}\right)$	2

The different fluid regimes (Table 4.5) require specific parameters to evaluate the mass transfer coefficient h_m $[\text{m s}^{-1}]$.

In this analysis, both fluid flows are assumed to be fully developed. The properties of helium are evaluated using CoolProp [39], while the diffusion coefficient is taken from [40], with a value of $1.72 \cdot 10^{-3} \text{ m}^2\text{s}^{-1}$. It is important to note that this coefficient corresponds to a temperature of 1000 K, whereas the helium properties are evaluated at 900 K. However, since tabulated values for the diffusion coefficient were only available at 500 K and 1000 K, the latter was selected as the most appropriate approximation.

The diameters of the purge gas and the coolant channels are, respectively $1 \cdot 10^{-2} \text{ m}$ and $1.6 \cdot 10^{-2} \text{ m}$.

For the purge gas, which operates in a laminar regime, the Sherwood number is constant:

$$Sh_D = 3.66$$

Conversely, for the coolant, the Sherwood number is determined using the Sieder-Tate correlation [15], adapted for mass transfer:

$$Sh_D = 0.027 Re_D^{4/5} Sc^{1/3} \left(\frac{\mu}{\mu_s}\right)^{0.14} = 20.8$$

The resulting mass transfer coefficients are summarized in Table 4.7.

Table 4.7: Mass transfer values implemented in the model.

Interaction	Mass transfer coefficient h_m
	$[\text{m s}^{-1}]$
Li_2TiO_3 and Coolant	2.24
Be and Purge gas	$6.3 \cdot 10^{-1}$

Based on the Monte Carlo N-Particle (MCNP) calculations presented in [37], the volumetric source is evaluated and we obtain the following value $1.6 \cdot 10^{19} \text{ m}^{-3}$.

4.1.2 2D model

The 2D model is shown in Figure 4.6, where materials, traps, and fluids are the same as those described in the previous sections.

- Due to the lack of a preferential horizontal transport direction, a zero-net-flux condition is assumed, equating the input and output at the lateral boundaries.
- The top and bottom boundary conditions are analyzed in two configurations: diffusion-dominant and recombination-dominant (see Table 4.7 and Table 4.6).
- A volumetric tritium source of $1.6 \cdot 10^{19} \text{ m}^{-3} \text{ s}^{-1}$ is applied to the Li_2TiO_3 region [37].

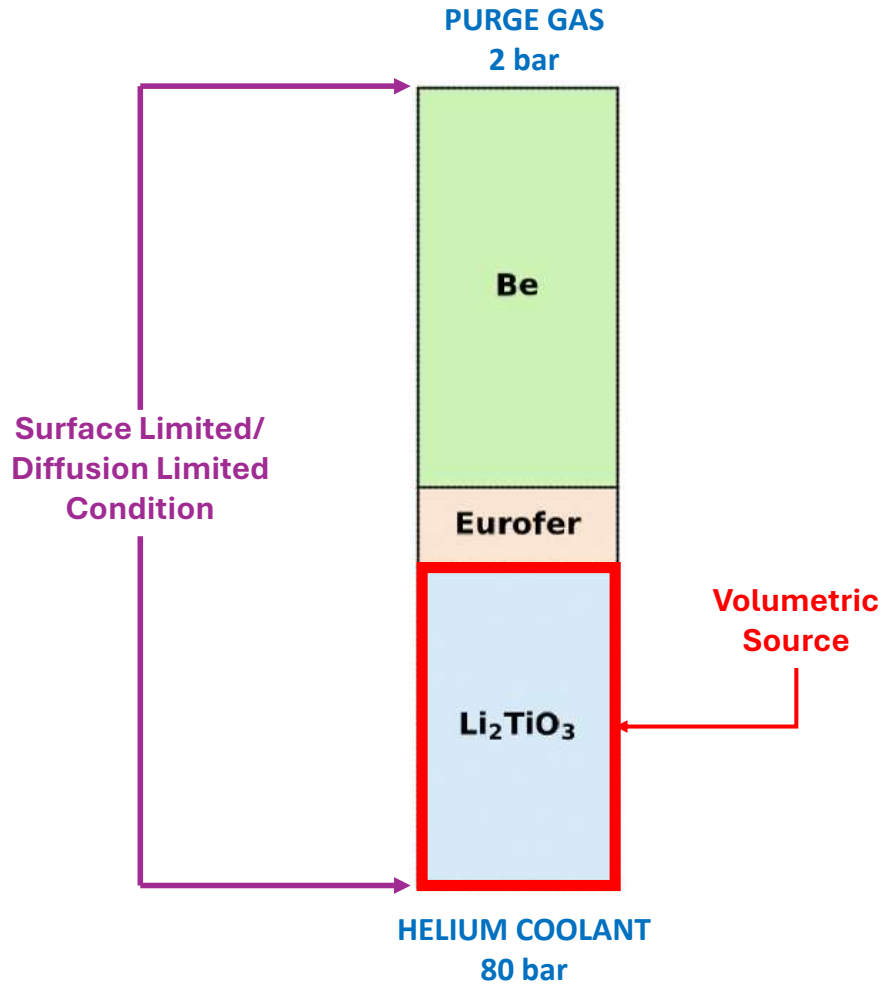


Figure 4.6: 2D model of simplified DEMO fuel-breeder pin concept.

A temperature distribution is applied to the model and depends solely on the y coordinate. The distribution is shown in Figure 4.7. In this model, the temperature distribution imposed does not take into account the heat released due to the tritium production in the breeder and in the neutron multiplier material.

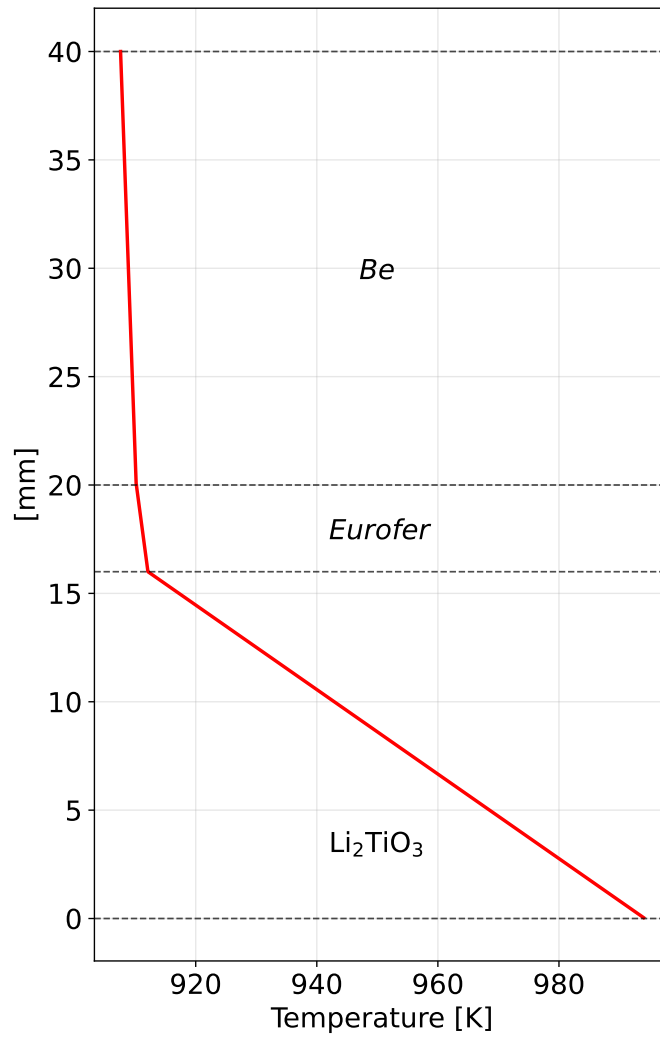


Figure 4.7: Temperature profile of the preconceptual design of the breeder fuel pin.

4.1.3 2D model results

These results illustrate the 2D distribution of the variable and a radial profile across the entire geometry at $x = 5$ mm.

The mobile concentration is shown in Figures 4.8 and 4.9. While Figure 4.8 displays the results for recombination boundary conditions, Figure 4.9 accounts for mass transfer effects at the boundaries.

Although both conditions exert a similar effect on the boundary in contact with the helium coolant, the diffusion-dominant mechanism is more effective at removing tritium from the Beryllium than the surface-limited case, specifically at the interface with the purge gas.

Figures 4.8 and 4.9 illustrate the theoretical aspects reported in Section 2.2, where concentration jumps occur between materials. This effect ensures the continuity of partial pressure.

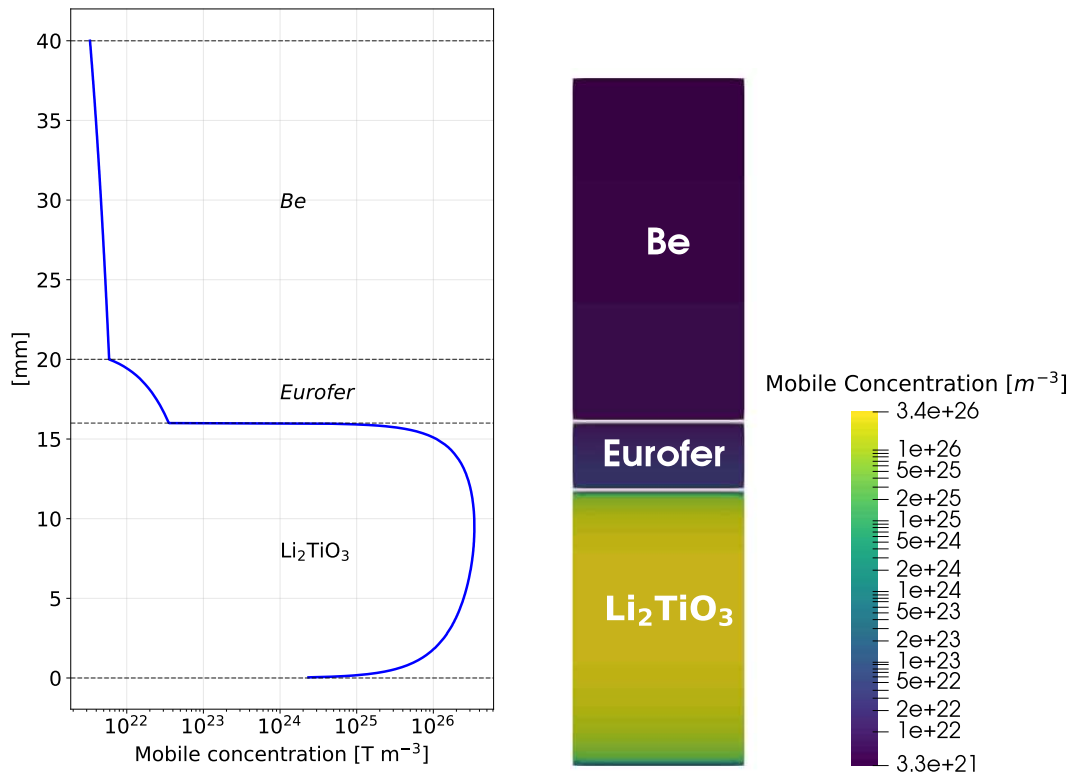


Figure 4.8: Preliminary 2D model and radial profile of mobile concentration considering the recombination boundary conditions.

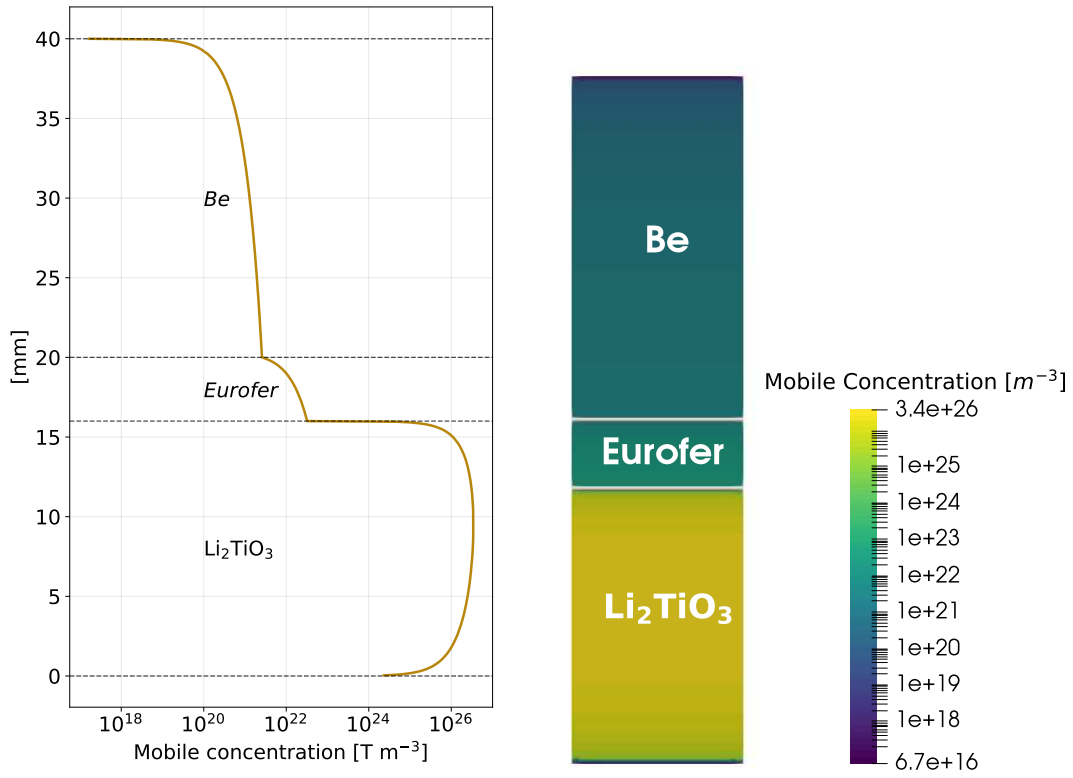


Figure 4.9: Preliminary 2D model and radial profile of mobile concentration considering the mass transfer boundary conditions.

The trapped concentrations are shown in Figures 4.10 and 4.11. Here again, the two boundary conditions, recombination and mass transfer, have different effects at the interface between Beryllium and the purge gas. In particular, a higher trapped concentration is observed near the boundary in the recombination case, whereas in the diffusion-limited case the more effective tritium removal reduces its concentration in this region.

The trapped tritium concentration is higher under surface-limited boundary conditions, whereas it is lower in the diffusion-limited regime. This discrepancy stems from the distinct concentration profiles established by each boundary condition.

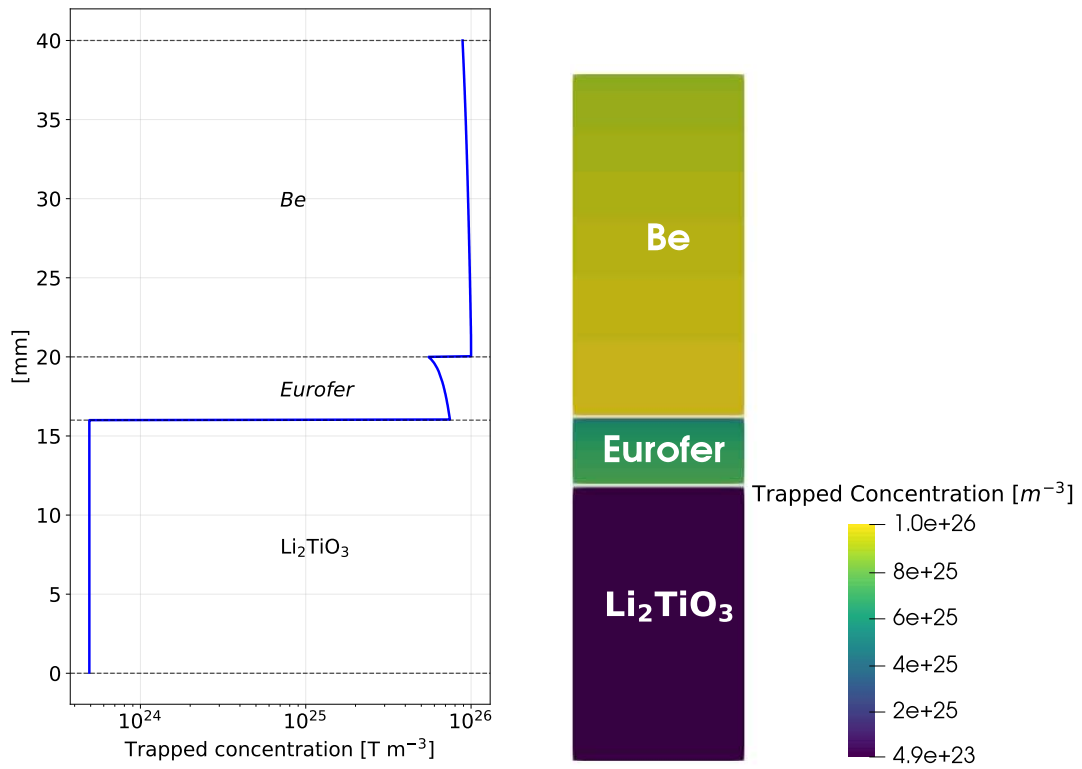


Figure 4.10: Preliminary 2D model and radial profile of trapped concentration considering the recombination boundary conditions.

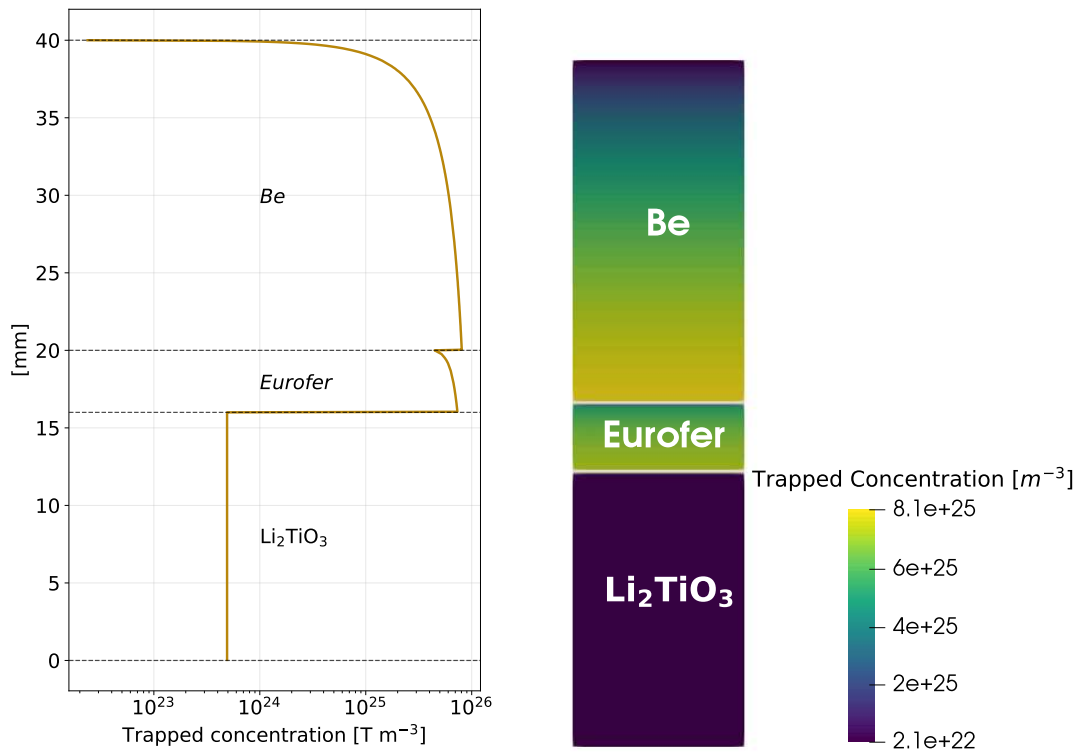


Figure 4.11: Preliminary 2D model and radial profile of trapped concentration considering the mass transfer boundary conditions.

The trapped tritium inventory within each material will now be examined.

The first material is the breeder Li_2TiO_3 the distribution and the profile is shown in Figures 4.12 and 4.13. In both cases, trap saturation is reached; indeed, the trapped concentration reaches the same value as the trap density reported in Table 4.2.

The second material analyzed is the structural steel (Eurofer); Figures 4.14 and 4.15 show the radial profile and the trapped concentration distribution. A higher concentration of trapped tritium is observed in the initial part of the domain compared to the final section. Since the temperature profile in the Eurofer is nearly constant (ranging from approximately 912 K to 910 K) Figure 4.7, this distribution might be driven by the local availability of mobile tritium or by subtle thermal effects on the detrapping rate.

The final material analyzed is the neutron multiplier (Beryllium); Figures 4.16 and 4.17 illustrate its radial profile and trapped concentration distribution. When recombination boundary conditions are applied, the trapped concentration is higher than in the other case; indeed, trap saturation is nearly reached.

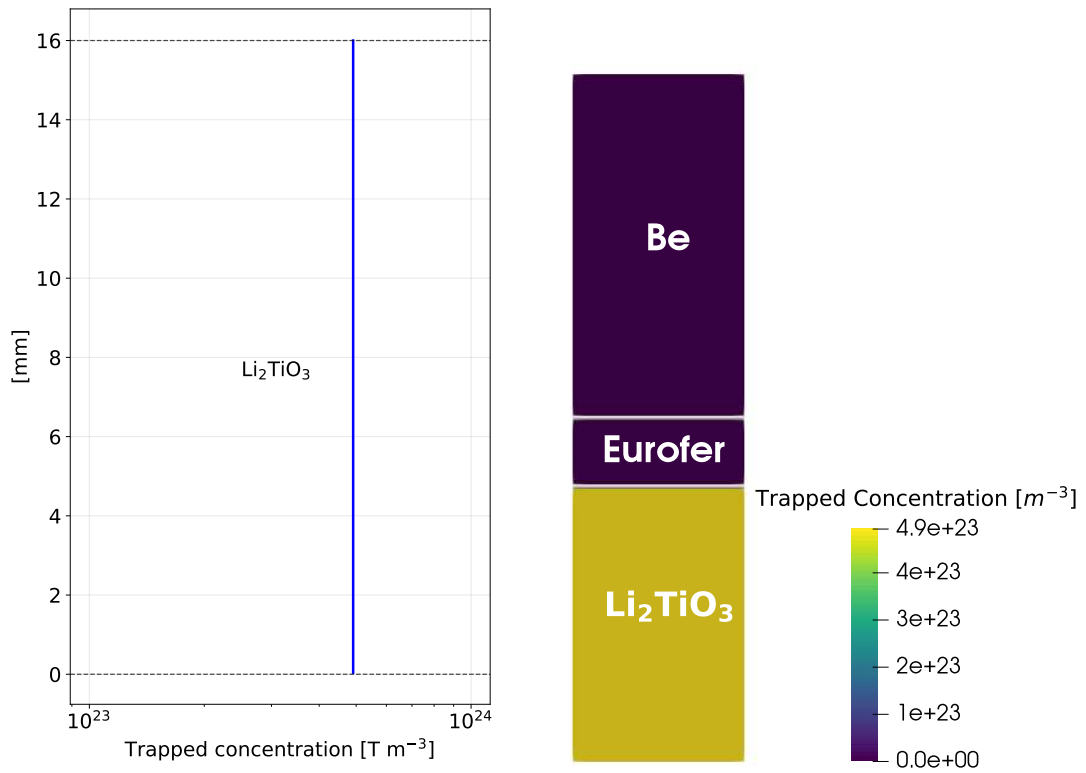


Figure 4.12: Preliminary 2D model and radial profile of trapped concentration in the breeder considering the recombination boundary conditions.

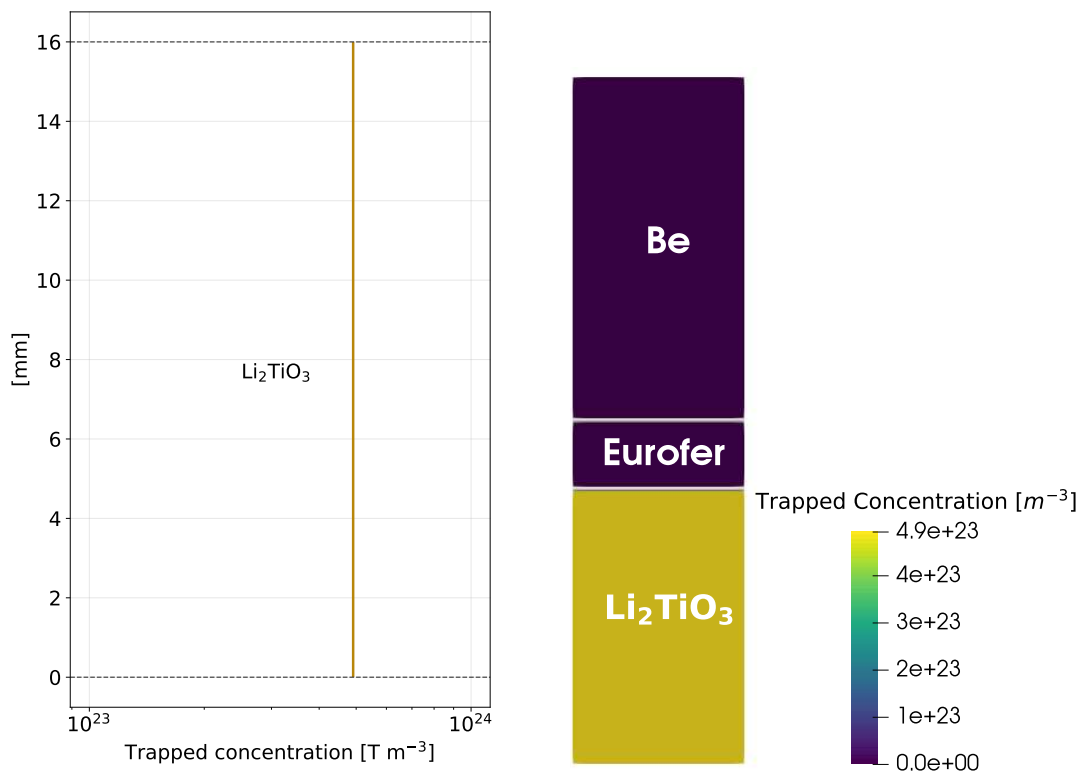


Figure 4.13: Preliminary 2D model and radial profile of trapped concentration in the breeder considering the mass transfer boundary conditions.

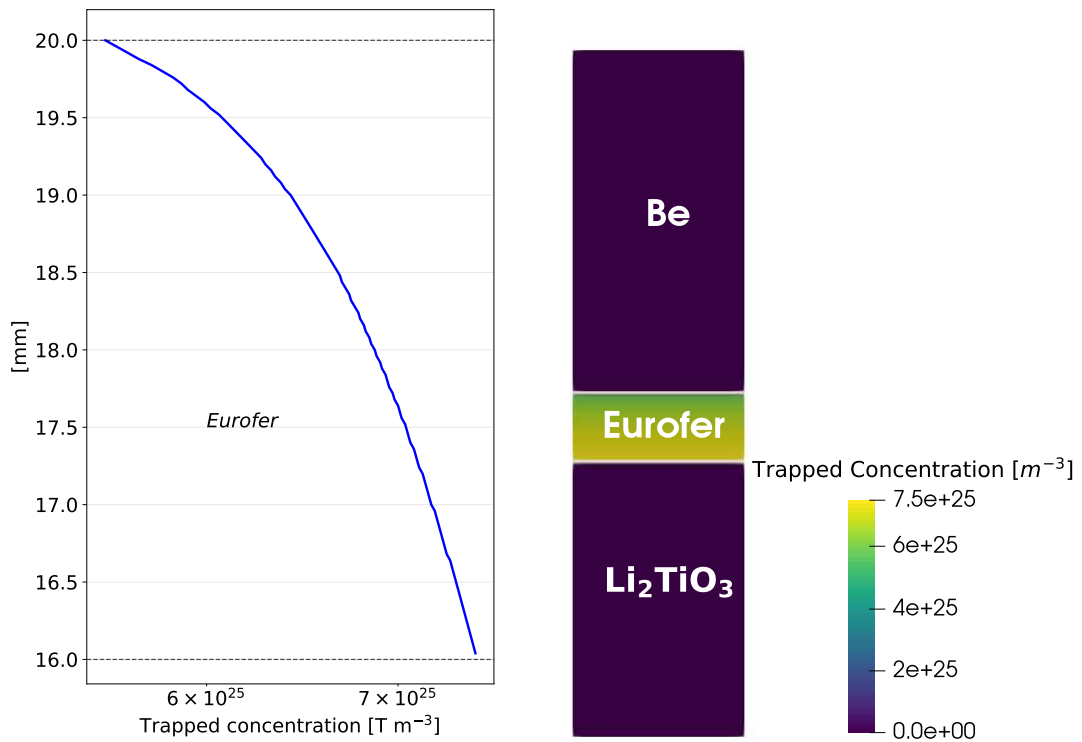


Figure 4.14: Preliminary 2D model and radial profile of trapped concentration in Eurofer considering the recombination boundary conditions.

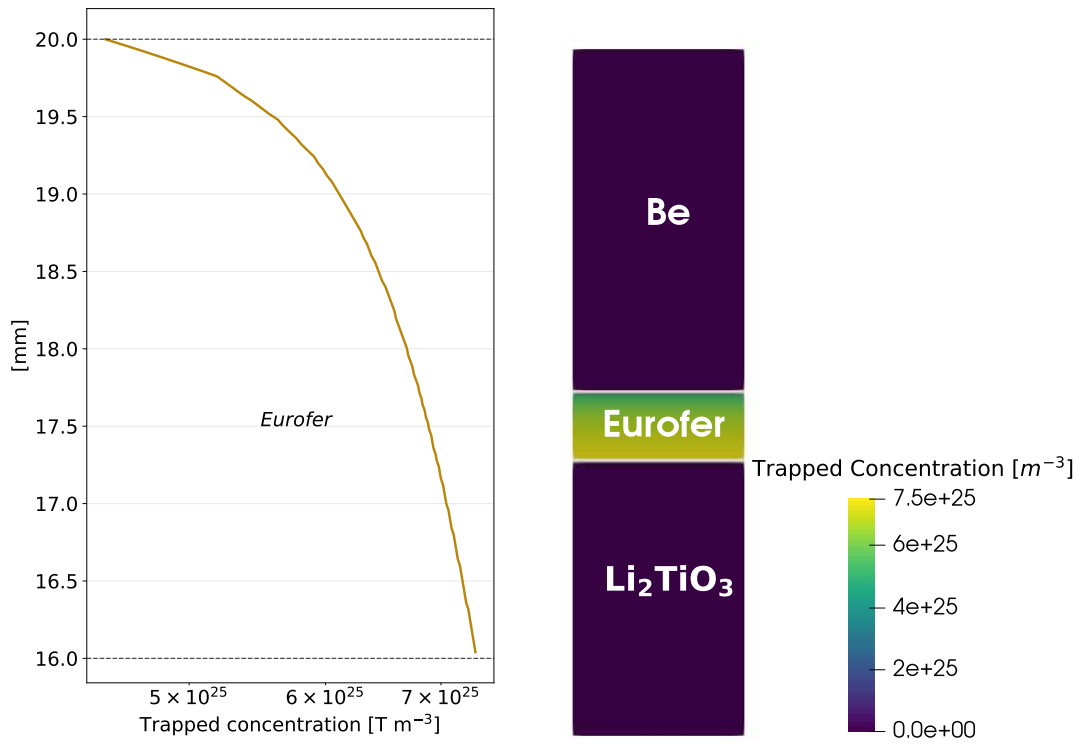


Figure 4.15: Preliminary 2D model and radial profile of trapped concentration in Eurofer considering the mass transfer boundary conditions.

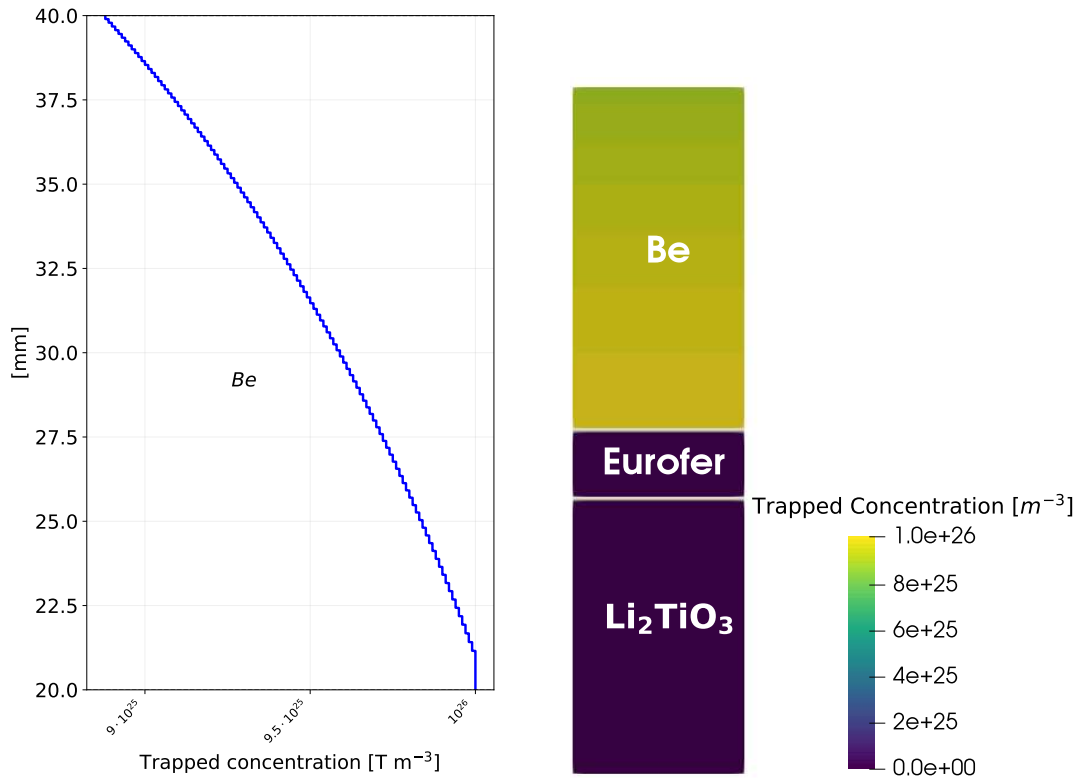


Figure 4.16: Preliminary 2D model and radial profile of trapped concentration in Beryllium considering the recombination boundary conditions.

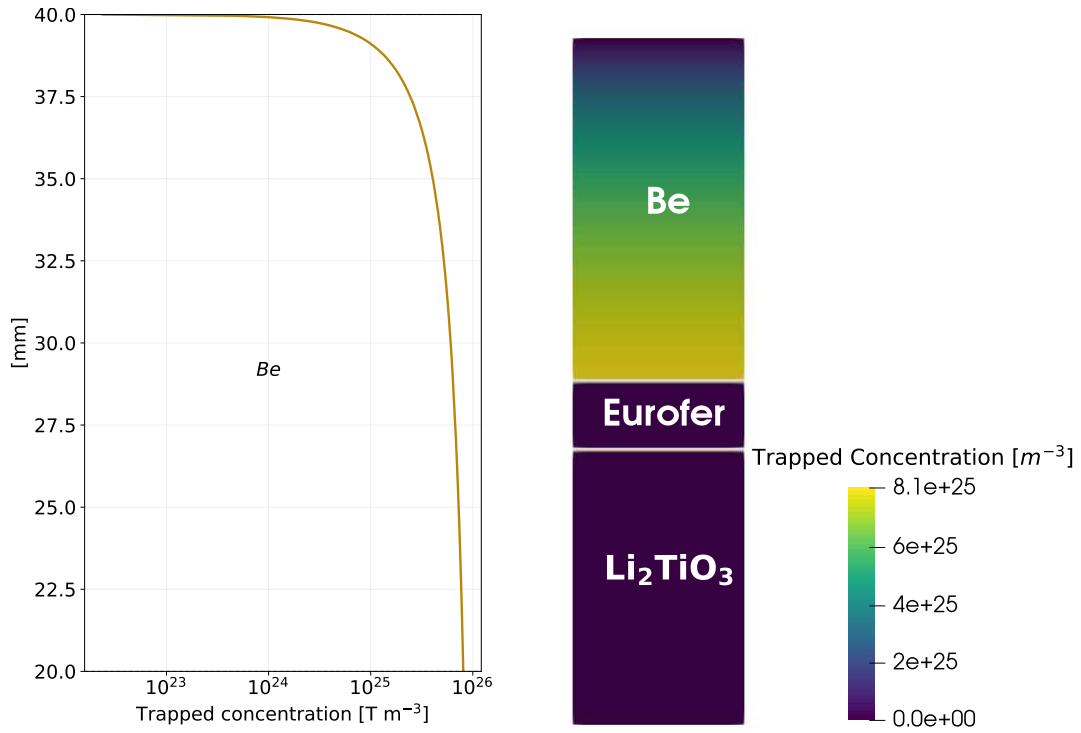


Figure 4.17: Preliminary 2D model and radial profile of trapped concentration in Beryllium considering the mass transfer boundary conditions.

The first mean value theorem for integrals is used to evaluate the mean concentration trapped in each material and activity per unit of volume. Tables 4.8 and 4.9 show the obtained value.

Table 4.8: Concentration and activity per unit of volume of trapped concentration with Surface limited condition.

Material	Concentration [g m ⁻³]	Activity [Bq m ⁻³]	Activity [Ci m ⁻³]
Li ₂ TiO ₃	2.46	$8.75 \cdot 10^{14}$	$2.37 \cdot 10^4$
Eurofer	$3.39 \cdot 10^2$	$1.21 \cdot 10^{17}$	$3.26 \cdot 10^6$
Beryllium	$4.78 \cdot 10^2$	$1.7 \cdot 10^{17}$	$4.6 \cdot 10^6$

Table 4.9: Concentration and activity per unit of volume of trapped concentration with Diffusion limited condition.

Material	Concentration [g m ⁻³]	Activity [Bq m ⁻³]	Activity [Ci m ⁻³]
Li ₂ TiO ₃	2.46	$8.75 \cdot 10^{14}$	$2.37 \cdot 10^4$
Eurofer	$3.26 \cdot 10^2$	$1.16 \cdot 10^{17}$	$3.13 \cdot 10^6$
Beryllium	$2.74 \cdot 10^2$	$9.74 \cdot 10^{16}$	$2.63 \cdot 10^6$

Comparing the mean trapped concentration for both the surface-limited and diffusion-limited configurations, the values are remarkably similar regardless of the boundary conditions applied. This suggests that the total trapped concentration is primarily governed by the bulk material properties and the volumetric source, rather than the specific surface regime.

4.2 Fuel-breeder pin concept

This section presents an analysis of mobile and trapped tritium within the final fuel-breeder pin concept. Figure 4.18 illustrates the design of the Helium Cooled Pebble Bed breeding blanket for DEMO.

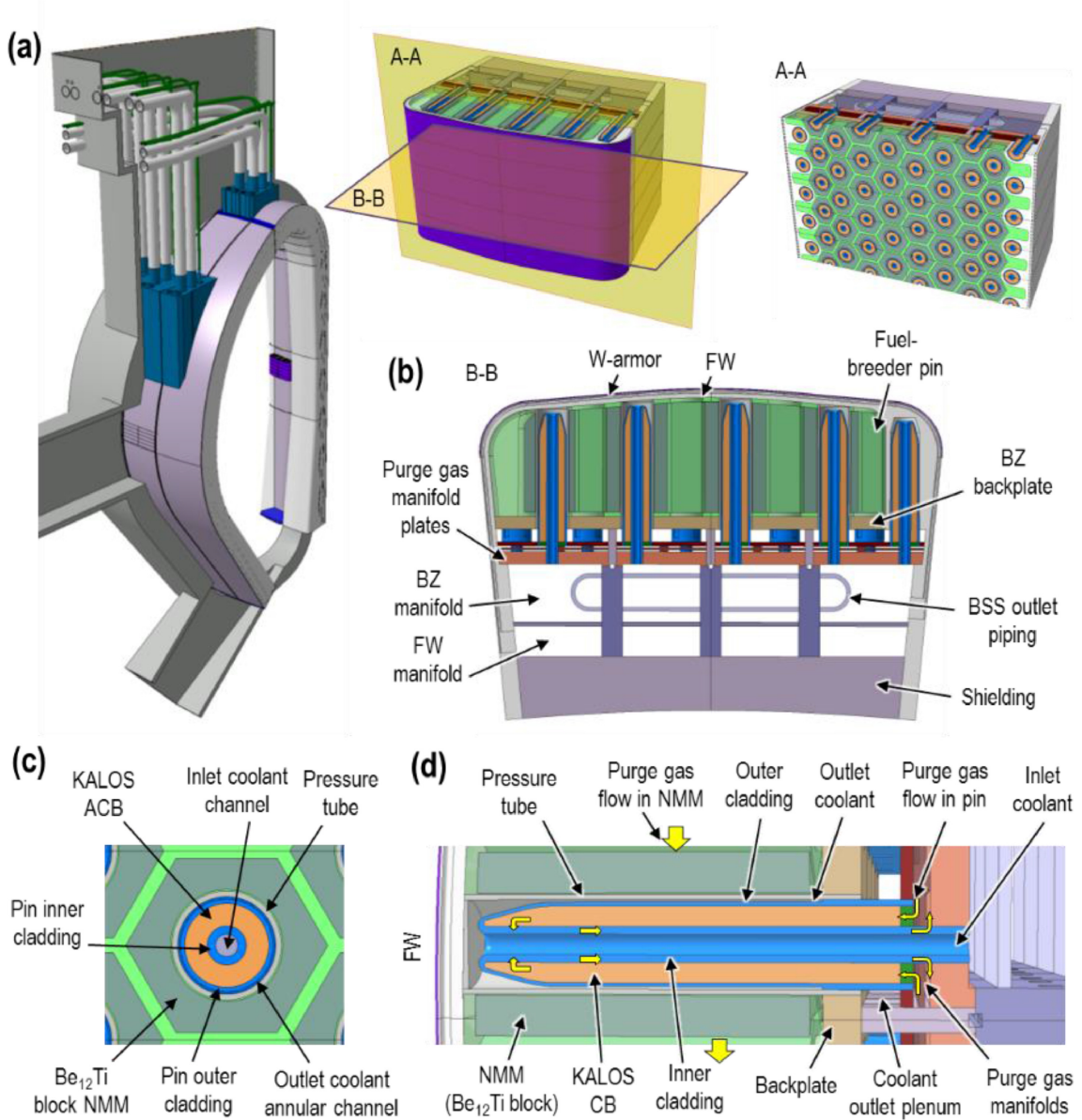


Figure 4.18: Fuel-breeder pin concept [41]. (a) 3D sketch of DEMO; (b) HCPB breeding blanket concept; (c) schematic cross-section of the pin; (d) longitudinal view of the fuel-breeder pin.

For the purpose of this thesis, we are interested in analyzing the fuel-breeder pin cross-section. The geometry is more complex than the model shown in Figure 4.4. In this configuration, two fluids operate simultaneously: helium and water. Both are used to remove tritium and cool the breeding blanket. However, as the heat removal capability of water is significantly higher than that of helium, this likely explains the

inclusion of water in the fuel-breeder pin design.

The implemented model is shown in Figure 4.19. As discussed in Section 4.1, the breeder material is Li_2TiO_3 , while the neutron multiplier material consists solely of Beryllium. The assembly is composed of several layers: an inner Eurofer layer (6 mm thick), a Li_2TiO_3 region with a thickness of 16 mm, and a second Eurofer layer (3 mm thick). These are followed by a 2 mm water channel, a pressure tube (assumed to be made of Eurofer) with a thickness of 4 mm, and finally an assumed 20 mm Beryllium layer [32].

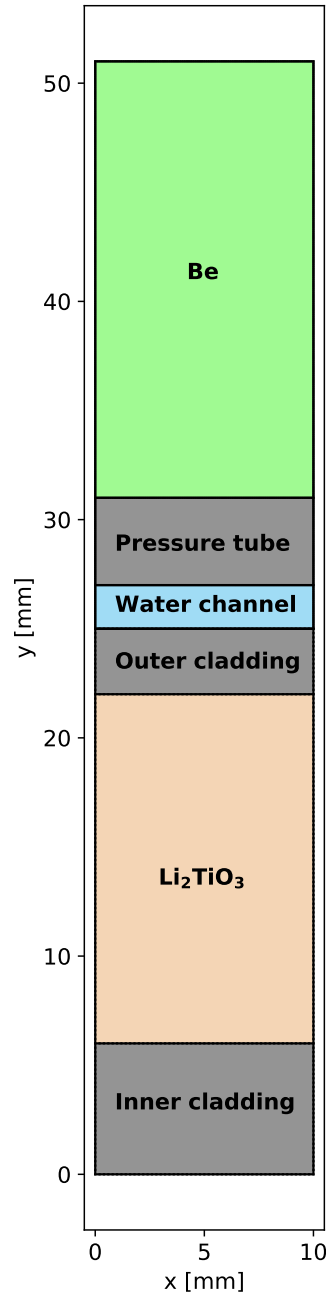


Figure 4.19: Model design of the Breeder-fuel pin. Eurofer ■, Breeder ■, Liquid water ■ and Neutron multiplier material ■.

4.2.1 Physical Parameters

The diffusivity and solubility of the materials utilized in this model are consistent with those from the preliminary study, with the addition of water. Table 4.10 summarizes the diffusion and solubility parameters for water. These are considered temperature-independent, as the values found in the literature refer to specific temperatures: 307 K for diffusivity and 302.47 K for solubility.

Table 4.10: Transport Properties of water.

Domain	Solubility [m ⁻³ Pa ⁻¹]	Diffusivity [m ² s ⁻¹]
Water [42] [43]	$7.54 \cdot 10^{-6}$	$9.27 \cdot 10^{-5}$

Traps are also included in this model. The corresponding trap parameters are consistent with those reported in Table 4.2 for Li₂TiO₃, Table 4.3 for each Eurofer layer and Table 4.4 for Beryllium.

The inclusion of a water channel between the two Eurofer layers requires an evaluation of both mass transfer and surface reaction phenomena.

Regarding surface reactions, the two primary processes involving tritium and water are recombination and isotopic exchange (see section 2.3.3). A significant challenge with isotopic exchange is its frequent deviation from Arrhenius-type behavior [44], which prevents its direct implementation in FESTIM. Consequently, only the recombination process is considered. The recombination parameters for the water channel (between the outer cladding and the pressure tube) are assumed to be identical to those reported for the interface between the water coolant and the inner cladding (Table 4.11). Similarly, the recombination reaction between the Beryllium neutron multiplier and the purge gas follows the parameters reported in Table 4.6

Table 4.11: Recombination equations for water.

Interaction	Recombination reaction [m ⁴ s ⁻¹]	Order
Water coolant and inner cladding [45]	$1.41 \cdot 10^{-26} \exp\left(\frac{2.57 \cdot 10^{-1} \text{ eV}}{k_B \cdot T}\right)$	2
Water channel and outer cladding [45]	$1.41 \cdot 10^{-26} \exp\left(\frac{2.57 \cdot 10^{-1} \text{ eV}}{k_B \cdot T}\right)$	2

Regarding mass transfer, the water properties for the water coolant and the water in the channel were evaluated at a pressure of 80 bar and a temperature of 548 K. This temperature was determined by subtracting a 20 K safety margin from the saturation temperature at 80 bar. The water velocities and their respective regimes are reported in Table 4.12. Although the water channel velocity corresponds to a turbulent regime, a relatively low value was chosen to minimize corrosion effects.

Table 4.12: Velocity parameters of the water fluids.

Fluid	Pressure [bar]	Velocity [m s ⁻¹]	Regime
Water Coolant [37]	80	2	Turbulent
Water Channel [37]	80	$5 \cdot 10^{-2}$	Turbulent

The mass transfer coefficients were evaluated as described in Section 2.3.3. Water properties were obtained from [39], while the Sherwood number was determined via the Sieder-Tate correlation [15]. Sherwood number and mass transfer coefficient values are reported in Table 4.13.

Table 4.13: Mass transfer values implemented in the model.

Interaction	Sherwood number	Mass transfer coefficient h_m [m s ⁻¹]
Inner cladding and water coolant	63.2	$3.66 \cdot 10^{-1}$
Water channel and outer cladding	1.09	$2.53 \cdot 10^{-2}$

The mass transfer coefficient for the purge gas is the same as that applied in the pre-conceptual model; these values are reported in Table 4.7.

4.2.2 2D model

Due to the complexity of the model, the whole domain has been divided into two separate domains, simulated independently. Figure 4.20 shows both domains and illustrates how the boundary conditions are applied. The first domain comprises the inner cladding, the Li_2TiO_3 , and the outer cladding; the second, instead, simulates the water, the pressure tube, and the neutron multiplier material. The same volumetric source term previously described in the pre-conceptual model (Section 4.1.2) is applied in this study.

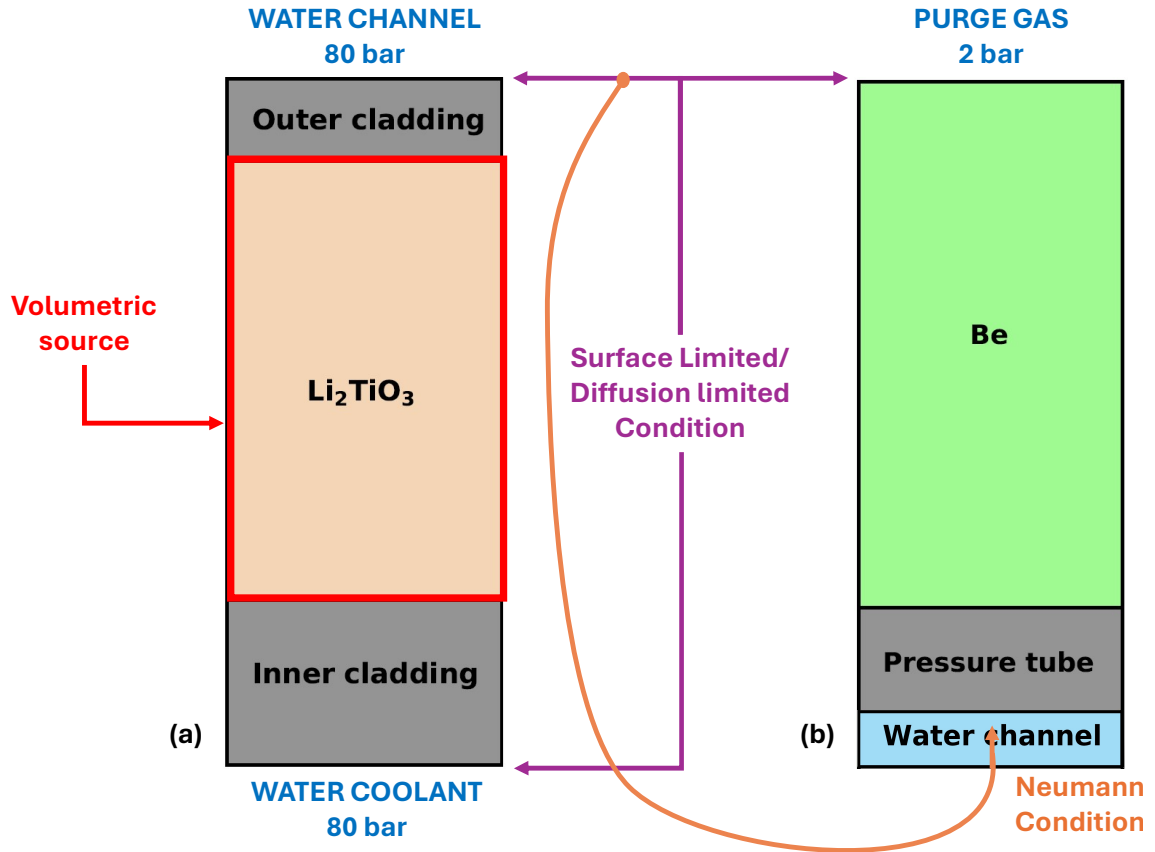


Figure 4.20: Decomposed simulation domains: (a) first domain including inner cladding, Li_2TiO_3 breeder, and outer cladding; (b) second domain comprising the water channel, pressure tube, and Beryllium multiplier.

The difficulty of this geometry lies in the modeling of the water channel. For this reason, the first domain is simulated using either a surface-dominant or a diffusion-dominant boundary condition on the outer cladding. After this simulation, an inlet boundary flux can be retrieved from the first simulation of each case (recombination and mass transfer) and applied to the water layer to simulate tritium permeation into the second domain. The Neumann boundary conditions applied in the second model are resumed in table 4.14.

Table 4.14: Neumann boundary conditions applied at the inlet of the second domain.

Boundary condition in the outer cladding	Neumann boundary condition value [m^{-2}]
Recombination	$1.0764 \cdot 10^{15}$
Mass transfer	$1.0715 \cdot 10^{15}$

This is a simplification of the physics involved in the system. In this case, mass transfer and recombination effects are not considered for the surface in contact with the pressure tube layer.

The boundary conditions for the external edges can be summarized as follows:

- **Coolant Water:** At 80 bar and 548 K, the water is in contact with the inner cladding, acting as both a coolant and a tritium carrier. Both surface-limited and diffusion-limited regimes are simulated.
- **Purge Gas:** At 2 bar and 900 K, the purge gas skims the Beryllium to remove any permeating tritium. These effects are also simulated for both surface-limited and diffusion-limited cases.
- **Horizontal Boundary Conditions:** It is assumed that there is no preferential horizontal direction for heat or mass transport. Consequently, where the flux entering the domain is equal to the flux exiting it.

The water in the channel follows the velocity profile shown in Figure 4.21, which represents a simplified turbulent flow. Consequently, the advection is explicitly included in the model.

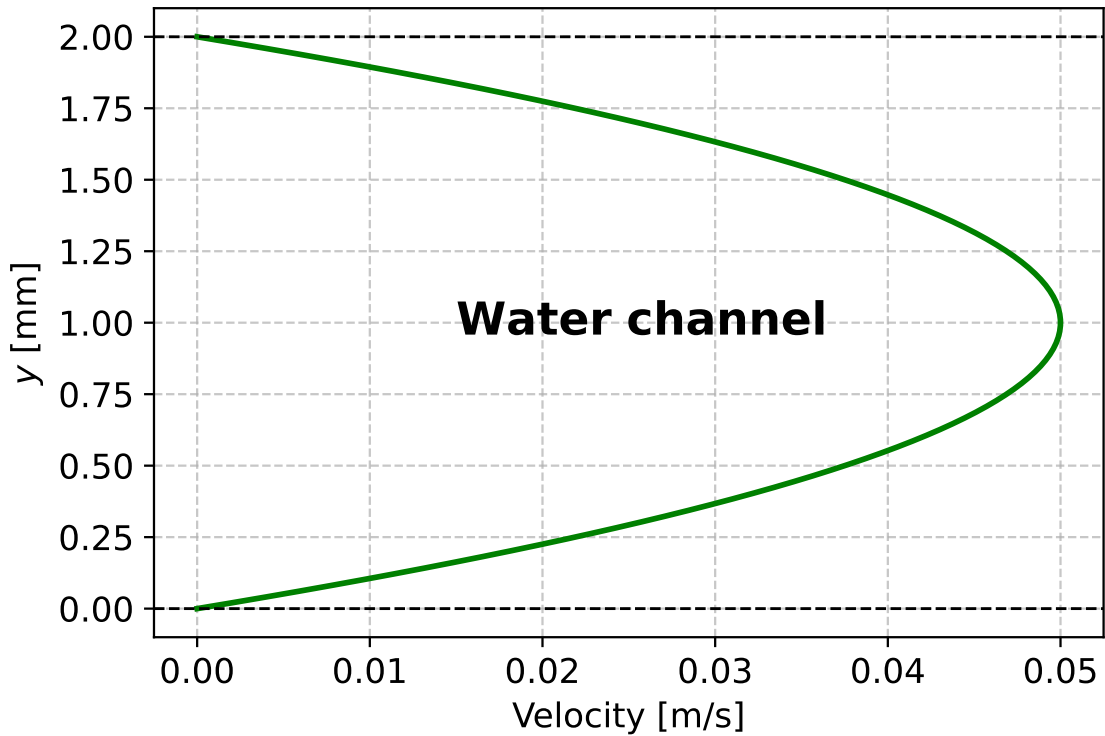


Figure 4.21: Velocity profile of the water in the channel.

In this model, nuclear heating is accounted for in both the Li_2TiO_3 and Beryllium layers. As shown in Figure 4.22, the temperature rise in the breeder material is primarily driven by the exothermic ${}^6\text{Li}(n,\alpha)$ reaction [1]. In the Beryllium multiplier, although the $(n,2n)$ reaction is endothermic, a significant temperature increase occurs due to neutron energy deposition through elastic scattering; these interactions transfer kinetic energy to the Beryllium nuclei, inducing lattice vibrations that result in sensible heat generation [1]. The assumed temperature profile does not account for potential thermo-mechanical effects on the materials.

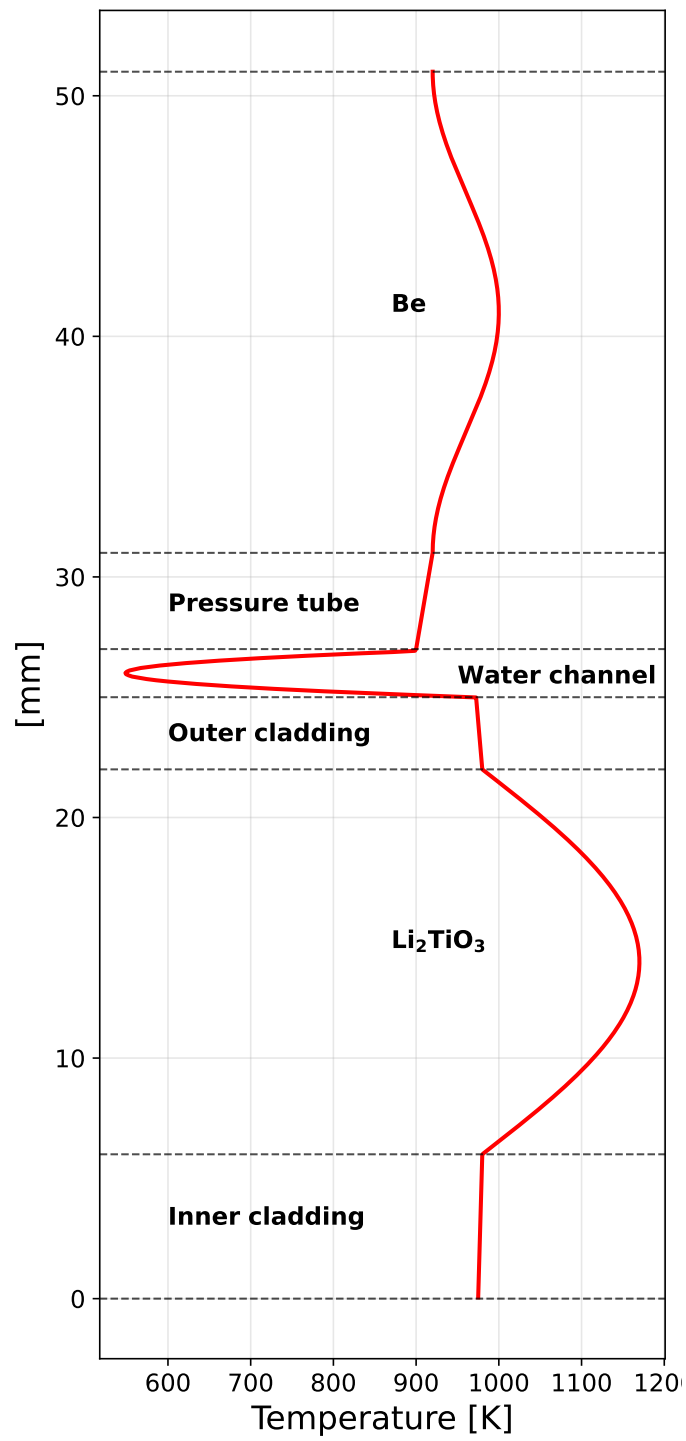


Figure 4.22: Temperature profile of the breeder fuel pin.

4.2.3 2D model results

This section is dedicated to the discussion of the results. Figures 4.23 and 4.24 illustrate the mobile concentration distribution within the domain. Specifically, Figure 4.23 displays the results for the surface-limited case, while Figure 4.24 reports the results for the diffusion-limited case.

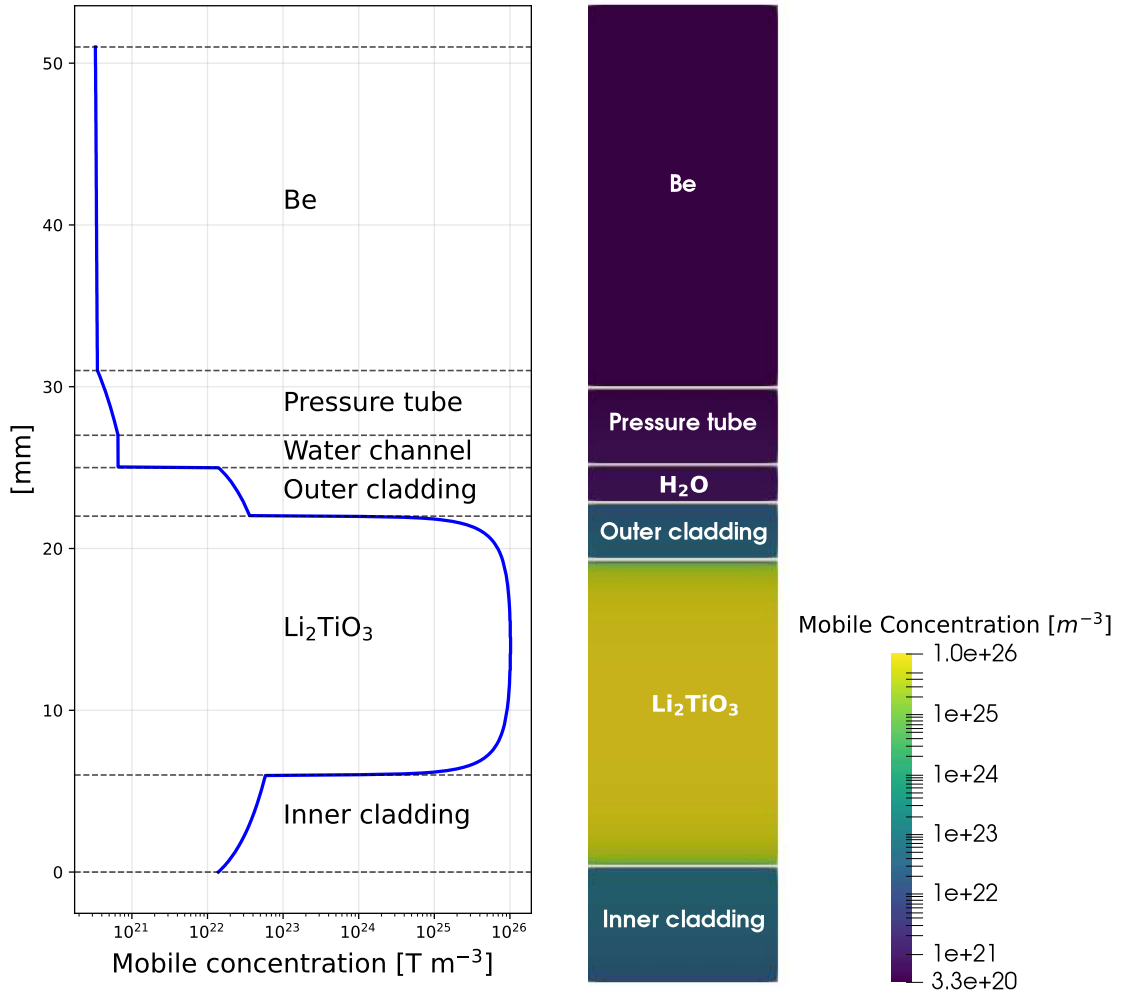


Figure 4.23: 2D model and radial profile of mobile concentration in the fuel breeder pin considering the recombination boundary conditions.

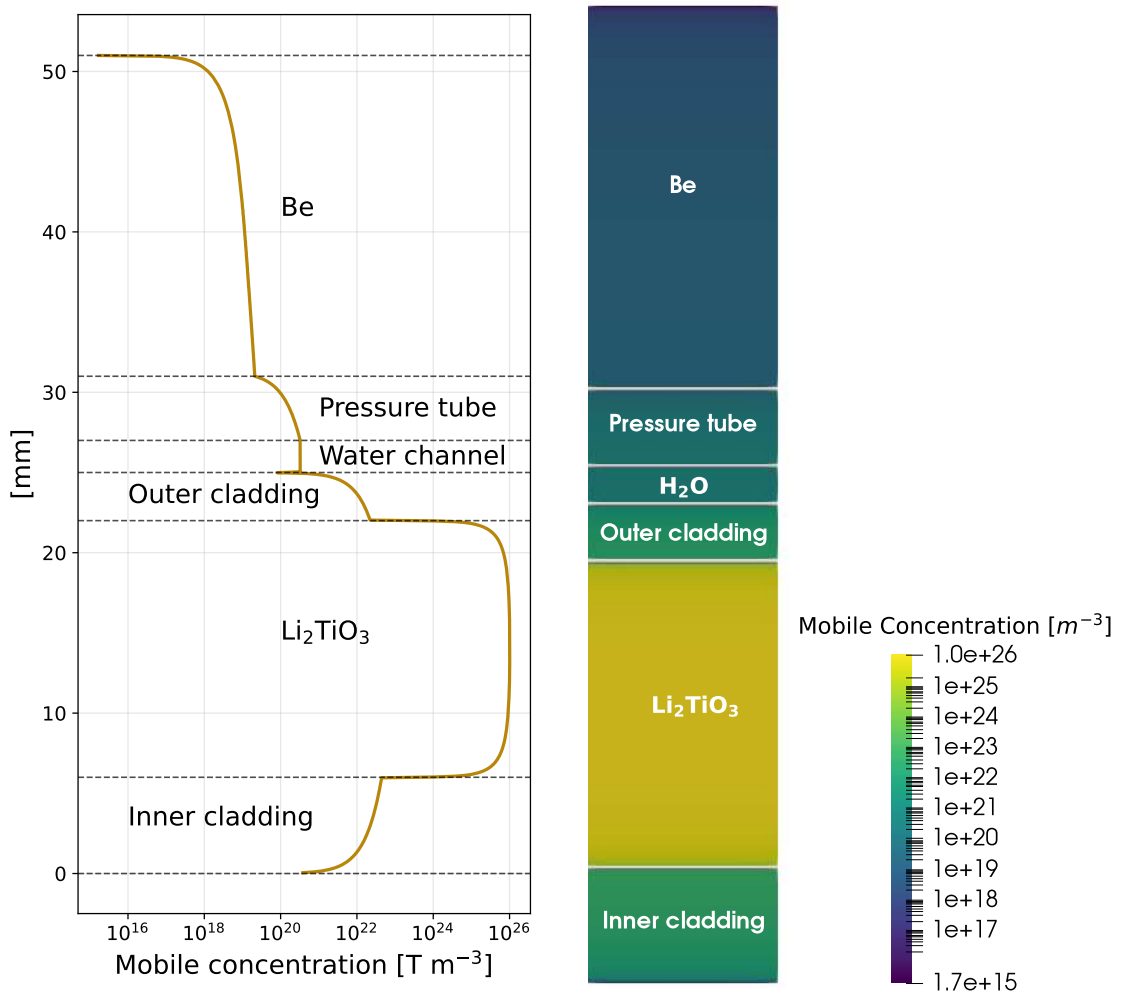


Figure 4.24: 2D model and radial profile of mobile concentration in the fuel breeder pin considering the mass transfer boundary conditions.

As discussed in Section 4.1.3, recombination effects are less effective at removing tritium from the domain boundaries compared to the use of a mass transfer coefficient. This difference in removal capability also impacts the distribution of mobile concentration within the domain.

When comparing Figure 3.12 with Figures 4.23 and 4.24, the mobile concentration in the DEMO fuel breeder pin is up to four times higher than in the ARC breeder case. This directly affects the carrier's ability to extract tritium from the breeding blanket. Specifically, the tritium extraction efficiency must be increased to prevent significant tritium losses.

This leads to further technical challenges: if water is used as a carrier, the tritium concentration must be strictly limited to satisfy radiation protection requirements. However, implementing a highly efficient Water Detritiation System (WDS) is particularly challenging due to the isotopic properties of tritium in water. Conversely, for helium, an accumulation of tritium increases the partial pressure within the gas mixture, potentially leading to enhanced tritium permeation through the cooling loop.

The trapped concentration distributions for both the diffusion-limited and recombination-limited cases are shown in Figures 4.25 and 4.26.

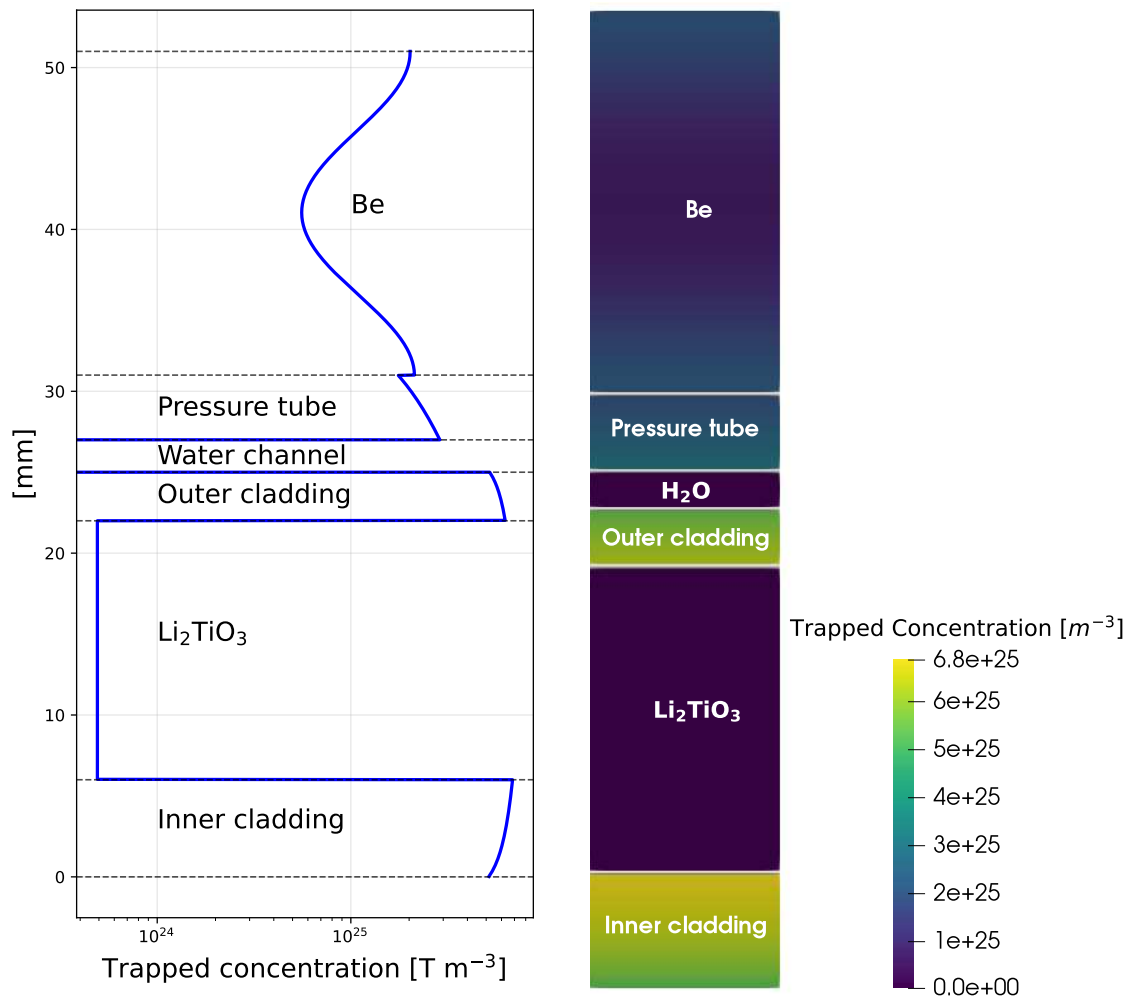


Figure 4.25: 2D model and radial profile of trapped concentration considering the recombination boundary conditions.

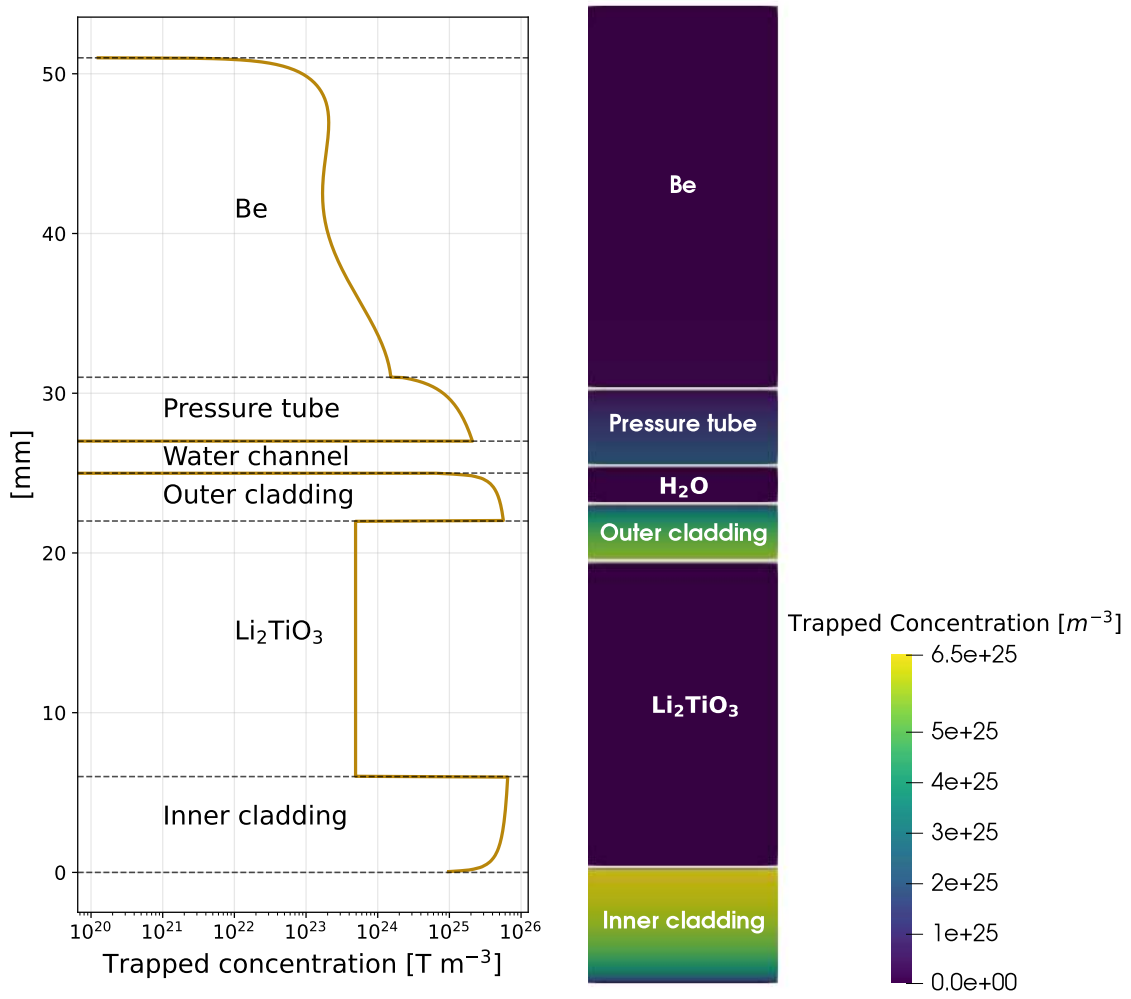


Figure 4.26: 2D model and radial profile of trapped concentration considering the diffusion limited case boundary conditions.

Both boundary conditions result in a comparable amount of trapped tritium within the fuel breeder pin. Since the water channel layer does not contain trapping sites, the trapped concentration in this region is zero.

A key observation is the saturation of traps within the breeder layer, which is primarily governed by the volumetric source term and the physical properties of the traps in both cases (Figures 4.29 and 4.30). This saturation enhances tritium migration toward the inner cladding (Figures 4.27, 4.28) and the outer cladding (Figures 4.31, 4.32). The residual tritium that is not removed and instead diffuses toward the water eventually reaches the pressure tube layer (Figures 4.33 and 4.34).

Finally, the mobile (untrapped) tritium enters the Beryllium layer (Figures 4.35 and 4.36), where it becomes trapped. The radial profiles in this region highlight the combined influence of the temperature distribution and the purge gas flow at the top of the domain.

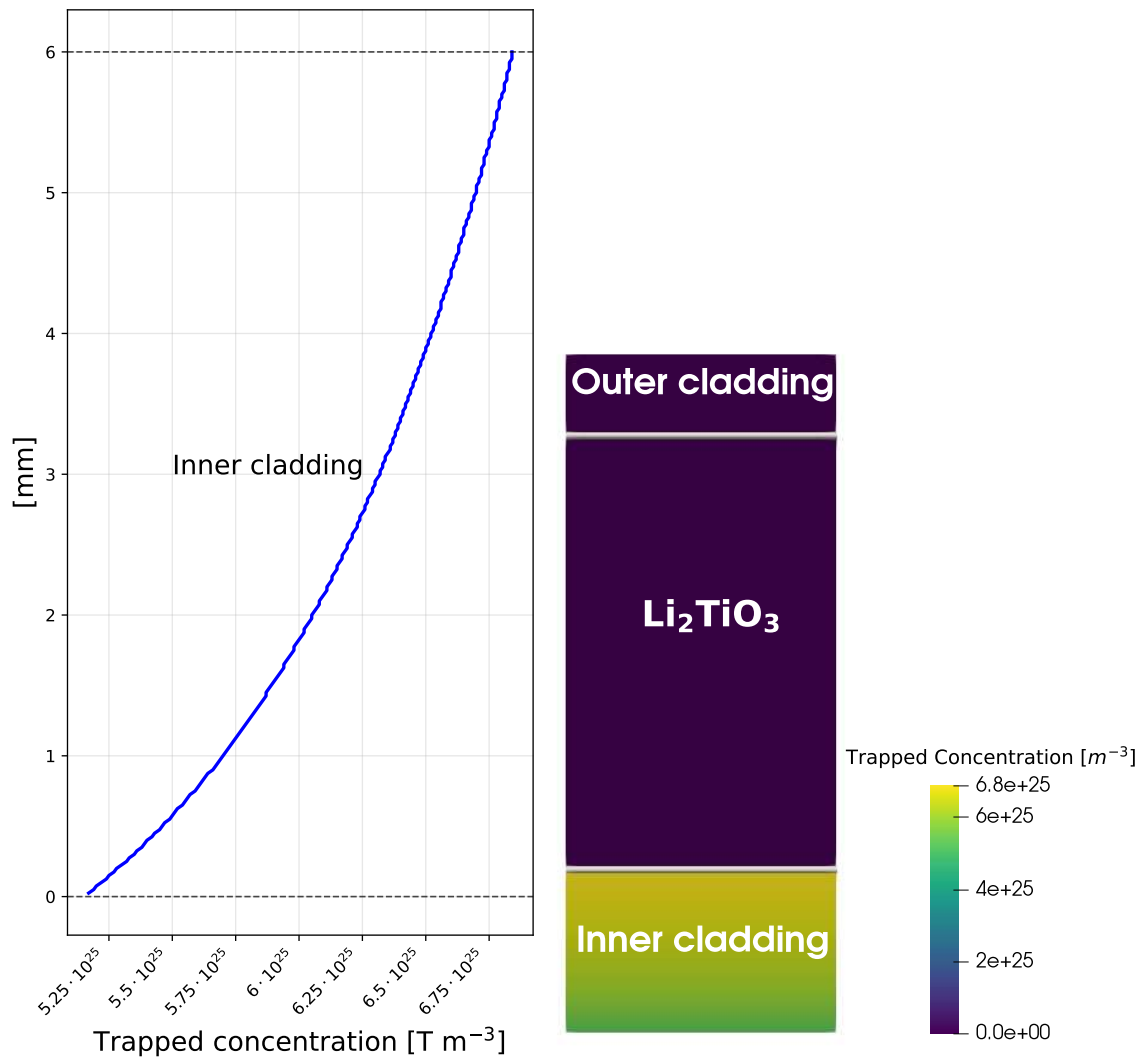


Figure 4.27: 2D model and radial profile of trapped concentration in the inner cladding layer considering the recombination boundary conditions.

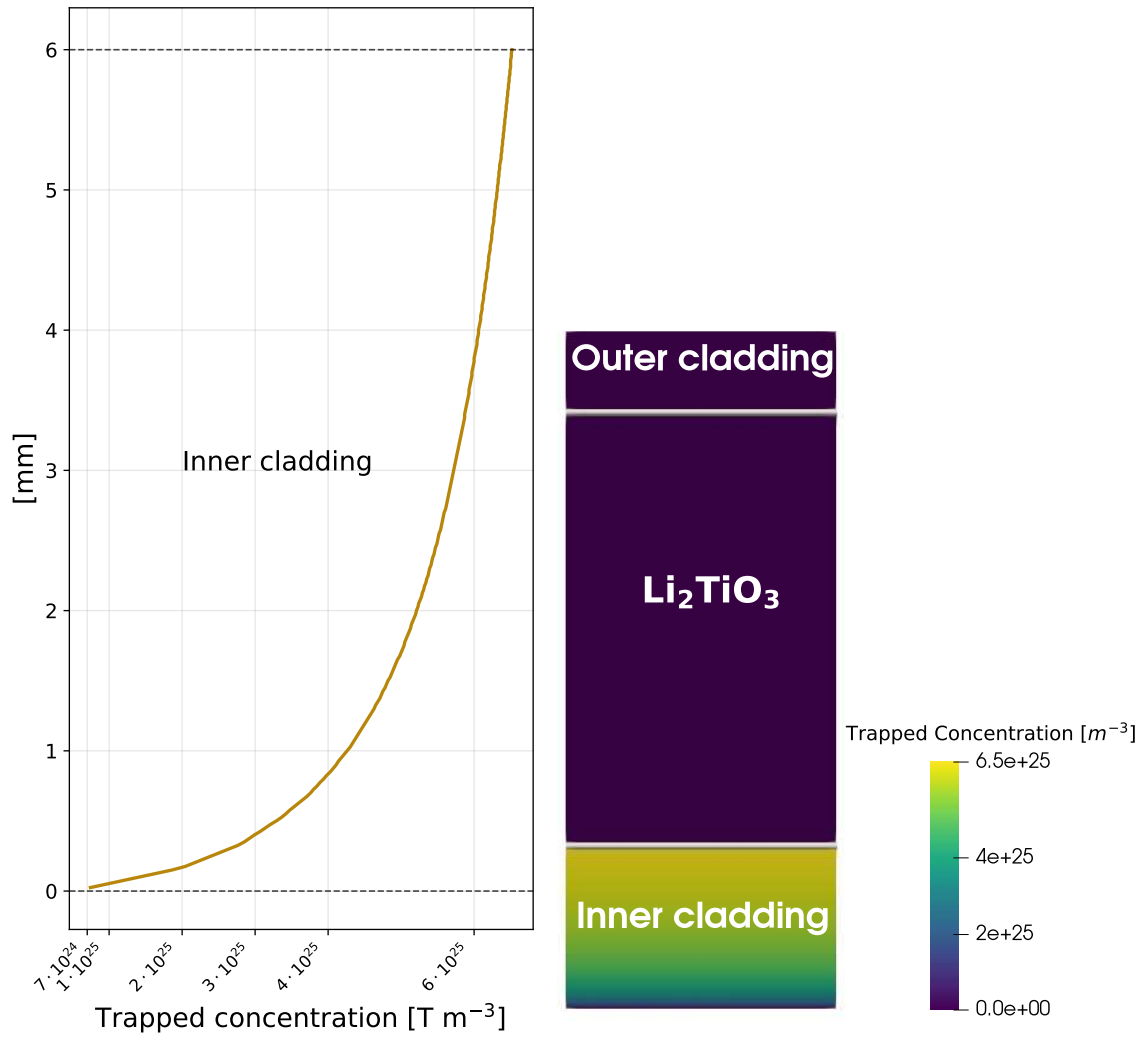


Figure 4.28: 2D model and radial profile of trapped concentration in the inner cladding layer considering the diffusion limited case boundary conditions.

Considering the temperature distribution in the inner cladding (Figure 4.22), thermal effects do not significantly influence the trapping and de-trapping behavior in either case (Figures 4.27, 4.28). This implies that the higher concentration of trapped tritium observed at the top of the layer is solely driven by the higher mobile tritium concentration entering the sub-domain, where it is immediately captured. Consequently, as the mobile concentration reaching the boundary gradually decreases, the trapping probability—and thus the resulting trapped inventory—is reduced accordingly.

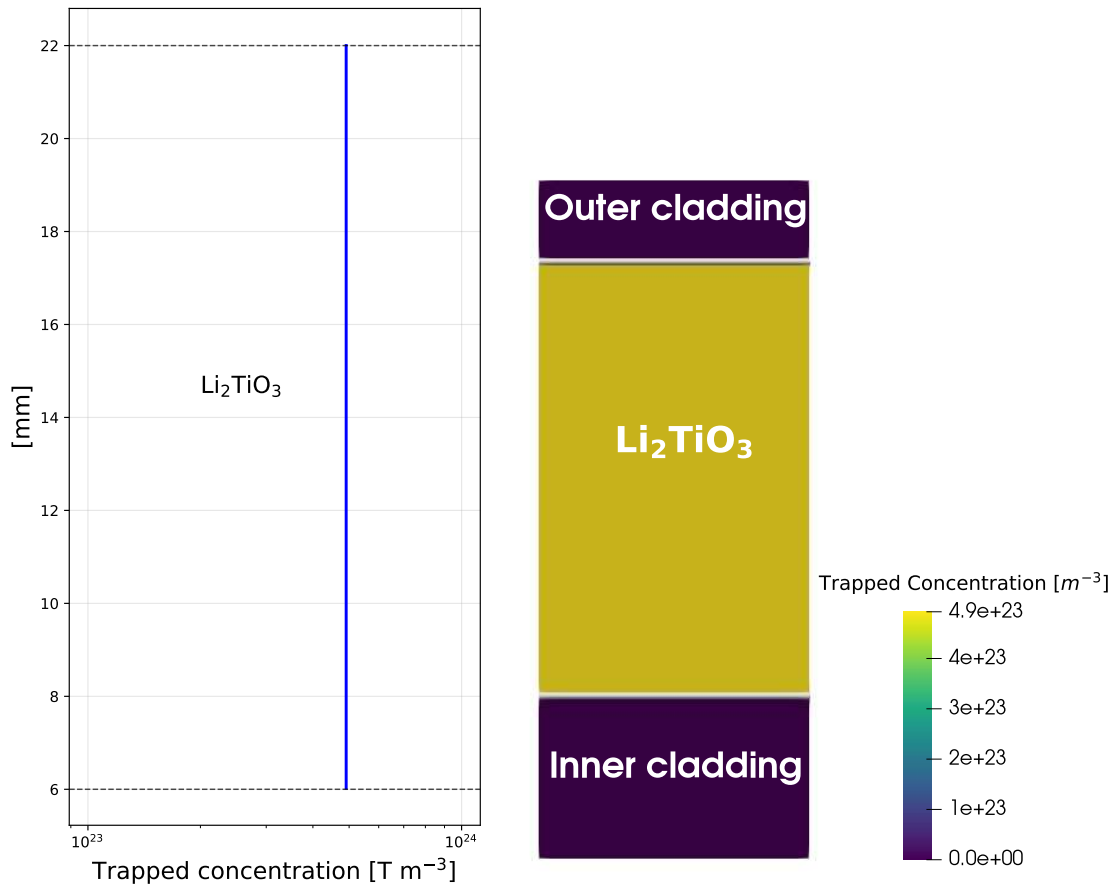


Figure 4.29: 2D model and radial profile of trapped concentration in the breeder considering the recombination boundary conditions.

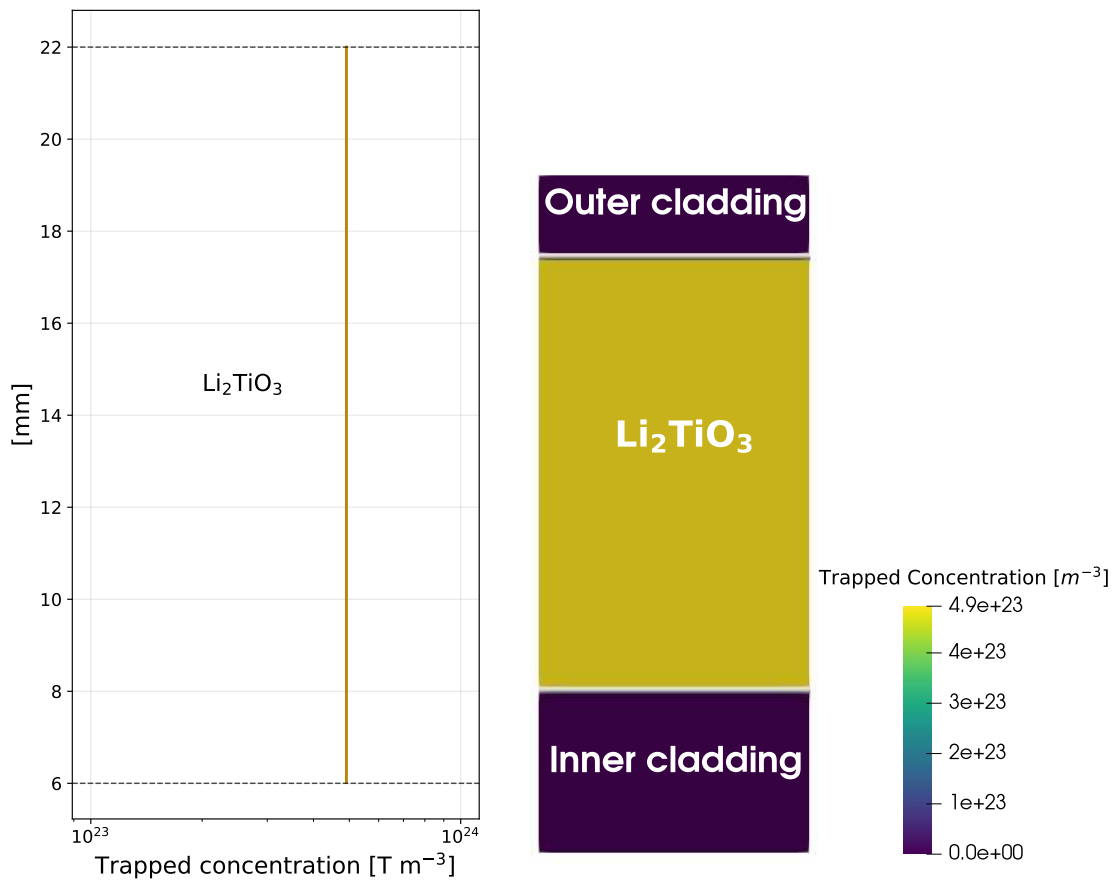


Figure 4.30: 2D model and radial profile of trapped concentration in the breeder considering the recombination boundary conditions.

As previously explained, trap saturation is achieved in both configurations. This phenomenon is also evident in the preliminary design model of the fuel-breeder pin (Figures 4.12 and 4.13).

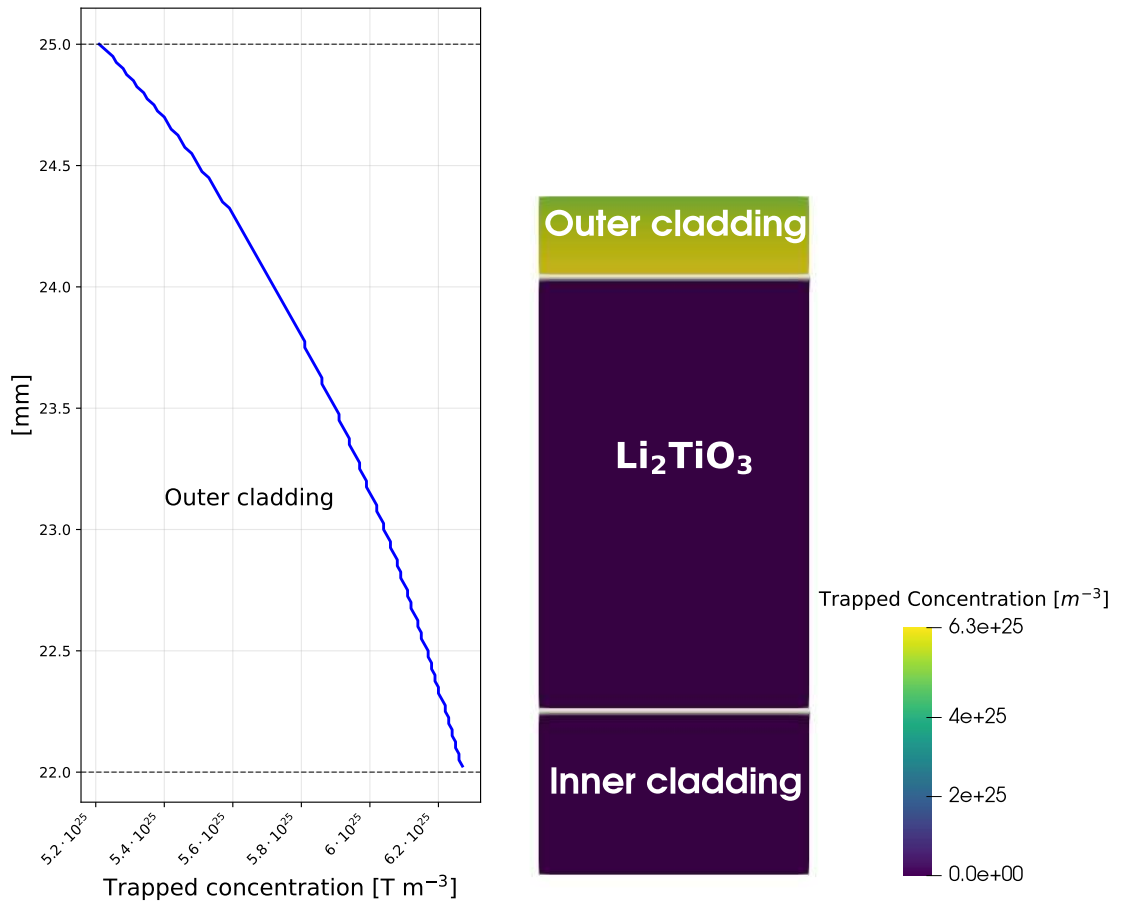


Figure 4.31: 2D model and radial profile of trapped concentration in outer cladding layer considering the recombination boundary conditions.

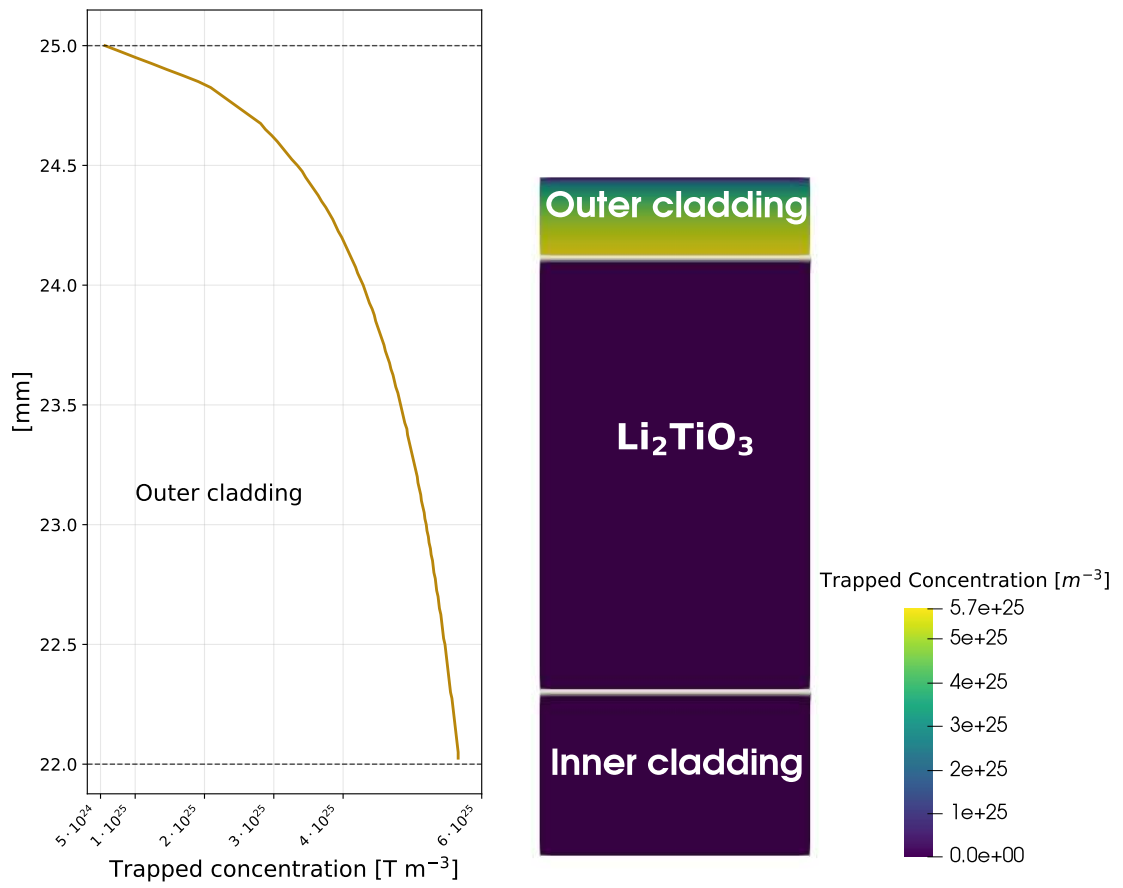


Figure 4.32: 2D model and radial profile of trapped concentration in the outer cladding layer considering the diffusion boundary conditions.

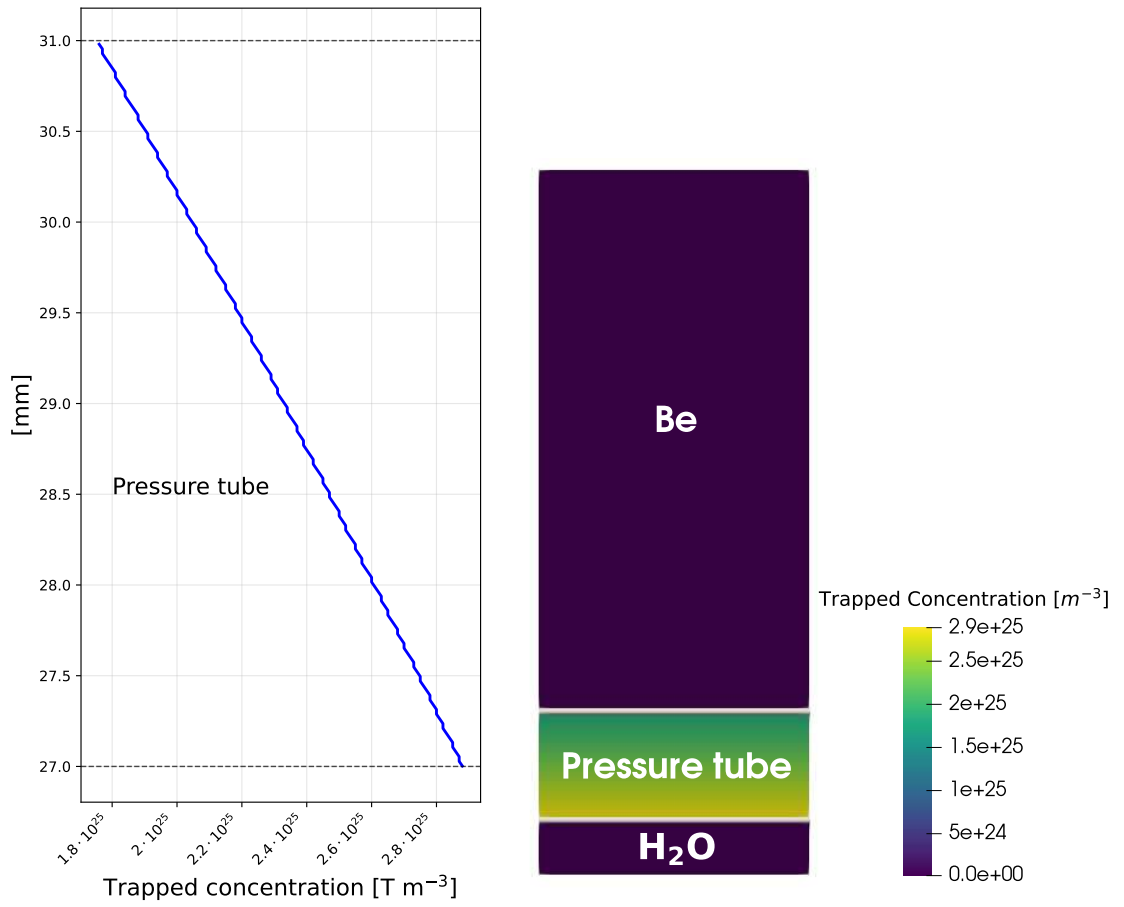


Figure 4.33: 2D model and radial profile of trapped concentration in pressure tube layer considering the recombination boundary conditions.

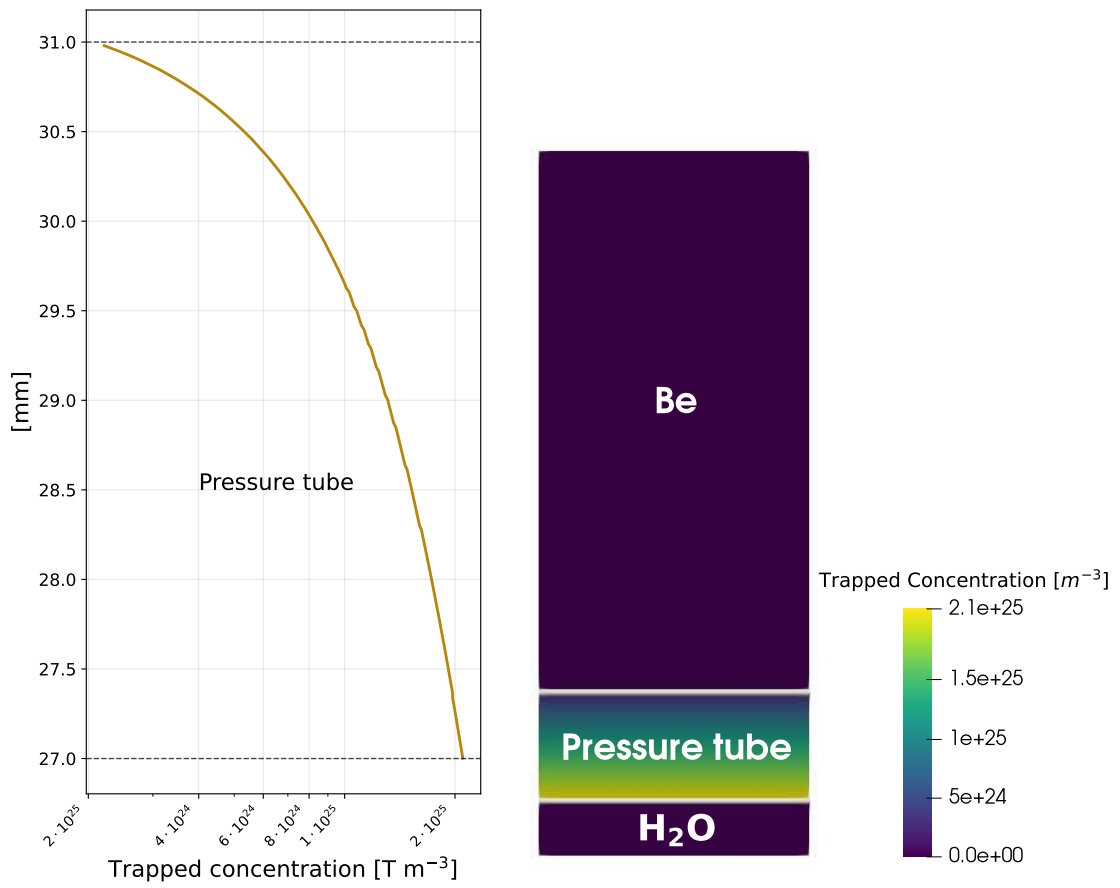


Figure 4.34: 2D model and radial profile of trapped concentration in pressure tube layer considering the mass transfer coefficient boundary conditions.

Regarding the trapped tritium in the outer cladding (Figures 4.31 and 4.32) and pressure tube layers (Figures 4.33 and 4.34), the 2D distributions reveal a gradient in the trapped concentration; however, the profiles show only a very slight reduction in concentration across the domain.

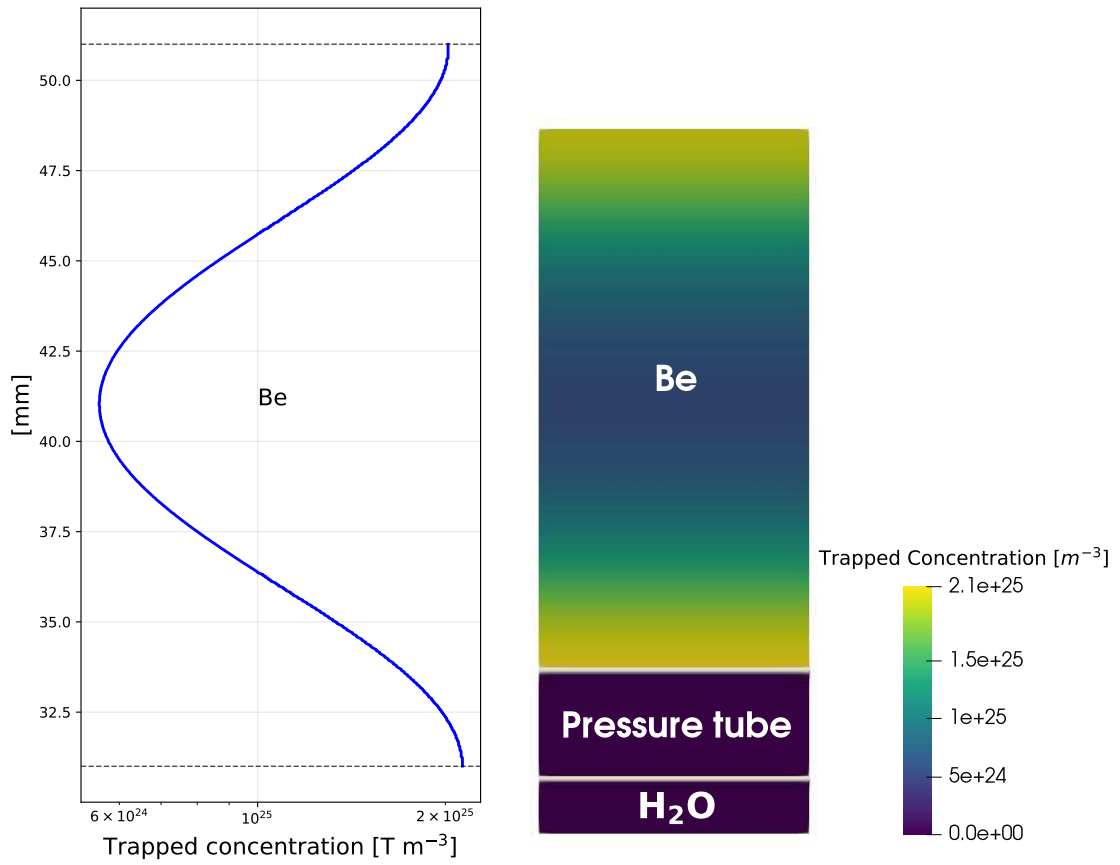


Figure 4.35: 2D model and radial profile of trapped concentration in Beryllium considering the recombination boundary conditions.

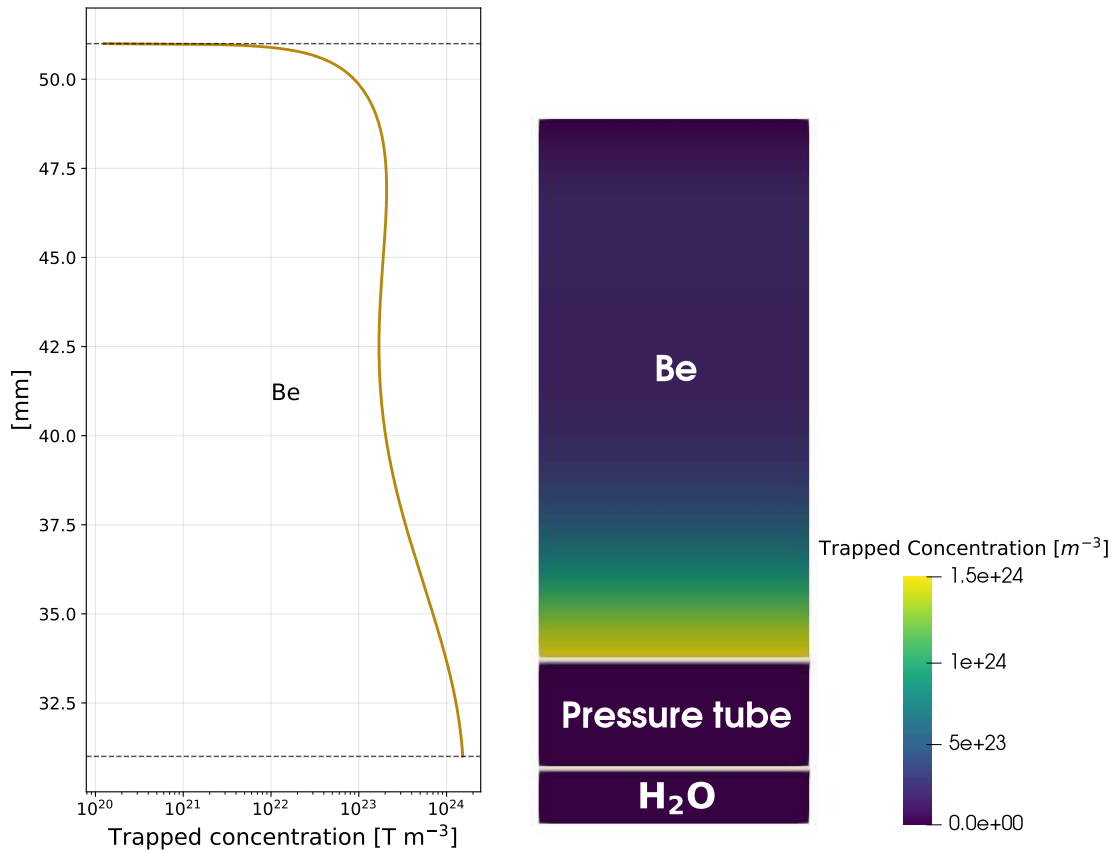


Figure 4.36: 2D model and radial profile of trapped concentration in Beryllium considering the recombination boundary conditions.

A particularly interesting effect can be observed in the Beryllium layer. The temperature distribution (Figure 4.22) strongly influences the trapped tritium in both configurations (Figures 4.35 and 4.36). Specifically, a decrease in trapped concentration is observed in regions of maximum temperature. This phenomenon occurs because the tritium atoms acquire sufficient thermal energy to overcome the binding energy of the traps. This behavior is consistent with the Arrhenius dependence of the de-trapping rate, where higher temperatures exponentially increase the probability of tritium release from trapping sites. Conversely, as tritium migrates into regions of lower temperature, the trapped concentration increases. This confirms that cooler zones act as preferential sinks for tritium accumulation, as the thermal energy is insufficient to sustain high de-trapping rates. The final consideration relates to the effect of the purge gas at the top of the layer. The reduction in mobile concentration near the boundary leads to a corresponding decrease in trapped tritium.

The average tritium concentration and the corresponding activity for each layer, considering the two different boundary conditions, are presented in Tables 4.15 and 4.16. A primary observation is that the larger amount of structural materials leads to a higher tritium trapped within the breeder blanket. In fact, when comparing the average trapped tritium in ARC (Table 3.6) with the trapped concentration in DEMO, the total amount of tritium and the resulting activity in the DEMO breeder are significantly higher than those in ARC. Furthermore, the comparison of the two boundary conditions, surface limited case and diffusion limited case, highlights a decrease in trapped tritium within the Beryllium layer for the diffusion-dominant case.

Table 4.15: Concentration and activity per unit of volume of trapped concentration with Surface limited condition in the fuel breeder pin concept.

Material	Concentration [g m ⁻³]	Activity [Bq m ⁻³]	Activity [Ci m ⁻³]
Inner cladding	$3.12 \cdot 10^2$	$1.11 \cdot 10^{17}$	$3 \cdot 10^6$
Li ₂ TiO ₃	2.46	$8.75 \cdot 10^{14}$	$2.37 \cdot 10^4$
Outer cladding	$2.93 \cdot 10^2$	$1.04 \cdot 10^{17}$	$2.82 \cdot 10^6$
Pressure tube	$1.17 \cdot 10^2$	$4.15 \cdot 10^{16}$	$1.12 \cdot 10^6$
Beryllium	$5.97 \cdot 10^2$	$2.13 \cdot 10^{16}$	$5.75 \cdot 10^5$

Table 4.16: Concentration and activity per unit of volume of trapped concentration with Diffusion limited condition in the fuel breeder pin concept.

Material	Concentration [g m ⁻³]	Activity [Bq m ⁻³]	Activity [Ci m ⁻³]
Inner cladding	$2.66 \cdot 10^2$	$9.46 \cdot 10^{16}$	$2.56 \cdot 10^6$
Li ₂ TiO ₃	2.46	$8.75 \cdot 10^{14}$	$2.37 \cdot 10^4$
Outer cladding	$2.22 \cdot 10^2$	$7.92 \cdot 10^{16}$	$2.14 \cdot 10^6$
Pressure tube	$6.3 \cdot 10^1$	$2.24 \cdot 10^{16}$	$6.06 \cdot 10^5$
Beryllium	2.08	$7.42 \cdot 10^{14}$	$2.01 \cdot 10^4$

Chapter 5

Conclusion

Given the scarcity of tritium and the necessity of breeding it within the plant itself, accurately modeling trapping phenomena is essential. Neglecting these effects results in an underestimation of the required TBR and can compromise the tritium self-sufficiency of the fusion reactor. The analysis of trapping phenomena was investigated with models representative of the ARC and DEMO breeding blanket concept. The developed models show coherence with existing literature results [4]. Initially, a 1D model was employed for the preliminary assessment of the blanket and its trapping mechanisms. Subsequently, a more sophisticated 2D model was implemented, incorporating the advection term within the FLiBe flow. While the fluid motion successfully removes a portion of the tritium produced within the fluid itself, this recovery is incomplete, due to the competition between the advective transport of the fluid and the permeation flux directed toward the surrounding solid boundaries.

Both 1D and 2D models highlight a critical feature of this blanket design: tritium generated in the FLiBe permeates into the structural materials, leading to significant trapping within the structural components. Simultaneously, a secondary tritium flux originating from the plasma permeates through the Tungsten first wall, resulting in further trapping within the first layer of the breeding blanket.

The focus of the study then shifts to the analysis of trapping phenomena within the DEMO breeding blanket. In this context, a ceramic configuration was selected to investigate the trapping effects, representing a novel contribution to the literature.

The first preliminary model is based on the pre-conceptual design of the fuel breeder pin. This model was used to understand the phenomenology of trapping in ceramic material.

Then, the design of the fuel breeder pin is modeled in a 2D domain. The primary focus is on the quantity of mobile tritium permeating through the various materials. A comparison between the DEMO and ARC breeding blanket concepts reveals that the mobile tritium concentration in the DEMO design is several orders of magnitude higher than that observed in the ARC configuration. This suggests a higher efficiency of a dual-function fluid (breeder and carrier) compared to a solid breeder with a dedicated fluid carrier.

The use of solid materials introduces a significant volume of structural and breeding media that inherently contain trapping sites due to neutron damage. Furthermore, the ceramic breeder itself exhibits a high trap density that can reach saturation levels. This phenomenon is detrimental to tritium recovery, as it constitutes a strong tritium sink.

A final, non-negligible aspect is the comparison of the total average tritium inventory trapped within the ARC and DEMO breeder blanket concepts. The extensive structural materials required for the DEMO design is responsible for a significantly higher quantity of trapped tritium compared to the ARC configuration. In contrast, the ARC design utilizes a reduced volume of structural materials, consequently leading to a lower overall tritium retention.

In conclusion, a liquid immersion blanket design appears more favorable in terms of minimizing total tritium retention as a result of its compact structural volume, while the ceramic breeder configuration provides a robust and well-understood technological solution, which is more susceptible to tritium trapping.

Future research should focus on the impact of neutron irradiation on material properties and trapping phenomena, while further refining studies on diffusion and solubility parameters. These data are fundamental to modeling trapping phenomena with greater precision and to elucidating the long-term evolution of the tritium inventory under operational conditions.

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